

Building and Environment

A Digital Twin-enabled Adaptive Optimization Framework for Balancing Holistic Comfort and Energy Efficiency in Underground Spaces

--Manuscript Draft--

Manuscript Number:	BAE-D-26-04271R2
Article Type:	Original Research Paper
Keywords:	Digital twin; Adaptive environmental management; Holistic comfort; Machine learning; Energy efficiency.
Corresponding Author:	Zijian Ye Beijing Institute of Fashion Technology CHINA
First Author:	Zijian Ye
Order of Authors:	Zijian Ye Tianlong Liang Yu Chen
Abstract:	Underground space environmental control often faces the dual challenge of insufficient passenger comfort and high system energy consumption. To balance holistic comfort and energy efficiency, this study proposes a digital twin-enabled adaptive optimization framework for underground environments. Environmental data were collected automatically through distributed thermal, acoustic, and lighting sensors, and a structured database was established. Field measurements, subjective comfort surveys, machine learning, and virtual simulations were integrated to construct a digital twin-enabled adaptive optimization framework balancing energy consumption and environmental comfort. A four-level early-warning and strategy-selection mechanism was evaluated in a calibrated simulation environment to coordinate comfort status and energy use under different operational scenarios. A case study was conducted at a Beijing subway station in a cold region to assess the potential effectiveness of the framework. Simulation results indicate that the strategy effectively reduced low-comfort periods across functional areas; uncomfortable periods in the station hall during summer decreased from 20.4% to 3.1%. Total annual energy consumption and operational carbon emissions both decreased by 15.3%, with lighting and ventilation contributing 32.5% and 22.8% of the energy savings, respectively. This study provides a methodological reference for comfort-oriented and low-carbon environmental management in existing underground transportation spaces.

Cover Letter

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**A Digital Twin Approach Balancing Holistic Comfort and Energy Efficiency for Proactive Environmental Control in Underground Spaces**” for consideration for publication in **Building and Environment**.

This study addresses a challenge in underground built environments: how to improve passenger comfort while reducing operational energy consumption. Subway stations are enclosed, high-density spaces where thermal conditions, lighting, acoustics, air quality, and passenger crowding interact dynamically. Conventional environmental control strategies based on fixed schedules or single-threshold triggers often have limited flexibility in responding to these dynamic conditions, resulting in insufficient comfort during peak periods and energy waste during off-peak periods.

To address this issue, we propose a digital twin-based proactive environmental control framework for underground spaces. The proposed approach combines field measurements, passenger comfort surveys, machine learning-based comfort evaluation, visualization modeling, and energy simulation to identify and optimize environmental states in underground spaces. Its effectiveness was validated through a case study of a Beijing subway station.

Our study offers several contributions to the field:

1. **Holistic comfort assessment in underground spaces:** The study integrates thermal, lighting, acoustic, air-quality, and passenger crowding data to evaluate overall passenger comfort beyond single-factor environmental assessment.
2. **Digital twin-based proactive control framework:** The proposed framework combines physical sensing, data management, machine learning evaluation, virtual simulation, and service-layer decision support to enable dynamic environmental state identification and optimization.
3. **Balance between comfort improvement and energy efficiency:** The case study demonstrates that the proposed strategy can reduce low-comfort periods across functional areas. In the summer station hall, the proportion of uncomfortable periods decreased from 20.4% to 3.1%.
4. **Quantified energy-saving and carbon-reduction effects:** The optimized strategy reduced total annual station energy consumption by 15.25% and carbon emissions by 15.32%, with lighting and ventilation systems contributing 32.45% and 22.83% of the energy savings, respectively.

We believe this work will be of great interest to the readership of *Building and Environment*, as it contributes to indoor environmental quality, occupant comfort, building performance simulation, energy-efficient operation, and low-carbon management in complex built environments. The findings provide both methodological insights and practical guidance for improving environmental control in existing underground transportation spaces.

This manuscript is an original work and has not been published or submitted for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Building and Environment*.

Thank you for your time and consideration. We look forward to hearing from you.

Sincerely yours,

Zijian Ye

(on behalf of all authors)

e-mail: zijianye@bift.edu.cn

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Response to Editor and Reviewers

Manuscript title: A Digital Twin-enabled Adaptive Optimization Framework for Balancing Holistic Comfort and Energy Efficiency in Underground Spaces

Manuscript ID: BAE-D-26-04271

Dear Editor and Reviewers,

We sincerely thank the Editor and Reviewers for their careful evaluation and constructive comments on the revised manuscript. In response to the second-round comments, we have further revised the manuscript to improve the methodological clarity, interpretation of the results, and scope of the proposed framework. The major revisions in this round include:

- (1) clarification of the meaning and implementation scope of the adaptive optimization framework, including the distinction among strategy generation, simulation-based performance evaluation, real-time monitoring, and operational decision support;
- (2) clarification of the treatment of comfort and energy objectives, with terminology related to multi-objective optimization revised to avoid implying a formal mathematical optimization formulation;
- (3) refinement of the interpretation and use of SHAP-based attribution, including clarification of its role in the comfort model and revision of the corresponding figure and terminology;
- (4) addition of a one-at-a-time sensitivity analysis to evaluate the robustness of the simulated energy-saving results under variations in key model inputs;
- (5) clarification of the spatial and temporal assumptions used in operational carbon accounting, together with an expanded discussion of simulation uncertainty, transferability, and future validation in the dedicated Limitations subsection; and
- (6) correction and refinement of figure and table presentation, including the formatting of Fig. 6 and the graphical issue in Fig. 12.

We are grateful to the Editor for the opportunity to further improve the manuscript and to the Reviewers for their detailed comments, which have helped us clarify the scope and strengthen the presentation of the study. All changes have been highlighted in the revised manuscript with tracked changes. Detailed point-by-point responses are provided below.

Response to Reviewer #6

The paper discussing and interesting application of digital twins for the evaluation and optimisation of multi-modal comfort in underground systems. The work introduces the proposed framework in detail and demonstrates its application in an existing underground station in Beijing, albeit only virtually as both energy savings and improved comfort conditions are only demonstrated through simulation and not implemented in practice and measured (the paper already states this as a limitation and future development in the discussion section).

I appreciate the high quality of the paper, likely helped by the previous round of reviews, and only have a few minor comments that might help further improve the work:

Comments 1: P16L319: Figure 6: the table part could be made clearer by a more robust formatting, including headers in each column (which are currently sitting at the bottom of the table)

Response 1:

Thank you for this helpful suggestion. We have revised the formatting of Fig. 6 by moving the variable labels to the top of the heatmap as column headers and adding a clearer header for the scenario column. The variable definitions are retained below the figure for reference. This revision improves the readability of the table without changing the underlying data.

Rewritten and added contents in the revised manuscript:

Location: Figure 6, lines 321~322.

Key revision:

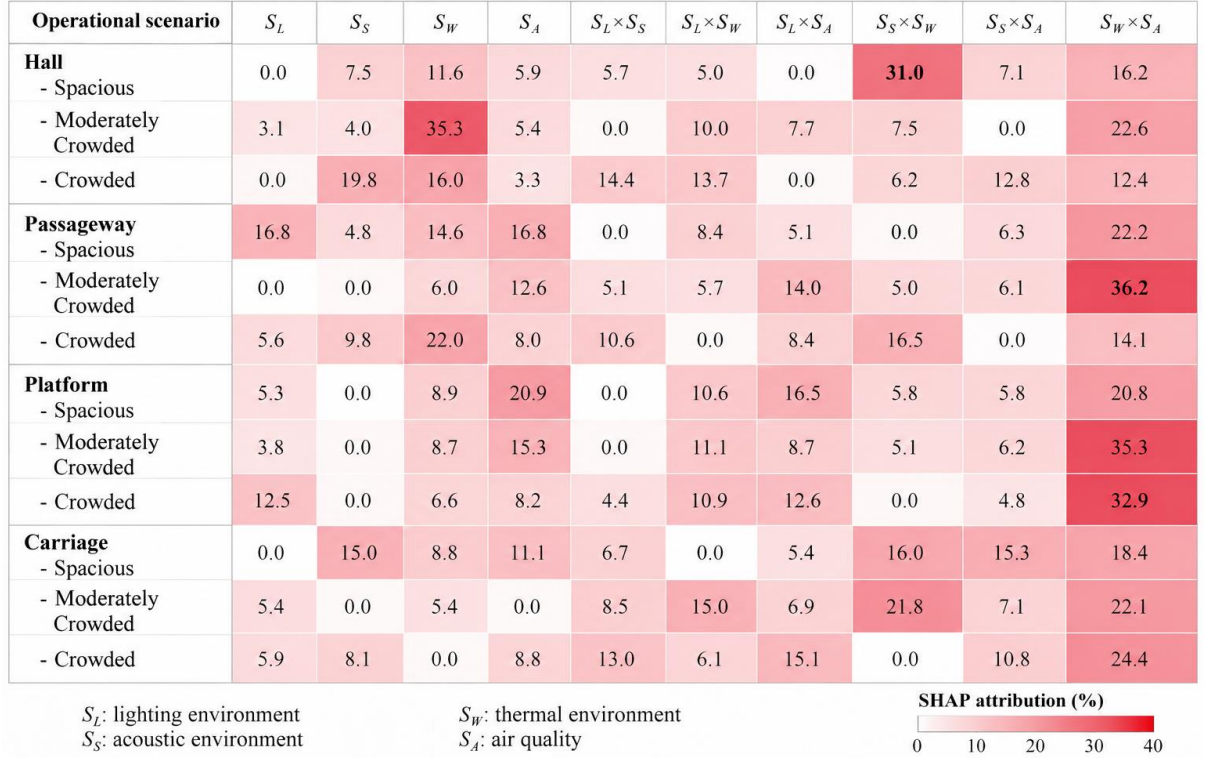


Fig. 6. SHAP attribution percentages of environmental factors under various operational scenarios.

Comments 2: P19L375: When introducing carbon emissions, and the use of carbon factors, a slightly more in depth discussion on carbon factor assumptions could be beneficial, including mention on the impact of using different local/national carbon factors, and the use of a constant factor, rather than a variable (hourly) one. This could be also mentioned as potential future work for the framework, as variable carbon factors can have significant impact on carbon savings.

Response 2:

Thank you for this helpful comment. We agree that the assumptions underlying the electricity emission factor should be stated more clearly. In the revised Section 3.6, we clarify that the Beijing provincial average electricity CO₂ emission factor (0.5554 kgCO₂/kWh) was adopted because the case station is located in Beijing. For comparison, the corresponding national and North China regional average factors are 0.5306 and 0.6361 kgCO₂/kWh, respectively. Different spatially aggregated factors affect the absolute carbon-emission estimate, while the relative reduction remains unchanged when the same constant factor is applied to both scenarios.

We further note that the present calculation uses an annual-average emission factor and therefore does not capture temporal variation in grid carbon intensity. The Limitations section has been revised to indicate that time-varying factors may influence estimated carbon savings when electricity use is redistributed across hours with different grid carbon intensities, and should be considered in future work.

Rewritten and added contents in the revised manuscript:

Location: section 3.6. *Virtual layer*, lines 379~388.

Key revision:

$$C_{\text{total}} = E_{\text{total}} \times EF_{\text{grid}} \quad (4)$$

Where C_{total} denotes annual total carbon emissions (kgCO₂), E_{total} denotes annual total electricity consumption (kWh), and the latest official Beijing provincial grid emission factor, $EF_{\text{grid}} = 0.5554$ kgCO₂/kWh, was adopted for operational carbon-emission accounting. The Beijing provincial factor was selected because the case station is located in Beijing; the corresponding national and North China regional average factors are 0.5306 and 0.6361 kgCO₂/kWh, respectively. Different local or national average factors would affect the absolute carbon-emission estimate but not the relative reduction when the same constant factor is applied to both scenarios. The annual-average factor adopted here does not capture temporal variations in grid carbon intensity.

Location: section 5.2. *Limitations*.

Key revision:

The energy and carbon results were also derived from scenario-based simulations and are sensitive to assumptions regarding passenger-flow schedules, HVAC setpoints, ventilation rates, lighting-power schedules, and the grid emission factor. In particular, the use of an annual-average constant electricity emission factor does not reflect hourly variations in grid carbon intensity. Time-varying emission factors may alter the estimated carbon benefits when adaptive operation shifts electricity demand across hours with different generation mixes [49]. Future work should therefore incorporate temporally resolved grid emission factors when evaluating carbon-oriented operational strategies. The carbon assessment covers operational emissions from station electricity

consumption and does not include embodied emissions associated with sensors, equipment, retrofit, or maintenance. Moreover, the case study represents one early-built transfer station in Beijing's cold climate. Application to other climates and underground-space typologies requires recalibration of meteorological boundaries, passenger-flow patterns, equipment characteristics, and comfort thresholds; thus, the transferability of the study lies primarily in its methodological workflow rather than in the reported numerical savings. Finally, the data-preprocessing procedures improve routine input quality but do not constitute fault-tolerant control. Future work should include systematic sensitivity analysis, multi-station field validation, fail-safe operation under sensor or communication failure, and testing under schedule disruptions and extreme weather.

References:

[49] L. Álvarez Flórez, T. Péan, J. Salom, Hourly based methods to assess carbon footprint flexibility and primary energy use in decarbonized buildings, *Energy Build.* 294 (2023) 113213. <https://doi.org/10.1016/j.enbuild.2023.113213>.

Comments 3: P28L509: Figure 12: the illuminance line is red for the first 30 minutes, it is unclear if this is a formatting error or if the color has significance.

Response 3:

Thank you for pointing this out. The red segment of the illuminance curve in Fig. 12 was an unintended plotting-format error caused by the data-format settings in Origin and had no physical or analytical significance. We have corrected Fig. 12 so that the illuminance curve is now displayed consistently in blue throughout the entire time series. This correction only affects the graphical presentation; the underlying data, analysis, and results remain unchanged.

Rewritten and added contents in the revised manuscript:

Location: section 4.2.2. *Identifying comfort weaknesses*, lines 529~531.

Key revision:

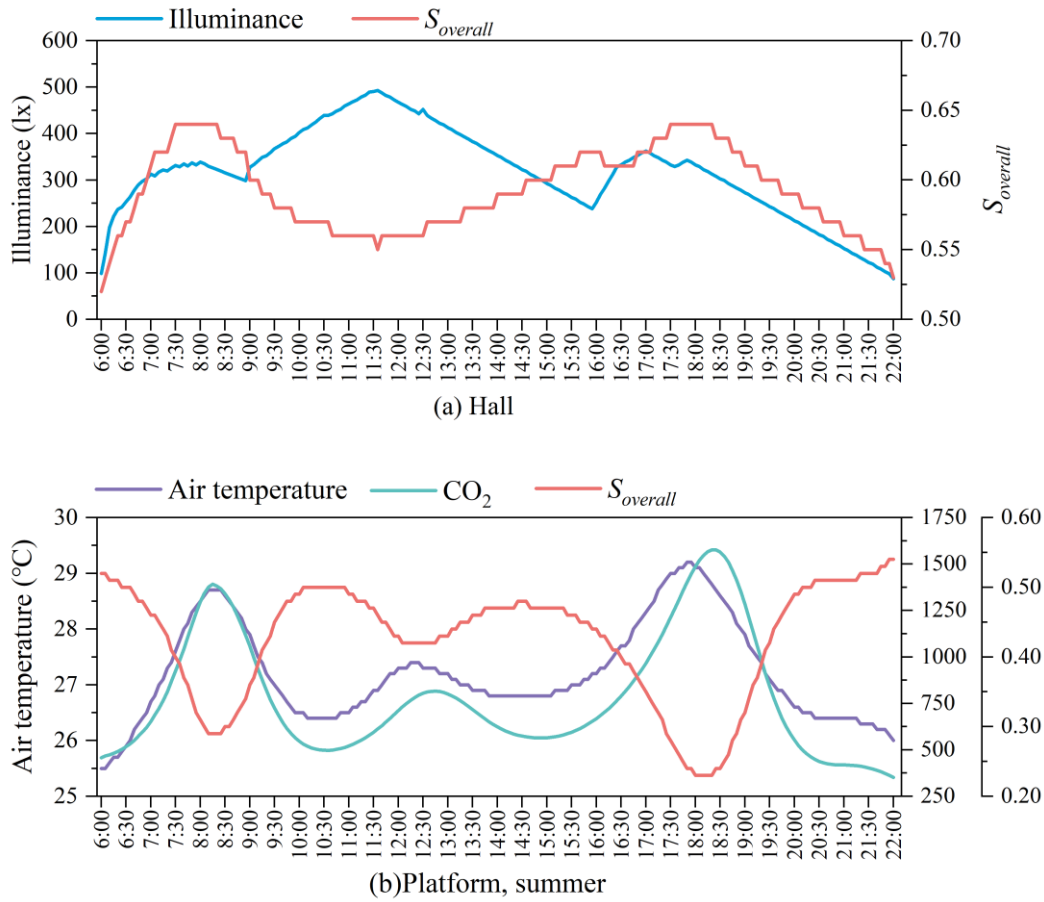


Fig. 12. Temporal evolution of environmental parameters and holistic comfort scores ($S_{overall}$) in functional zones during operation hours. (a) Station hall; (b) Platform in summer.

Comments 4: P36L662-682: I would suggest separating those paragraphs in a "limitations" section (which can be a subsection of the "discussion" section) for improved clarity.

Response 4:

Thank you for this helpful suggestion. We agree that separating the limitations from the general discussion improves the clarity and organization of the manuscript. Accordingly, Section 5 has been reorganized into two subsections: Section 5.1, "Main findings and implications," which discusses the principal findings and their relationship to previous studies, and Section 5.2, "Limitations," which consolidates the validation boundaries, simulation assumptions, transferability considerations, and future validation needs of the proposed framework.

Rewritten and added contents in the revised manuscript:

Location: section 5.1. Main findings and implications; section 5.2. Limitations

Response to Reviewer #7

The authors tried to reply the comments raised by several reviewers, where many parts were replied well. Still, my concerns are as follows:

Comments 1: The "digital twin" terminology of this manuscript (even title) is stronger than the actual implementation, EnergyPlus simulation-based work which is not validated in a real space. Deep worrying point. So that, one more revision is needed, with more cautious claims, explicit uncertainty analysis, and a clearer distinction between simulated potential and demonstrated operational performance.

Response 1:

Thank you for this important comment. We agree that the implementation scope and the distinction between simulation-estimated potential and demonstrated operational performance should be stated more clearly. The proposed framework is not based solely on synthetic simulation data: environmental and passenger-flow information was obtained through field monitoring at real subway stations, subjective comfort data were collected through on-site surveys, and the EnergyPlus model was calibrated against measured environmental data. On this basis, the digital-twin framework integrates real-time state sensing, context-aware comfort assessment, graded optimization-strategy generation, calibrated simulation-based performance evaluation, and operator-oriented decision support within a unified workflow.

In the present study, the comfort and energy effects of the recommended operation strategies were evaluated in the calibrated virtual environment rather than through long-term intervention in the operating station. Direct deployment in an operating metro system would require integration with existing station-control interfaces, operational authorization, fail-safe verification, and extended field testing, which represent a subsequent implementation stage beyond the present framework-validation scope. Accordingly, we have revised the Abstract, Section 3.7, Section 4.4, Fig. 16, Discussion, and Conclusions to distinguish field-based monitoring and decision support

from simulation-based performance evaluation and to describe the reported benefits consistently as simulated performance potential.

In addition, an explicit one-at-a-time sensitivity analysis was conducted using the calibrated OpenStudio/EnergyPlus model and the original weather file. Passenger density, ventilation rate, and lighting power were varied by $\pm 10\%$, and cooling/heating setpoints by ± 1 °C. The annual energy-saving rate remained between 14.7% and 15.8%, compared with 15.3% in the reference simulation, and the overall energy-saving tendency remained consistent across all tested cases. These results have been added to Section 4.2.1 and Appendix Table B.5, with the remaining uncertainty discussed in Section 5.2.

The principal contribution of the study therefore lies in establishing and testing an integrated digital-twin-enabled methodological framework that connects real-world multi-source sensing, multi-domain comfort assessment, passenger-flow context, adaptive strategy generation, and calibrated energy simulation, rather than in claiming completed automatic control deployment in an operating station.

Rewritten and added contents in the revised manuscript:

Location: Abstract, lines 18~24.

Key revision:

A four-level early-warning and strategy-selection mechanism was evaluated in a calibrated simulation environment to coordinate comfort status and energy use under different operational scenarios. A case study was conducted at a Beijing subway station in a cold region to assess the potential effectiveness of the framework. Simulation results indicate that the strategy effectively reduced low-comfort periods across functional areas; uncomfortable periods in the station hall during summer decreased from 20.4% to 3.1%.

Location: Section 3.7 Decision Service Layer (SS), lines 397~420.

Key revision:

The decision service layer (S_S) transforms the comfort assessment and energy simulation results from the virtual simulation layer (U_{VE}) into environmental control strategies for operational management. As shown in Fig. 8, this layer takes real-time environmental parameters, passenger

crowding, overall comfort scores, and system energy consumption as inputs, and evaluates the current operational status and corresponding response levels based on three types of constraints: safety limits, equipment operational ranges, and comfort thresholds. Accordingly, the adaptive optimization process is achieved by selecting context-dependent environmental control strategies according to the identified operational states and generating corresponding scenarios for subsequent performance evaluation [40].

As shown in Fig. 9, the service layer jointly considers two performance goals: enhancing passenger comfort and reducing system energy consumption. Their coordination follows the graded decision logic shown in Fig. 8, in which safety and equipment constraints are prioritized, followed by comfort status and passenger-flow conditions. When environmental parameters exceed safety or equipment limits, the system enters a red emergency mode; when the overall comfort score ranges from 0.42 to 0.68, a yellow warning mode is triggered; when comfort requirements are met but passenger flow is high, a blue dynamic balance mode is activated; and when passenger flow is low and the environment is stable, a green energy-saving mode is adopted. Based on these graded results, the service layer further matches response strategies for subsystems such as ventilation, air conditioning, lighting, and broadcasting, as illustrated in Fig. 9, providing a strategic basis for scenario-based adaptive optimization simulations in subsequent case studies. These response strategies constitute the adaptive optimization scenarios, in which alternative environmental configurations are evaluated through subsequent comfort and energy simulations to identify suitable operation strategies under different conditions [41].

References:

- [40] A. Fuller, Z. Fan, C. Day, C. Barlow, Digital twin: enabling technologies, challenges and open research, IEEE Access 8 (2020) 108952–108971. <https://doi.org/10.1109/ACCESS.2020.2998358>.
- [41] A. Afram, F. Janabi-Sharifi, Theory and applications of HVAC control systems – a review of model predictive control (MPC), Build. Environ. 72 (2014) 343–355. <https://doi.org/10.1016/j.buildenv.2013.11.016>.

Location: section 4.4, lines 606~633.

Key revision:

Original title: Digital twin-based intelligent environmental control

Revised title: Digital twin-enabled visualization and decision support

The digital twin visualization platform serves as the terminal interface of the adaptive environmental management framework, integrating multi-dimensional sensor data, comfort-assessment results, and recommended optimization strategies. The platform visualizes monitored environmental states and comfort scores, while the comfort and energy performance of the recommended optimization strategies are evaluated in the calibrated virtual simulation layer and subsequently presented to operators.

Through a high-fidelity digital interface, operators can visually monitor the dynamic evolution of overall comfort scores across the entire station ($S_{overall}$) and visualize the real-time comfort state and the recommended adaptive operation strategies generated by the module (U_{AO}) under complex fluctuating loads. This supports the transition of environmental management from fixed-schedule operation to adaptive decision support.

As shown in Fig. 16, the platform consolidates monitored data and comfort information from the station hall, platform, transfer passage, and carriage, enabling operators to examine environmental states and identify spatial differences across functional zones. Recommended optimization strategies and their corresponding simulation-based performance information are displayed through the interface to support operational decision-making. This integrated mode allows complex environmental conditions to be quantified using intuitive ($S_{overall}$) indicators and enables operators to identify weak points in environmental quality in each sub-zone. Historical monitoring records can further be used to examine temporal environmental variations and support adaptive management under changing passenger-flow conditions.

Additionally, the visualization system supports decision-making for balancing passenger comfort and energy efficiency. By displaying sub-system energy consumption, it supports the coordinated management of passenger comfort and system energy consumption. This digital twin-based full-process monitoring supports adaptive decision-making under passenger-flow fluctuations and facilitates smarter regulation of complex environmental comfort and energy consumption in underground spaces.

Location: Figure 16, lines 634~635.

Key revision:

Original caption: Fig. 16. Digital twin visualization dashboard for intelligent environmental regulation across different metro zones.

Revised caption: Fig. 16. Digital twin visualization dashboard for environmental monitoring and decision support across different metro zones.

Location: Section 6. conclusions, lines 727~743.

Key revision:

(ii) The simulation-based adaptive optimization results suggest simultaneous improvements in modeled comfort and energy performance. Simulation results indicate that the proposed strategy not only substantially reduces uncomfortable periods across all spaces during winter and summer, for example, the proportion of periods with $S_{overall} < 0.42$ in the summer station hall decreases from 20.4% to 3.1%, but also reduces the annual total station energy consumption and carbon emissions by 15.3%. Lighting (32.5%) and ventilation (22.8%) contribute the majority of energy savings, demonstrating that significant energy-saving potential can be achieved through operational strategy optimization without hardware modifications.

(iii) The digital twin framework supports adaptive environmental management by linking real-time monitoring and comfort-state identification with graded strategy generation and simulation-based evaluation. The calibrated EnergyPlus model provides a basis for estimating seasonal thermal-load tendencies and evaluating alternative operation scenarios, while the four-level warning mechanism translates comfort and operational constraints into actionable management strategies. Together, these components support adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework.

Location: Section 4.2.1. Simulation model analysis, lines 483~484; Appendix B.5, lines 958~959.

Key revision:

All error metrics were well below the allowable limits defined by ASHRAE Guideline 14 ($|NMBE| < 10\%$, $CV(RMSE) < 30\%$) [42], indicating close agreement between simulated and measured air temperatures during the calibration periods and supporting the use of the calibrated

model for subsequent evaluation. To further examine the robustness of the simulated energy-saving results, a one-at-a-time sensitivity analysis was conducted using the calibrated OpenStudio/EnergyPlus model and the original weather file. Passenger density, ventilation rate, and lighting power were varied by $\pm 10\%$, while cooling and heating setpoints were varied by $\pm 1\text{ }^{\circ}\text{C}$, with all other inputs held constant. Across the tested cases, the annual energy-saving rate ranged from 14.7% to 15.8%, compared with 15.3% in the reference simulation, with ventilation rate producing the largest variation, as shown in Table B.5. The energy-saving tendency remained consistent across all tested input variations.

Table B.5
One-at-a-time sensitivity analysis of key simulation inputs.

Simulation input	Variation	Annual energy-saving rate (%)	Change from reference (percentage points)
Reference case	—	15.25	—
Passenger density/ internal load	−10%	14.83	−0.42
	+10%	15.69	+0.44
Ventilation rate	−10%	14.73	−0.52
	+10%	15.80	+0.55
Lighting power	−10%	14.94	−0.31
	+10%	15.54	+0.30
Cooling setpoint	−1 °C	15.27	+0.03
	+1 °C	15.22	−0.03
Heating setpoint	−1 °C	15.23	−0.02
	+1 °C	15.27	+0.02

Response to Reviewer #8

The manuscript addresses a relevant and practically important topic and presents an integrated framework combining digital-twin technologies, multi-domain environmental comfort

assessment, machine learning, real-time sensing, and energy simulation for underground transportation spaces. The study is generally well structured and demonstrates considerable effort in integrating multiple environmental and operational dimensions. However, several conceptual and methodological aspects require clarification to ensure that the terminology and conclusions are fully aligned with the methodology presented. My major comments are as follows:

Comments 1: Clarification of the "adaptive optimization" concept

The manuscript repeatedly describes the proposed framework as an "adaptive optimization" approach. However, based on the presented methodology, the framework appears to employ a four-level context-dependent decision mechanism that maps environmental conditions, comfort states, and passenger-flow information to predefined HVAC, ventilation, and lighting strategies. It is not sufficiently clear whether a formal optimization procedure is involved. The authors are therefore encouraged to clarify what specifically constitutes "optimization" within the proposed framework. If no explicit optimization formulation or solution procedure is employed, terminology such as adaptive decision support, context-aware control, or scenario-based adaptive management may more accurately characterize the proposed approach.

Response 1:

Thank you for this valuable comment. We agree that the optimization mechanism of the proposed framework requires further clarification. In the revised manuscript, we have clarified that the adaptive optimization process is implemented through the decision service layer (SS). Specifically, the layer integrates environmental states, passenger-flow conditions, comfort levels, energy consumption, safety constraints, and equipment limitations to generate context-dependent operation strategies. These strategies are subsequently evaluated through simulation-based scenarios to assess their potential effects on comfort improvement and energy performance. The corresponding descriptions have been revised in Section 3.7.

Rewritten and added contents in the revised manuscript:

Location: Section 3.7 Decision Service Layer (SS), lines 397~420.

Key revision:

The decision service layer (S_s) transforms the comfort assessment and energy simulation results from the virtual simulation layer (U_{VE}) into environmental control strategies for operational management. As shown in Fig. 8, this layer takes real-time environmental parameters, passenger crowding, overall comfort scores, and system energy consumption as inputs, and evaluates the current operational status and corresponding response levels based on three types of constraints: safety limits, equipment operational ranges, and comfort thresholds. Accordingly, the adaptive optimization process is achieved by selecting context-dependent environmental control strategies according to the identified operational states and generating corresponding scenarios for subsequent performance evaluation [40].

As shown in Fig. 9, the service layer jointly considers two performance goals: enhancing passenger comfort and reducing system energy consumption. Their coordination follows the graded decision logic shown in Fig. 8, in which safety and equipment constraints are prioritized, followed by comfort status and passenger-flow conditions. When environmental parameters exceed safety or equipment limits, the system enters a red emergency mode; when the overall comfort score ranges from 0.42 to 0.68, a yellow warning mode is triggered; when comfort requirements are met but passenger flow is high, a blue dynamic balance mode is activated; and when passenger flow is low and the environment is stable, a green energy-saving mode is adopted. Based on these graded results, the service layer further matches response strategies for subsystems such as ventilation, air conditioning, lighting, and broadcasting, as illustrated in Fig. 9, providing a strategic basis for scenario-based adaptive optimization simulations in subsequent case studies. These response strategies constitute the adaptive optimization scenarios, in which alternative environmental configurations are evaluated through subsequent comfort and energy simulations to identify suitable operation strategies under different conditions [41].

References:

- [40] A. Fuller, Z. Fan, C. Day, C. Barlow, Digital twin: enabling technologies, challenges and open research, IEEE Access 8 (2020) 108952–108971. <https://doi.org/10.1109/ACCESS.2020.2998358>.
- [41] A. Afram, F. Janabi-Sharifi, Theory and applications of HVAC control systems – a review of model predictive control (MPC), Build. Environ. 72 (2014) 343–355. <https://doi.org/10.1016/j.buildenv.2013.11.016>.

Comments 2: Clarification of the "multi-objective optimization" claim.

The manuscript considers both passenger comfort and energy consumption as important performance objectives. However, simultaneous consideration of multiple objectives does not necessarily constitute multi-objective optimization. The manuscript does not clearly present an explicit optimization formulation or a systematic trade-off between comfort and energy objectives. Therefore, statements such as "real-time adaptive multi-objective optimization," particularly in the Conclusions, should either be further justified based on the presented methodology or moderated to more accurately reflect the scope of the framework.

Response 2:

Thank you for this important clarification. As further explained in our response to Comment 1, the revised manuscript now specifies how adaptive optimization is implemented through scenario-specific strategy generation and simulation-based performance evaluation. We further agree that the simultaneous consideration of passenger comfort and energy consumption should not, by itself, be described as formal multi-objective optimization. Previous reviews of building-performance optimization characterize multi-objective optimization in terms of explicit objective functions, decision variables, constraints, and optimization procedures for resolving trade-offs among competing objectives [1].

Accordingly, we have moderated the related terminology while retaining the concept of adaptive optimization clarified in Comment 1. In Section 3.7, "dual objectives" has been revised to "two performance goals," and the manuscript now explicitly describes how comfort and energy performance are coordinated through the graded decision logic: safety and equipment constraints are prioritized, comfort-oriented responses are applied when comfort is insufficient, and energy-saving responses are activated when comfort requirements are satisfied under low-load conditions.

Most importantly, the statement "real-time adaptive multi-objective optimization" in the Conclusions has been removed and replaced with "adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework." This revision distinguishes the joint consideration of comfort and energy performance from formal multi-objective optimization.

References:

- [1] T. Al Mindeel, E. Spentzou, M. Eftekhari, Energy, thermal comfort, and indoor air quality: Multi-objective optimization review, Renewable and Sustainable Energy Reviews 202 (2024) 114682. <https://doi.org/10.1016/j.rser.2024.114682>.

Rewritten and added contents in the revised manuscript:

Location: section 3.7. Service layer, lines 406~409.

Key revision:

As shown in Fig. 9, the service layer jointly considers two performance goals: enhancing passenger comfort and reducing system energy consumption. Their coordination follows the graded decision logic shown in Fig. 8, in which safety and equipment constraints are prioritized, followed by comfort status and passenger-flow conditions.

Location: section 6. Conclusions, lines 741~743.

Key revision:

Together, these components support adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework.

Comments 3: Clarification of the role of SHAP in the overall comfort model.

The use of machine learning and SHAP analysis provides a useful means of identifying context-dependent relationships between environmental factors and overall comfort. However, the manuscript also uses normalized SHAP-derived contributions as scenario-specific weights in the overall comfort equation. Since SHAP primarily provides model attribution and interpretability information, the conceptual rationale for translating these attribution values into quantitative comfort weights should be explained more clearly. In addition, SHAP-based contributions should be consistently interpreted as model-specific feature attribution rather than physical or causal contributions to passenger discomfort. This distinction is particularly important when reporting values such as the 35.3% contribution of the thermal-air-quality interaction.

Response 3:

Thank you for this important comment. We agree that the previous wording did not sufficiently distinguish SHAP-based model attribution from physical or causal contributions to

passenger discomfort. In the revised manuscript, the role of SHAP has therefore been clarified in Sections 3.5.2 and 3.5.3. SHAP is now consistently described as a model-attribution method, and the normalized mean absolute SHAP values used in the overall comfort calculation are described as model-informed relative aggregation weights rather than physical or causal comfort contributions. Corresponding statements in the Results, Discussion, and Conclusions have also been revised. In particular, the previously reported 35.3% value is now described as the SHAP attribution of the thermal–air-quality interaction among the displayed environmental terms, rather than as a causal contribution to passenger discomfort.

感谢您这条重要的意见。我们认同原文表述未能充分区分基于 SHAP 的模型归因与造成乘客不适的物理或因果影响。因此，在修改稿中，我们在 3.5.2 节和 3.5.3 节阐明了 SHAP 的作用。文中统一将 SHAP 描述为一种模型归因方法，同时将整体舒适度计算中所使用的归一化 SHAP 平均绝对值，定义为由模型得到的相对聚合权重，而非对舒适度产生作用的物理或因果贡献。结果、讨论与结论部分的相关表述也已同步修改。其中，原先给出的 35.3% 这一数值，现描述为在展示的各类环境因素中，热环境与空气质量交互项的 SHAP 归因结果，而非引发乘客不适的因果贡献。

Rewritten and added contents in the revised manuscript:

Location: Section 3.5.2. *Dynamic comfort evaluation framework*, lines 310~318.

Key revision:

To enhance model interpretability, SHAP attribution analysis was applied to the trained machine-learning model. SHAP assigns feature-attribution values to individual model predictions, while aggregated mean absolute SHAP values can be used to characterize model-level feature and interaction importance. In this study, these attribution results were used to identify the relative importance of environmental features and selected interaction terms under different operational scenarios, with the results illustrated in Fig. 6. This analysis framework helps explain how environmental conditions and their interactions are associated with overall comfort under different operational scenarios, providing methodological support for generating precise quantitative evaluation benchmarks in the digital twin system.

Location: section 3.5.3. *Dynamic evaluation model and grading standards*, lines 328~332.

Key revision:

For each operational scenario, the mean absolute SHAP values of the selected comfort features and interaction terms were normalized to obtain scenario-specific relative aggregation weights for the overall comfort calculation. This use of SHAP-derived weights provides a model-informed representation of the relative importance learned by the fitted comfort model.

Location: figure 6, lines 321~322.

Key revision:

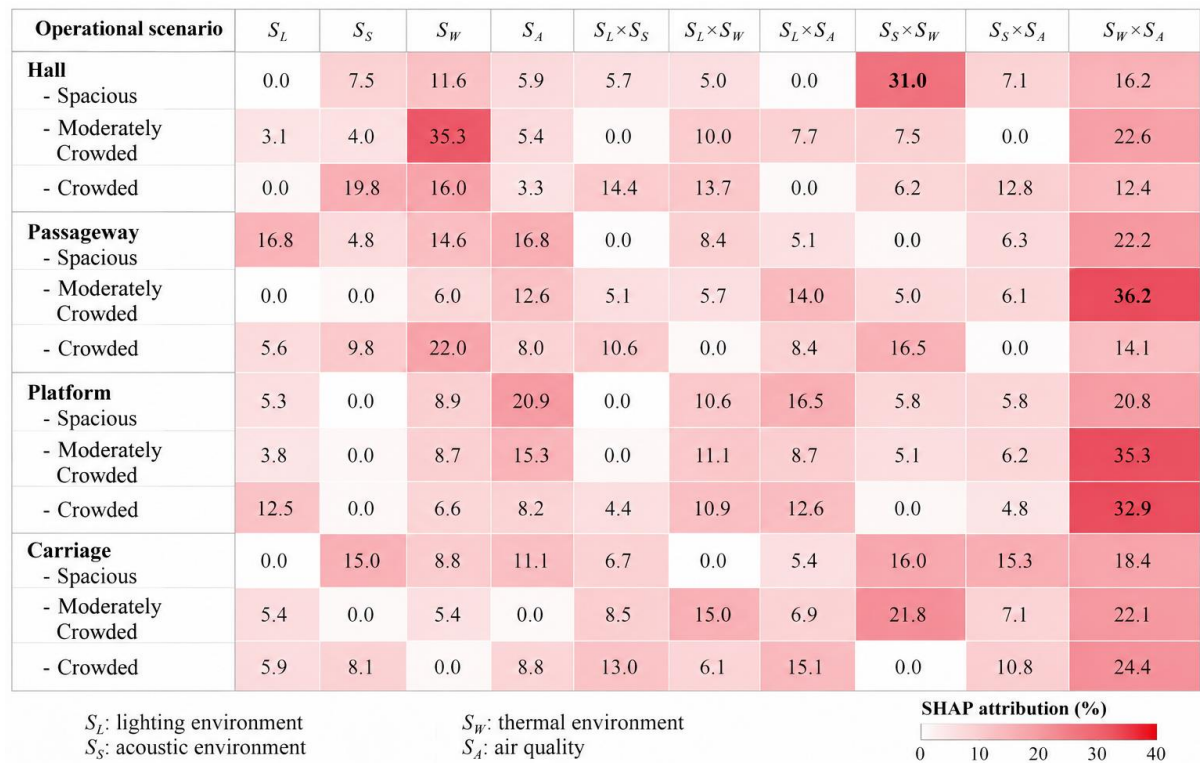


Fig. 6. SHAP attribution percentages of environmental factors under various operational scenarios.

Location: section 4.2.2. Identifying comfort weaknesses, lines 513~517.

Key revision:

Using SHAP dynamic weight attribution, during summer peak periods, the interaction of thermal environment and air quality showed the highest SHAP attribution, accounting for 35.3% in the summer-peak platform scenario. Air temperature reaches 28.7°C with 67.2% relative humidity and a CO₂ concentration of 1391 ppm, frequently causing overall comfort scores to fall below 0.42.

Location: section 4.3.1. Generation of adaptive optimization strategies, lines 578~581.

Key revision:

The SHAP-informed prioritization strategy was used to identify the environmental features with relatively high model-attribution importance under each scenario, thereby supporting targeted adjustment of HVAC, ventilation, and lighting settings while contributing to simulated energy and emission reductions.

Location: section 6. Conclusions, lines 721~726.

Key revision:

(i) Overall subway passenger comfort results from the combined effects of multiple environmental factors and operational conditions. The interaction between thermal environment and air quality showed the highest SHAP attribution during summer peak periods, reaching 35.3%. The weights of the environmental dimensions dynamically adjust according to passenger crowding. Based on calculated comfort thresholds of 0.42 and 0.68, four-level warning grades and corresponding response strategies were established.

Comments 4: Clarification of the scope of the digital-twin framework.

The proposed framework combines real-time environmental sensing and visualization with comfort assessment, simulation-based evaluation, and operator-oriented decision support. The manuscript would benefit from a clearer distinction between these functions. In particular, real-time monitoring and decision support should be distinguished from simulation-based scenario evaluation and automatic closed-loop environmental control. Clarifying these boundaries and using consistent terminology throughout the title, abstract, methodology, discussion, and conclusions would provide a more precise description of the capabilities and contribution of the proposed digital-twin framework.

Response 4:

Thank you for this helpful comment. We have revised the manuscript to clarify the functional scope of the proposed digital-twin framework. The revised text now distinguishes real-time environmental monitoring and comfort-state identification, adaptive strategy generation,

simulation-based performance evaluation, and operator-oriented decision support. Corresponding terminology has been revised in the Abstract, methodology, Section 4.4, Discussion, Fig. 16, and Conclusions to ensure consistency. Equipment-level closed-loop feedback is further identified as a direction for future development.

Rewritten and added contents in the revised manuscript:

Location: Abstract, lines 18~24.

Key revision:

A four-level early-warning and strategy-selection mechanism was evaluated in a calibrated simulation environment to coordinate comfort status and energy use under different operational scenarios. A case study was conducted at a Beijing subway station in a cold region to assess the potential effectiveness of the framework. Simulation results indicate that the strategy effectively reduced low-comfort periods across functional areas; uncomfortable periods in the station hall during summer decreased from 20.4% to 3.1%.

Location: Keywords, lines 29.

Key revision:

Original keyword: Adaptive environmental control

Revised keyword: Adaptive environmental management

Location: Figure 1, lines 203~204.

Key revision:

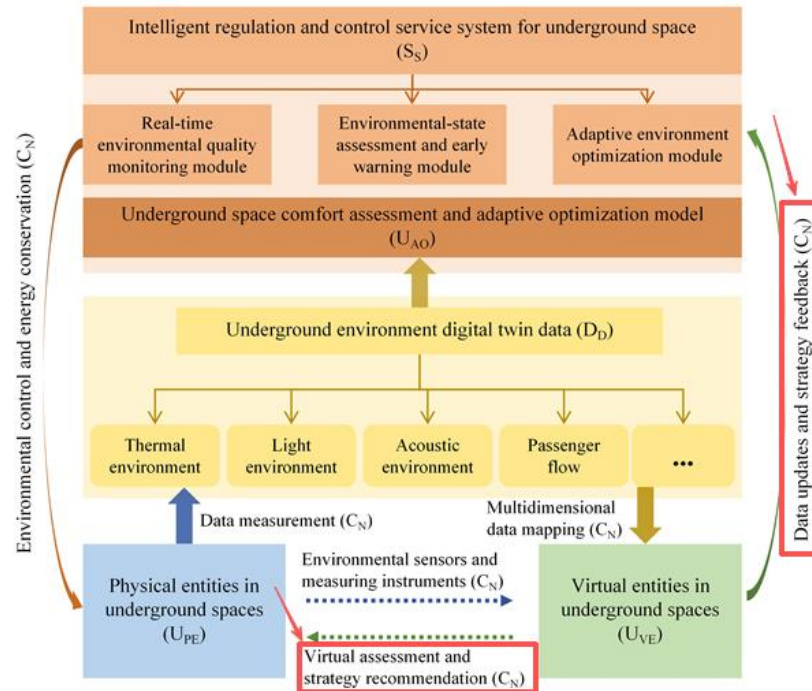


Fig. 1. Hybrid model of multidimensional comfort in underground space based on digital twin.

Location: Figure 2, lines 211~213.

Key revision:

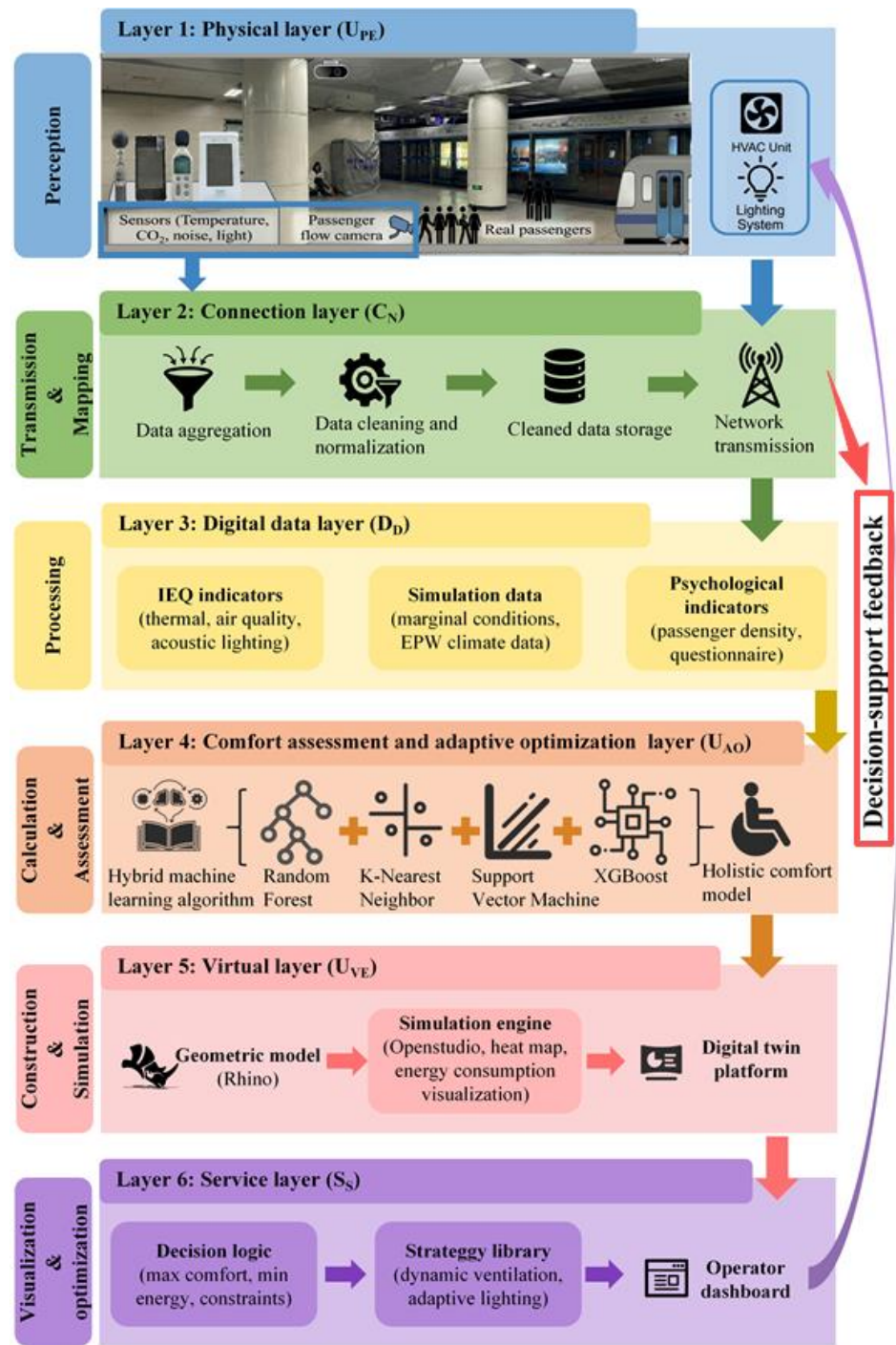


Fig. 2. System framework for environmental regulation and comfort improvement in underground spaces.

Location: section 4.4, lines 606~633.

Key revision:

Original title: Digital twin-based intelligent environmental control

Revised title: Digital twin-enabled visualization and decision support

The digital twin visualization platform serves as the terminal interface of the adaptive environmental management framework, integrating multi-dimensional sensor data, comfort-assessment results, and recommended optimization strategies. The platform visualizes monitored environmental states and comfort scores, while the comfort and energy performance of the recommended optimization strategies are evaluated in the calibrated virtual simulation layer and subsequently presented to operators.

Through a high-fidelity digital interface, operators can visually monitor the dynamic evolution of overall comfort scores across the entire station ($S_{overall}$) and visualize the real-time comfort state and the recommended adaptive operation strategies generated by the module (U_{AO}) under complex fluctuating loads. This supports the transition of environmental management from fixed-schedule operation to adaptive decision support.

As shown in Fig. 16, the platform consolidates monitored data and comfort information from the station hall, platform, transfer passage, and carriage, enabling operators to examine environmental states and identify spatial differences across functional zones. Recommended optimization strategies and their corresponding simulation-based performance information are displayed through the interface to support operational decision-making. This integrated mode allows complex environmental conditions to be quantified using intuitive ($S_{overall}$) indicators and enables operators to identify weak points in environmental quality in each sub-zone. Historical monitoring records can further be used to examine temporal environmental variations and support adaptive management under changing passenger-flow conditions.

Additionally, the visualization system supports decision-making for balancing passenger comfort and energy efficiency. By displaying sub-system energy consumption, it supports the coordinated management of passenger comfort and system energy consumption. This digital twin-based full-process monitoring supports adaptive decision-making under passenger-flow fluctuations and facilitates smarter regulation of complex environmental comfort and energy consumption in underground spaces.

Location: Figure 16, lines 634~635.

Key revision:

Original caption: Fig. 16. Digital twin visualization dashboard for intelligent environmental regulation across different metro zones.

Revised caption: Fig. 16. Digital twin visualization dashboard for environmental monitoring and decision support across different metro zones.

Location: Section 5.1. Main findings and implications, lines 681~683.

Key revision:

Accordingly, the framework integrates real-time environmental monitoring and comfort-state identification with adaptive strategy generation, simulation-based performance evaluation, and operator-oriented decision support.

Location: Section 5.2. Limitations, lines 709~713.

Key revision:

Future work should extend the present one-at-a-time sensitivity analysis to joint parameter uncertainty and interactions among multiple inputs, together with multi-station field validation, equipment-level closed-loop feedback, fail-safe operation under sensor or communication failure, and testing under schedule disruptions and extreme weather.

Location: Section 6. conclusions, lines 727~743.

Key revision:

(ii) The simulation-based adaptive optimization results suggest simultaneous improvements in modeled comfort and energy performance. Simulation results indicate that the proposed strategy not only substantially reduces uncomfortable periods across all spaces during winter and summer, for example, the proportion of periods with $S_{overall} < 0.42$ in the summer station hall decreases from 20.4% to 3.1%, but also reduces the annual total station energy consumption and carbon emissions by 15.3%. Lighting (32.5%) and ventilation (22.8%) contribute the majority of energy savings, demonstrating that significant energy-saving potential can be achieved through operational strategy optimization without hardware modifications.

(iii) The digital twin framework supports adaptive environmental management by linking real-time monitoring and comfort-state identification with graded strategy generation and simulation-based evaluation. The calibrated EnergyPlus model provides a basis for estimating seasonal thermal-load tendencies and evaluating alternative operation scenarios, while the four-level warning mechanism translates comfort and operational constraints into actionable management strategies. Together, these components support adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework.

Highlights

- A digital twin framework enables proactive subway environmental control.
- Holistic comfort is evaluated using sensor, survey, and flow data.
- Machine learning and SHAP reveal dynamic comfort drivers across scenarios.
- The strategy reduces summer hall discomfort from 20.4% to 3.1%.
- Annual energy use and carbon emissions fall by 15.25% and 15.32%.

A Digital Twin-enabled Adaptive Optimization Framework for Balancing Holistic Comfort and Energy Efficiency in Underground Spaces

Zijian Ye ^{a, b, c*}, Tianlong Liang ^{a,c}, Yu Chen ^a

^a School of Art and Design, Beijing Institute of Fashion Technology, Beijing 100029, China

^b Key Laboratory of Urban Underground Engineering of Ministry of Education, Beijing Jiaotong University,
Beijing 100044, China

^c These authors contributed equally to this work and should be considered co-first authors.

* Corresponding author: E-mail: zijianye@bift.edu.cn

Abstract

Underground space environmental control often faces the dual challenge of insufficient passenger comfort and high system energy consumption. To balance holistic comfort and energy efficiency, this study proposes a digital twin-enabled adaptive optimization framework for underground environments. Environmental data were collected automatically through distributed thermal, acoustic, and lighting sensors, and a structured database was established. Field measurements, subjective comfort surveys, machine learning, and virtual simulations were integrated to construct a digital twin-enabled adaptive optimization framework balancing energy consumption and environmental comfort. A four-level early-warning and strategy-selection mechanism was evaluated in a calibrated simulation environment to coordinate comfort status and energy use under different operational scenarios. A case study was conducted at a Beijing subway station in a cold region to assess the potential effectiveness of the framework. Simulation results indicate that the strategy effectively reduced low-comfort periods across functional areas; uncomfortable periods in the station hall during summer decreased from 20.4% to 3.1%. Total annual energy consumption and operational carbon emissions both decreased by 15.3%, with lighting and ventilation contributing 32.5% and 22.8% of the energy savings, respectively. This study provides a

methodological reference for comfort-oriented and low-carbon environmental management in existing underground transportation spaces.

Keywords: Digital twin; Adaptive environmental management; Holistic comfort; Machine learning; Energy efficiency.

1. Introduction

Rapid urbanization has intensified surface traffic congestion, making subway systems a critical backbone in major cities for alleviating transportation pressure [1,2]. However, explosive growth in passenger flow poses challenges for underground stations, which are enclosed environments. High-density crowds and confined structures lead to heat and humidity accumulation, degraded air quality, and noise pollution, threatening passenger comfort and health [3]. Additionally, HVAC and lighting systems consume substantial energy to maintain indoor conditions, significantly impacting operating costs [4].

To improve the environmental quality of underground spaces, numerous studies have focused on the assessment and control of indoor environmental quality (IEQ). Many scholars have conducted in-depth investigations into individual physical factors, such as thermal conditions, lighting, air quality, or acoustic environment, and have established baseline comfort thresholds for these indicators [5]. For instance, Fang et al. analyzed measured data and survey responses from subway platforms in hot and humid regions, identifying an optimal comfort temperature range of 21.6–28.8 °C [6]. Hu et al. emphasized lighting comfort in underground spaces, showing that the optimal illuminance levels for long-term and short-term occupants are greater than 300 lx and 494 lx, respectively, which helps reduce the incidence of sick building syndrome (SBS) [7].

Despite the maturity of single-factor studies, current assessment systems have two notable limitations. First, they lack a holistic perspective on overall comfort. Environmental psychology research indicates that human perception of the environment is the result of complex interactions between multi-dimensional physical stimuli (sound, light, thermal, and air quality) and psychological factors such as crowding and perceived safety [8,9]. Focusing on individual environmental dimensions alone may not fully reflect passengers' overall

satisfaction. Second, existing models generally overlook the dynamic influence of operational contexts. Most models are constructed under static experimental conditions and do not fully consider the real-time impact of high dynamic passenger flow on comfort thresholds. For example, Liu et al. found that individuals' thermal tolerance changes significantly under crowded conditions, highlighting the need to further examine comfort differences under dynamic operational scenarios [10].

Recent studies further confirm that these limitations are not isolated issues. Multi-domain IEQ research has shown that the interactions among thermal, acoustic, visual, and air-quality domains remain insufficiently quantified, especially when overall satisfaction is considered [11]. Multi-domain simulation studies have also indicated that validated models integrating all IEQ domains are still limited [12]. In parallel, occupant-centric control research has emphasized that building environmental regulation should respond to occupancy information rather than rely solely on predefined schedules [13]. This issue is particularly pronounced in subway stations, where passenger flow is highly uneven and dynamic; passenger-flow information therefore needs to be integrated with building energy modeling to evaluate occupancy-adaptive HVAC strategies [14]. Although recent digital twin studies have demonstrated the potential of real-time sensing, virtual simulation, and decision support for built-environment management [15–18], most applications still focus on monitoring, prediction, facility management, or single-domain environmental control rather than an integrated framework that jointly considers multi-domain comfort, passenger-flow context, and energy performance. Therefore, a methodological gap remains in developing a digital twin-enabled framework that links multi-source sensing, context-aware comfort evaluation, and simulation-based energy optimization for underground transportation spaces.

To address this gap, this study proposes a digital twin-enabled adaptive optimization framework for underground space environmental management, aiming to coordinate overall passenger comfort with system energy consumption. The methodology combines field measurements, survey assessments, machine learning evaluation, visualization modeling, and energy consumption simulation to develop strategies for identifying and optimizing

environmental states in underground spaces. The effectiveness of the proposed framework is assessed through energy consumption simulations of a real-world case. The main objectives of this study are: (i) to integrate multi-source data including thermal, lighting, acoustic, air quality, and passenger crowding metrics in underground spaces; (ii) to develop a system framework that merges geometric modeling, digital twin visualization, and performance simulation; and (iii) to apply the framework to a representative underground transfer station and, through a comparative analysis of conventional and optimized control strategies, evaluate the potential benefits of the proposed digital twin framework.

2. Literature review

This section reviews the progress and remaining gaps in digital twin-enabled environmental management for underground transportation spaces, focusing on three aspects: (1) digital twins in underground environments, (2) holistic comfort assessment under dynamic operational contexts, and (3) integrated comfort–energy optimization and decision support.

2.1. Digital twins in underground environments

Digital twin research has gradually evolved from virtual representations of individual industrial assets toward integrated systems involving physical entities, virtual models, data exchange, connection mechanisms, and service-oriented functions. The five-dimensional digital twin architecture proposed by Tao et al. provides a structured representation of these components through the physical entity, virtual entity, data, connection, and service dimensions [19]. Semeraro et al. further identified continuous physical–virtual synchronization, multi-source data integration, model updating, and decision support as central characteristics of digital twins, while also noting that inconsistent definitions and limited interoperability remain common implementation barriers [20]. In the architecture, engineering, construction, operation, and facility-management fields, existing DT applications have mainly supported lifecycle information management, construction monitoring, facility operation, and asset maintenance, with relatively limited integration across the complete operational workflow [21].

Within underground environments, digital twin applications have primarily concentrated on construction safety, structural condition assessment, and infrastructure monitoring. Ye et al. developed a digital twin approach for tunnel-construction safety warning and management by integrating construction information, monitoring data, and virtual visualization [22]. Du et al. proposed an intelligent underground-infrastructure monitoring system based on spatio-temporal data, demonstrating how heterogeneous sensing records can be organized and visualized to support infrastructure-state identification [23]. These studies establish important technical foundations for underground-space sensing, data management, and virtual representation. However, their principal objects are structural or construction states rather than passenger-oriented environmental quality, and the interaction between comfort evaluation, operational context, and station energy performance is generally outside their scope.

2.2. Holistic comfort assessment under dynamic occupancy conditions

Holistic environmental comfort requires the simultaneous consideration of thermal, visual, acoustic, and air-quality stimuli. A critical review by Chinazzo et al. found substantial heterogeneity in the experimental design, terminology, analysis, and reporting of multi-domain indoor-environment studies, which restricts comparison and generalization across different settings [24]. Mamulova et al. similarly showed that the practical scalability of multi-domain comfort models is limited by differences in model scope, resolution, portability, and data requirements [25]. These limitations are particularly relevant to underground transportation spaces, because passenger exposure is transient, functional zones differ substantially, and environmental conditions change rapidly with train operation and crowding.

Occupancy-responsive environmental control provides a possible means of addressing this temporal variability. Esrafilian-Najafabadi and Haghighat showed that occupancy-based HVAC control can use current or historical occupancy information to adjust system schedules and setpoints, but its effectiveness depends on occupancy-model accuracy, sensing resolution, and the selected control logic [26]. A side-by-side experimental study by Kong et al. further

demonstrated that occupancy sensing can support simultaneous energy savings and the maintenance of thermal comfort and perceived air quality, while also showing that control performance is sensitive to sensor accuracy [27]. For subway stations, passenger-flow simulation has been coupled with building energy modeling to assess occupancy-adaptive HVAC strategies [14], and data-driven models have been developed to improve dynamic cooling-load prediction under changing station conditions [28]. Nevertheless, these studies mainly address occupancy, thermal load, ventilation, or HVAC energy use. Their outputs are not generally connected to an overall comfort model that simultaneously represents thermal, acoustic, lighting, air-quality, functional-zone, and crowding effects.

2.3. Digital twin-enabled comfort and energy management

Recent studies have begun to extend digital twins from infrastructure monitoring toward occupant-related building operation. Abdelrahman et al. incorporated BIM-derived spatial-temporal proximity data into digital twins to predict individual thermal preferences, demonstrating the potential of DT-supported personal comfort modeling [29]. Hosamo et al. integrated BIM, real-time sensor data, occupant feedback, and Bayesian networks to identify discomfort conditions and examine the relationship between equipment operation and occupant comfort [30]. Qian et al. developed an intelligent DT platform combining IoT, BIM, and machine learning for indoor-air-quality monitoring, rapid prediction, visualization, and feedback-oriented management [31]. ElArwady et al. also combined digital twins and machine-learning models to estimate indoor thermal comfort, illustrating the value of linking virtual building representations with data-driven comfort prediction [32].

Collectively, these studies demonstrate that digital twins can integrate sensing, virtual models, occupant-related data, environmental prediction, and operational decision support. However, the reviewed applications generally emphasize a particular component, such as personal thermal preference, indoor air quality, equipment diagnosis, or thermal-comfort prediction. Underground-space DT studies, meanwhile, remain primarily focused on construction and infrastructure states. A methodological gap therefore remains between three research streams: underground digital-twin monitoring, multi-domain and context-sensitive

comfort assessment, and occupancy-responsive energy management. The proposed M_{UDT} framework is positioned at this intersection. It extends the five-dimensional DT architecture by incorporating a dedicated comfort-evaluation and adaptive-optimization component, thereby linking multi-source sensing, operational-context identification, holistic comfort assessment, calibrated performance simulation, and graded decision support within a unified methodological workflow for underground transportation spaces.

3. Research methods

3.1. Design framework of the platform (M_{UDT})

The five-dimensional model proposed by Tao et al. was selected because its physical entity, virtual entity, data, connection, and service dimensions align closely with the sensing, simulation, data integration, and decision-support requirements of underground environmental control [19]. Compared with application-specific BIM-IoT or monitoring-oriented architectures, this model provides clearer cross-layer relationships and greater extensibility. Therefore, to achieve precise evaluation and adaptive optimization of underground space environments, this study builds a hybrid model for multidimensional comfort in underground spaces (M_{UDT}), integrating multi-dimensional coupled characteristics of acoustic, lighting, thermal, and air quality factors with dynamic passenger flow patterns. The resulting six-dimensional model is expressed in Equation (1) as follows:

$$M_{UDT} = (U_{PE}, U_{VE}, D_D, U_{AO}, S_S, C_N) \quad (1)$$

where U_{PE} represents the physical entity of the underground space; U_{VE} is the high-fidelity mapping of the physical entity in the digital space, encompassing not only geometric models but also physics-based performance simulation models for real-time environment state simulation; D_D represents the multi-source core data and knowledge base; C_N denotes the communication and data processing mechanisms connecting all components; U_{AO} is the core extended dimension of the model, representing the comfort evaluation and adaptive optimization algorithms, serving as the computational core; and S_S refers to the visual decision service interface.

As shown in Fig. 1, the M_{UDT} model is based on the collaborative relationships among the physical entity, virtual model, data layer, connection mechanism, evaluation and optimization algorithms, and service interface. Environmental and passenger flow data from the physical space are collected, cleaned, and temporally and spatially matched through the connection layer, then managed in the data layer. The virtual layer characterizes the underground environment using geometric and performance simulation models. The evaluation and optimization algorithm layer further computes comfort levels and energy performance, with results fed back to the service layer to support environmental control strategy formulation.

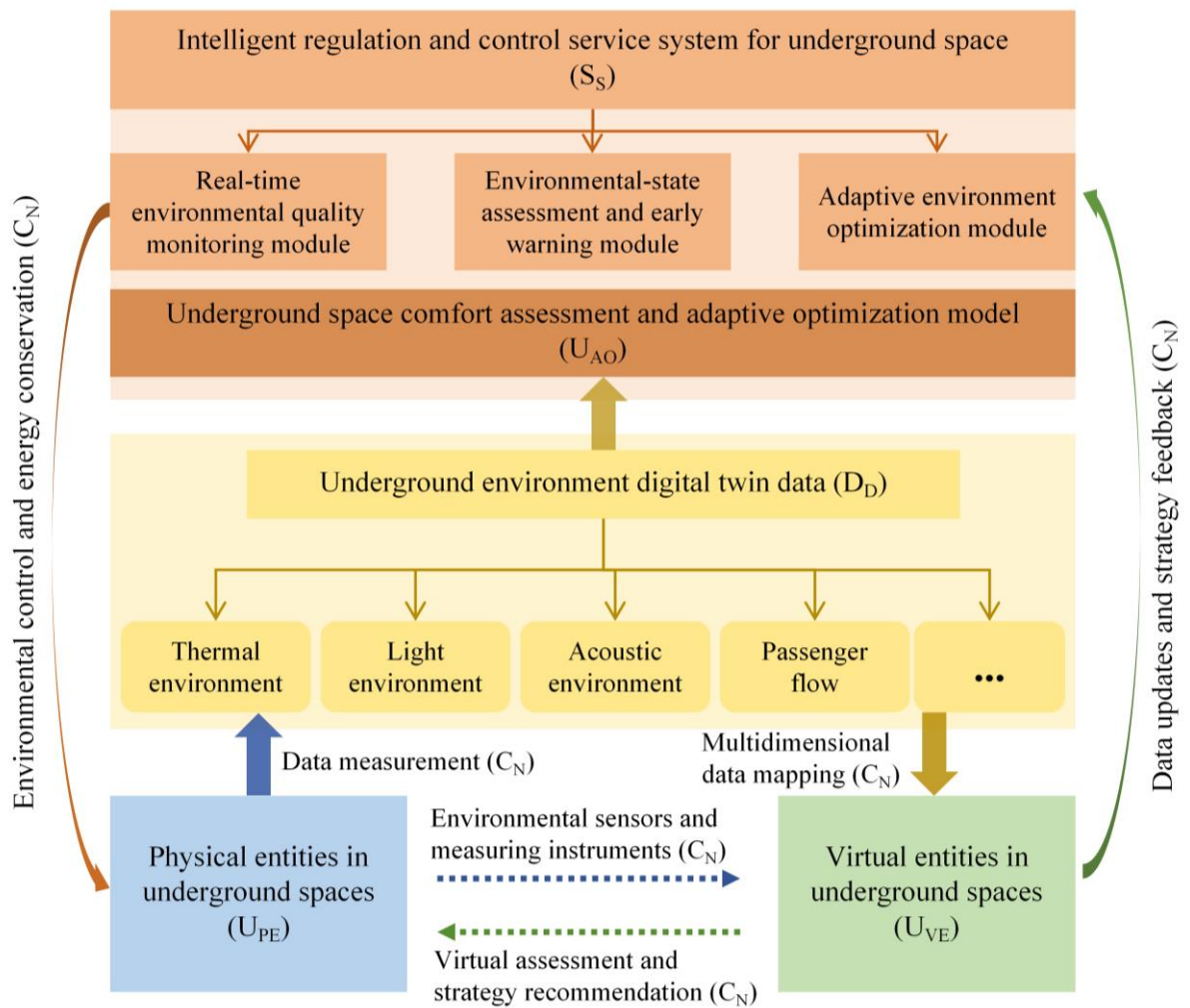


Fig. 1. Hybrid model of multidimensional comfort in underground space based on digital twin.

Based on this theoretical model, a hierarchical and modular six-layer digital twin system architecture is further developed, as shown in Fig. 2, translating the components of Equation

207 (1) into interoperable technical modules that support data acquisition, comfort assessment,
208 energy simulation, optimization strategy generation, and case-based evaluation. Construction
209 methods for each layer are detailed in Sections 3.2–3.7.
210

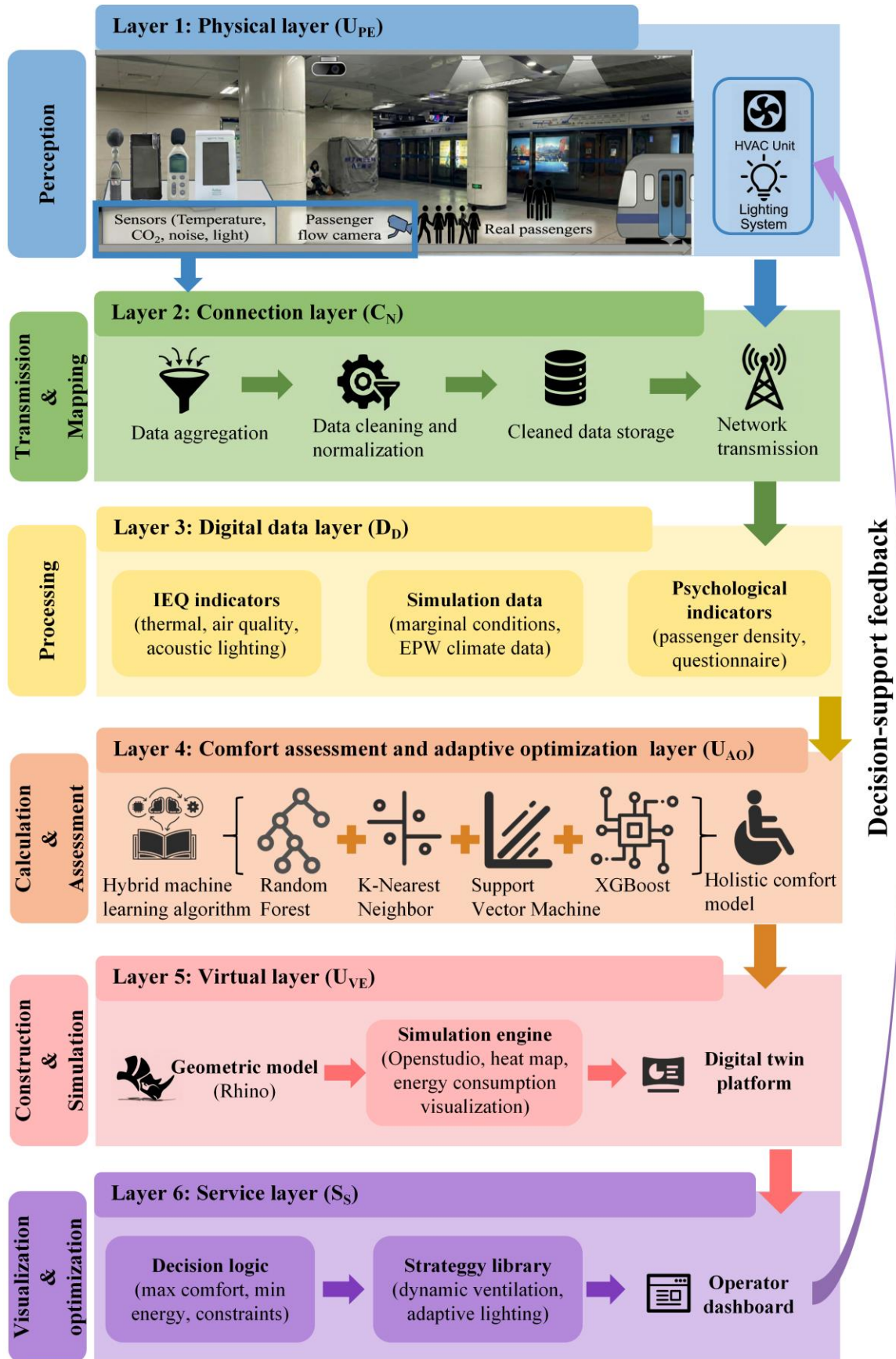


Fig. 2. System framework for environmental regulation and comfort improvement in underground spaces.

3.2. Physical layer

The physical perception layer (U_{PE}) provides fundamental data support for the M_{UDT} model, enabling high-precision mapping of environmental fields and dynamic passenger flow through a sensor network. Sensor deployment follows the guidelines of the *Code for Design of Metro* [33] and *Acoustical Requirement and Measurement on Station Platform of Urban Rail Transit* [34], covering four functional zones: station hall, platform, transfer corridor, and carriage, while also synchronously collecting outdoor meteorological data for reference. Measurement points are deployed on a 10–15 m grid, positioned at least 0.5 m away from HVAC outlets, heat sources, and walls, with monitoring heights corresponding to standard positions for human breathing, visual, and auditory perception. Passenger flow terminals are ceiling-mounted above gates and carriages, collecting data from a vertical perspective to avoid occlusion.

Based on existing specifications of physical environmental monitoring instruments, a detailed layout and deployment method for station sensors are proposed, as illustrated in Fig. 3. Detailed parameters and equipment specifications of each sensor are listed in Appendix Table B.1.

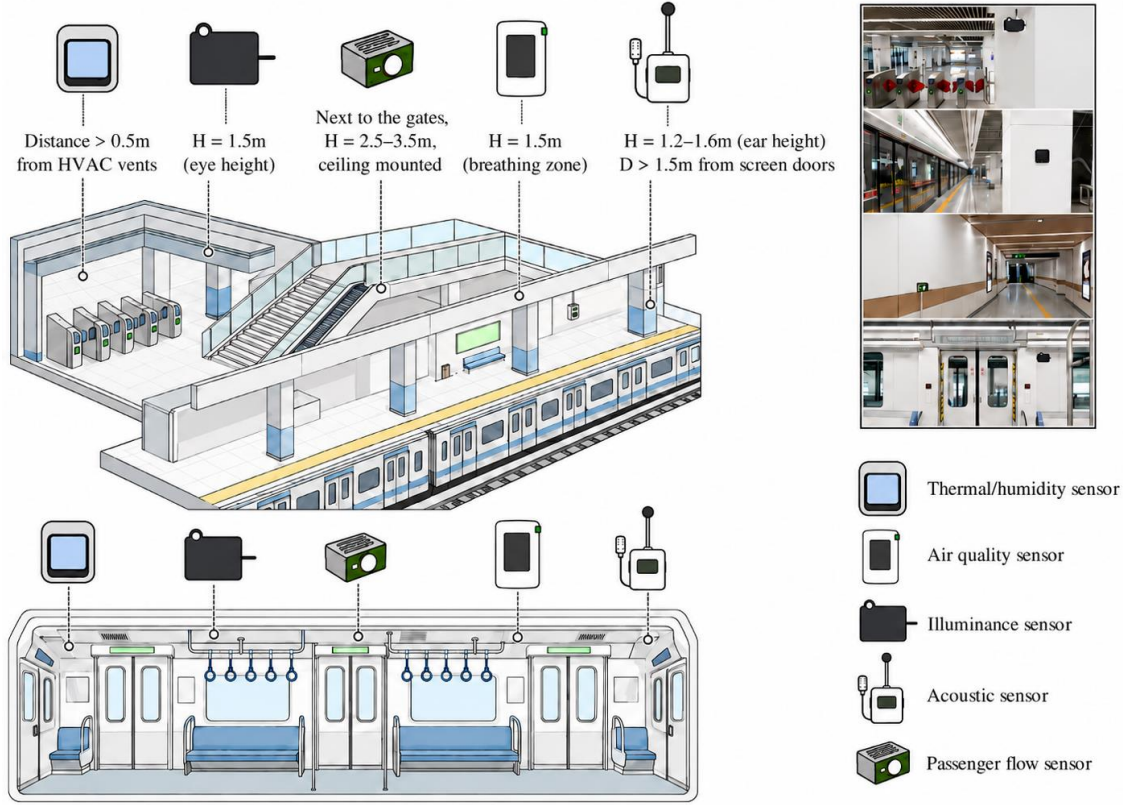


Fig. 3. Deployment strategy for monitoring instruments.

3.3. Connection layer

The connection layer (C_N) receives data collected from the Physical Perception Layer (U_{PE}) and converts it into standardized data that can be stored and accessed in the digital twin data layer (D_D) [35]. Environmental sensors and ToF passenger flow terminals deployed in the underground space are connected to on-site acquisition terminals according to their measurement point identifiers. Data are uploaded via wireless Wi-Fi to regional gateways and then transmitted through the station's local area network to the backend database.

To ensure consistency across multi-source data, the connection layer standardizes timestamps, functional zone identifiers, and sensor type fields, and performs unit conversions, outlier detection, missing data imputation, and format normalization. To further evaluate the data quality of the connection layer, the raw monitoring records were checked before being used for comfort assessment and simulation input. Missing records accounted for 1.8% of the environmental-monitoring dataset, and outliers identified by physical range checks and the 3σ criterion accounted for 0.7%. Short gaps no longer than three consecutive sampling intervals

were filled using linear interpolation, whereas longer gaps were replaced using the median value of the same functional zone and corresponding operating period. For real-time environmental and passenger flow data, the connection layer continuously writes updates to the database at fixed intervals. Non-real-time data, such as survey responses, geometric models, and meteorological boundary conditions, are linked using unified identifiers with temporal or spatial fields.

Thus, the connection layer not only facilitates the transmission of physical perception data to the digital twin platform but also establishes correspondences among physical measurement points, database records, and virtual model nodes, providing reliable input for subsequent environmental state identification, comfort evaluation, and adaptive optimization strategy generation.

3.4. Digital data layer

The digital data layer (D_D) is responsible for storing and managing multi-source heterogeneous data [36]. Based on the post-processed transmitted data, this layer implements multi-source data storage through a relational database (MySQL) and cloud storage solutions. For large-volume, unstructured static data, such as 3D geometric model files of subway stations and electromechanical equipment, the study indexes and manages them via link addresses, with the actual files stored in the cloud.

For real-time environmental and passenger flow data, as shown in Appendix Fig. A.1, structured storage and management are carried out using MySQL, supporting retrieval and backtracking of extensive historical monitoring data. By implementing unified temporal and spatial coding mapping, the system supports real-time visualization of underground space environmental states and identification of comfort weaknesses [23], laying a solid data foundation for generating adaptive environmental control strategies.

3.5. Comfort evaluation and adaptive optimization layer

3.5.1. Data sources

To construct an underground space environmental comfort assessment model, this study establishes a foundational dataset based on three types of data: on-site environmental

monitoring, passenger flow statistics, and subjective passenger surveys. Field monitoring was conducted at 11 representative Beijing subway stations, covering both winter heating and summer cooling periods. Monitored variables included thermal and humidity conditions, lighting, acoustic environment, air quality, and passenger crowding. Sensors were deployed following the principles described in Section 3.2, and the grading rules for passenger crowding are shown in Appendix Fig. A.2.

Subjective comfort data were collected simultaneously via on-site questionnaires. The surveys were designed in accordance with ISO 10551:2019 [37] and ASHRAE Standard 55 [38], using a five-level Likert scale (-2 to +2) to quantify passengers' satisfaction with thermal, lighting, acoustic, air quality, and overall environmental conditions, as illustrated in Fig. 4. After lie-detection question checks, invalid responses were removed, and response time filtering was applied, resulting in 581 valid samples with a response rate of 81.7%. The respondent profile was characterized by a high proportion of young and frequent metro users. Specifically, 79% of respondents were 30 years old or younger, 60% were female, and 80% were medium-to-high frequency passengers taking at least three trips per week. These valid samples provided reliable data support for subsequent machine learning model training, dynamic comfort assessment, and optimization strategy generation.

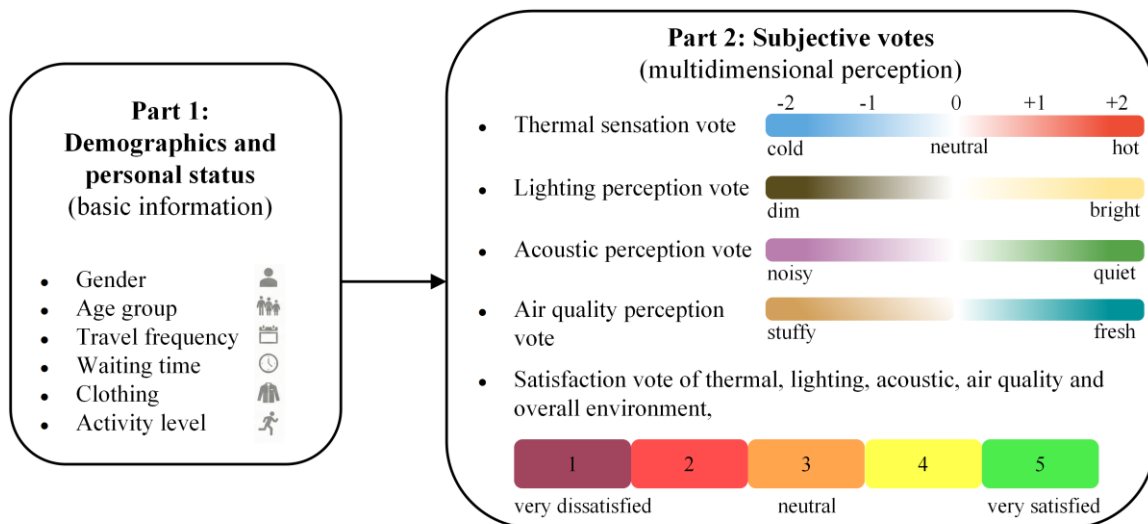


Fig. 4. Logic structure of the subjective questionnaire for passenger comfort assessment.

3.5.2. Dynamic comfort evaluation framework

To achieve precise environmental perception inference within the digital twin system, an operation scenario-adaptive comfort evaluation and decision-support framework (U_{AO}) was developed. This framework incorporates spatial functional attributes (e.g., platform, station hall) and real-time passenger crowding as key situational variables in the assessment logic.

At the computational architecture level, the system integrates machine learning algorithms such as Random Forest (RF) and Extreme Gradient Boosting (XGBoost), as shown in Fig. 5. Using synchronized subjective and objective data, passenger subjective comfort scores are used as training labels, while monitored environmental indicators, space types, and passenger crowding are input features to establish the mapping between environmental states and overall comfort. The comfort model was trained using a stratified 8:2 training–test split based on overall comfort class, functional zone, and crowding level. To address class imbalance, SMOTE was applied only to the training set. RF and XGBoost models were compared using five-fold stratified cross-validation, with macro-F1 as the primary evaluation metric. The validation procedure and model performance are summarized in Appendix Table B.2.

To enhance model interpretability, SHAP attribution analysis was applied to the trained machine-learning model. SHAP assigns feature-attribution values to individual model predictions, while aggregated mean absolute SHAP values can be used to characterize model-level feature and interaction importance. In this study, these attribution results were used to identify the relative importance of environmental features and selected interaction terms under different operational scenarios, with the results illustrated in Fig. 6. This analysis framework helps explain how environmental conditions and their interactions are associated with overall comfort under different operational scenarios, providing methodological support for generating precise quantitative evaluation benchmarks in the digital twin system.

Comfort Assessment and Adaptive Optimization Model (U_{AO})

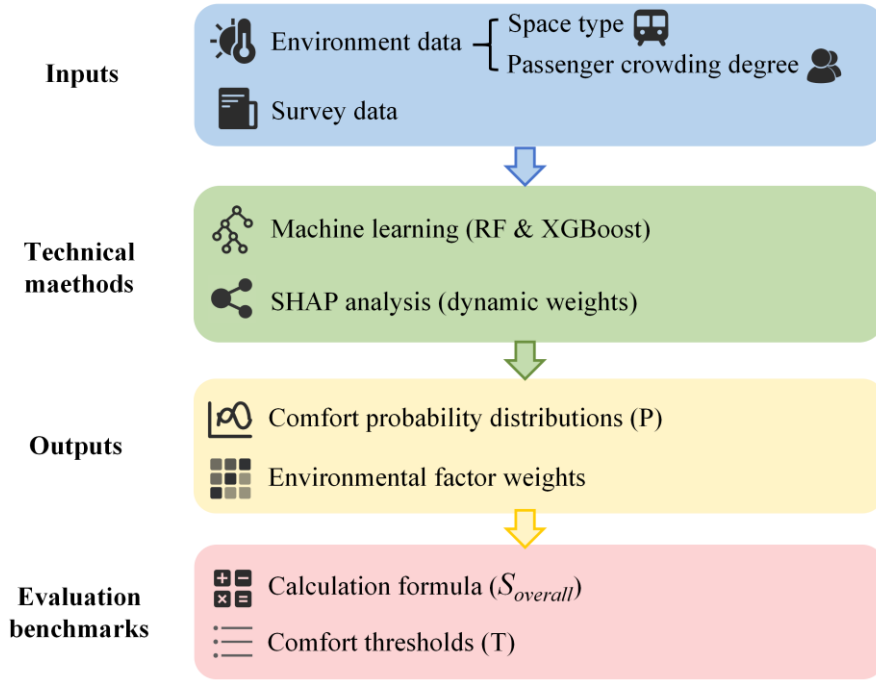


Fig. 5. Comfort evaluation framework.

Operational scenario	S_L	S_S	S_W	S_A	$S_L \times S_S$	$S_L \times S_W$	$S_L \times S_A$	$S_S \times S_W$	$S_S \times S_A$	$S_W \times S_A$
Hall										
- Spacious	0.0	7.5	11.6	5.9	5.7	5.0	0.0	31.0	7.1	16.2
- Moderately Crowded	3.1	4.0	35.3	5.4	0.0	10.0	7.7	7.5	0.0	22.6
- Crowded	0.0	19.8	16.0	3.3	14.4	13.7	0.0	6.2	12.8	12.4
Passageway										
- Spacious	16.8	4.8	14.6	16.8	0.0	8.4	5.1	0.0	6.3	22.2
- Moderately Crowded	0.0	0.0	6.0	12.6	5.1	5.7	14.0	5.0	6.1	36.2
- Crowded	5.6	9.8	22.0	8.0	10.6	0.0	8.4	16.5	0.0	14.1
Platform										
- Spacious	5.3	0.0	8.9	20.9	0.0	10.6	16.5	5.8	5.8	20.8
- Moderately Crowded	3.8	0.0	8.7	15.3	0.0	11.1	8.7	5.1	6.2	35.3
- Crowded	12.5	0.0	6.6	8.2	4.4	10.9	12.6	0.0	4.8	32.9
Carriage										
- Spacious	0.0	15.0	8.8	11.1	6.7	0.0	5.4	16.0	15.3	18.4
- Moderately Crowded	5.4	0.0	5.4	0.0	8.5	15.0	6.9	21.8	7.1	22.1
- Crowded	5.9	8.1	0.0	8.8	13.0	6.1	15.1	0.0	10.8	24.4

S_L : lighting environment
 S_S : acoustic environment

S_W : thermal environment
 S_A : air quality

SHAP attribution (%)
 0 10 20 30 40

Fig. 6. SHAP attribution percentages of environmental factors under various operational scenarios.

3.5.3. Dynamic evaluation model and grading standards

The overall comfort state was learned using nonlinear machine-learning models, in which the input features included sub-dimensional comfort scores, operational context

variables, and interaction terms among comfort dimensions. To support rapid comfort-state estimation in the digital twin platform, the trained model was further interpreted using SHAP analysis. For each operational scenario, the mean absolute SHAP values of the selected comfort features and interaction terms were normalized to obtain scenario-specific relative aggregation weights for the overall comfort calculation. This use of SHAP-derived weights provides a model-informed representation of the relative importance learned by the fitted comfort model. Based on the scenario-specific feature weights output by the machine learning framework as shown in Fig. 6, the overall comfort score is calculated as follows:

$$S_{overall} = \sum_{i=1}^4 w_{i,c} S_i + \sum_{i<j} w_{ij,c} S_i S_j \quad (2)$$

Where $S_{overall}$ denotes the overall comfort score; S_i denotes the sub-dimensional comfort score; $S_i S_j$ denotes the interaction term between two comfort dimensions; $w_{i,c}$ and $w_{ij,c}$ denote the scenario-specific normalized weights; and c denotes the operational context defined by functional zone and crowding level.

To support physical-parameter range extraction at the sub-dimensional level, a comfort-probability criterion of $P \geq 0.6$ was used to identify parameter intervals associated with a relatively high probability of being perceived as comfortable. For the overall comfort score $S_{overall}$, which serves as the primary decision variable, the classification thresholds were determined separately. Based on the distributions of $S_{overall}$ in the training set, Gaussian fitting and intersection analysis were applied to adjacent subjective evaluation classes, with the Gaussian probability density function given in Equation (3), yielding the optimal classification thresholds $T1 = 0.42$, $T2 = 0.68$. These mathematical boundaries classify environmental states into three levels: “uncomfortable” ($S_{overall} < 0.42$), “neutral” ($0.42 \leq S_{overall} \leq 0.68$) and “comfortable” ($S_{overall} > 0.68$). The class-wise goodness-of-fit summary for the Gaussian fitting is reported in Appendix Table B.3, and the classification performance of the Gaussian-derived thresholds on the test set is presented in Appendix Table

B.4. This validation provides a solid grading basis for subsequent operation-scenario decision support.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (3)$$

Where x represents the overall comfort score $S_{overall}$. By solving the intersection equations of the adjacent evaluation level distribution curves, $f_i(x) = f_{i+1}(x)$, the optimal classification thresholds T_n are obtained.

3.6. Virtual layer

The virtual simulation layer (U_{VE}) provides physical simulation support for the Comfort Evaluation and Adaptive Optimization Layer (U_{AO}). This layer replicates the geometric structure of the subway station in the digital space and maps thermal and energy performance. The output environmental and energy consumption data are used to complement and validate the comfort evaluation results.

Based on CAD drawings of the station, 3D geometric models were constructed in Rhino, with thermal zoning applied to station halls, platforms, and transfer corridors in accordance with thermal consistency requirements, while non-critical building components were appropriately simplified. The geometric model is then imported into OpenStudio, where building envelope thermal parameters are defined according to current national standards, such as *Design Standard for Energy Efficiency of Public Buildings* [39], and on-site measured data. Soil temperature boundary conditions are set to accurately represent the thermal exchange characteristics between underground structures and surrounding soil.

Using the Grasshopper parametric platform and the Honeybee plugin, the model integration and simulation workflow is constructed. Typical local meteorological data, dynamic passenger loads, lighting, and equipment operation parameters are input as boundary conditions. EnergyPlus is used as the simulation engine to perform hourly simulations throughout the year, producing key indicators such as air temperature, mean radiant temperature, and HVAC system energy consumption. The annual operational carbon

emissions were calculated from the simulated site electricity consumption using the following equation:

$$C_{\text{total}} = E_{\text{total}} \times EF_{\text{grid}} \quad (4)$$

Where C_{total} denotes annual total carbon emissions (kgCO₂), E_{total} denotes annual total electricity consumption (kWh), and the latest official Beijing provincial grid emission factor, $EF_{\text{grid}} = 0.5554$ kgCO₂/kWh, was adopted for operational carbon-emission accounting.

The Beijing provincial factor was selected because the case station is located in Beijing; the corresponding national and North China regional average factors are 0.5306 and 0.6361 kgCO₂/kWh, respectively. Different local or national average factors would affect the absolute carbon-emission estimate but not the relative reduction when the same constant factor is applied to both scenarios. The annual-average factor adopted here does not capture temporal variations in grid carbon intensity.

The simulation workflow is illustrated in Fig. 7. Simulation results are fused with comfort evaluation model data within the digital twin platform, presenting the spatial distribution of environmental parameters and comfort in 3D visualization, providing direct data support for the generation and evaluation of adaptive environmental management strategies.

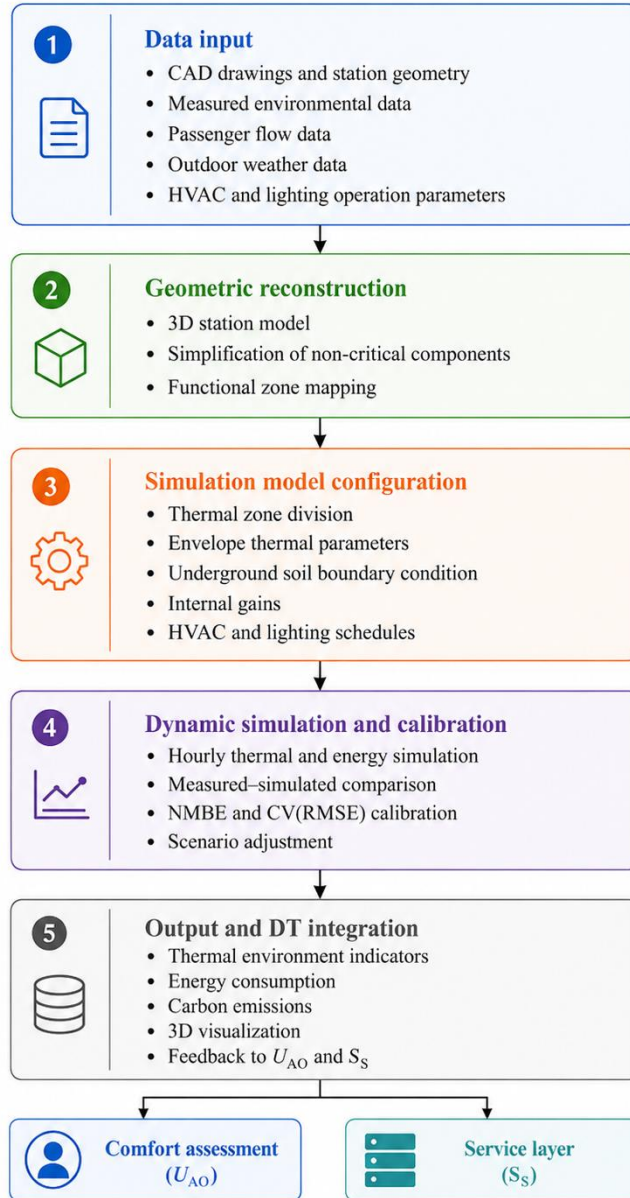


Fig. 7. Technical route for energy and comfort simulation.

3.7. Service layer

The decision service layer (S_S) transforms the comfort assessment and energy simulation results from the virtual simulation layer (U_{VE}) into environmental control strategies for operational management. As shown in Fig. 8, this layer takes real-time environmental parameters, passenger crowding, overall comfort scores, and system energy consumption as inputs, and evaluates the current operational status and corresponding response levels based on three types of constraints: safety limits, equipment operational ranges, and comfort thresholds. Accordingly, the adaptive optimization process is achieved by selecting

context-dependent environmental control strategies according to the identified operational states and generating corresponding scenarios for subsequent performance evaluation [40].

As shown in Fig. 9, the service layer jointly considers two performance goals: enhancing passenger comfort and reducing system energy consumption. Their coordination follows the graded decision logic shown in Fig. 8, in which safety and equipment constraints are prioritized, followed by comfort status and passenger-flow conditions. When environmental parameters exceed safety or equipment limits, the system enters a red emergency mode; when the overall comfort score ranges from 0.42 to 0.68, a yellow warning mode is triggered; when comfort requirements are met but passenger flow is high, a blue dynamic balance mode is activated; and when passenger flow is low and the environment is stable, a green energy-saving mode is adopted. Based on these graded results, the service layer further matches response strategies for subsystems such as ventilation, air conditioning, lighting, and broadcasting, as illustrated in Fig. 9, providing a strategic basis for scenario-based adaptive optimization simulations in subsequent case studies. These response strategies constitute the adaptive optimization scenarios, in which alternative environmental configurations are evaluated through subsequent comfort and energy simulations to identify suitable operation strategies under different conditions [41].

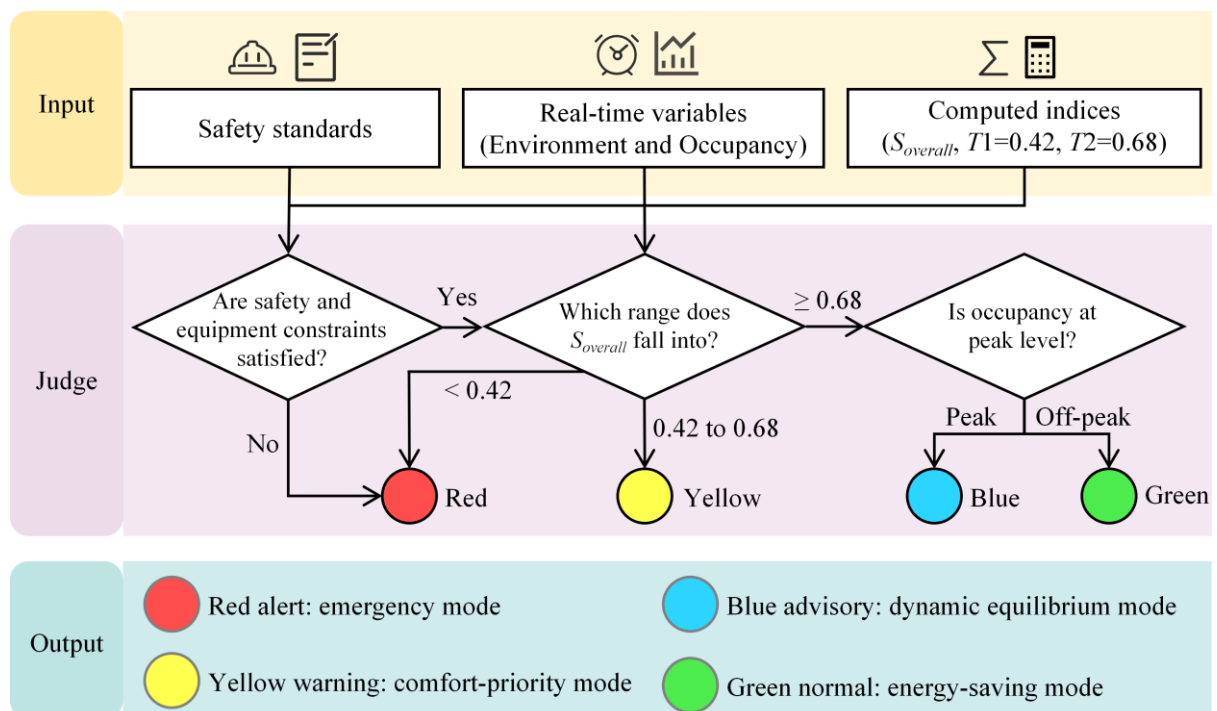


Fig. 8. Decision framework for balancing safety, comfort, and energy consumption in subway stations.

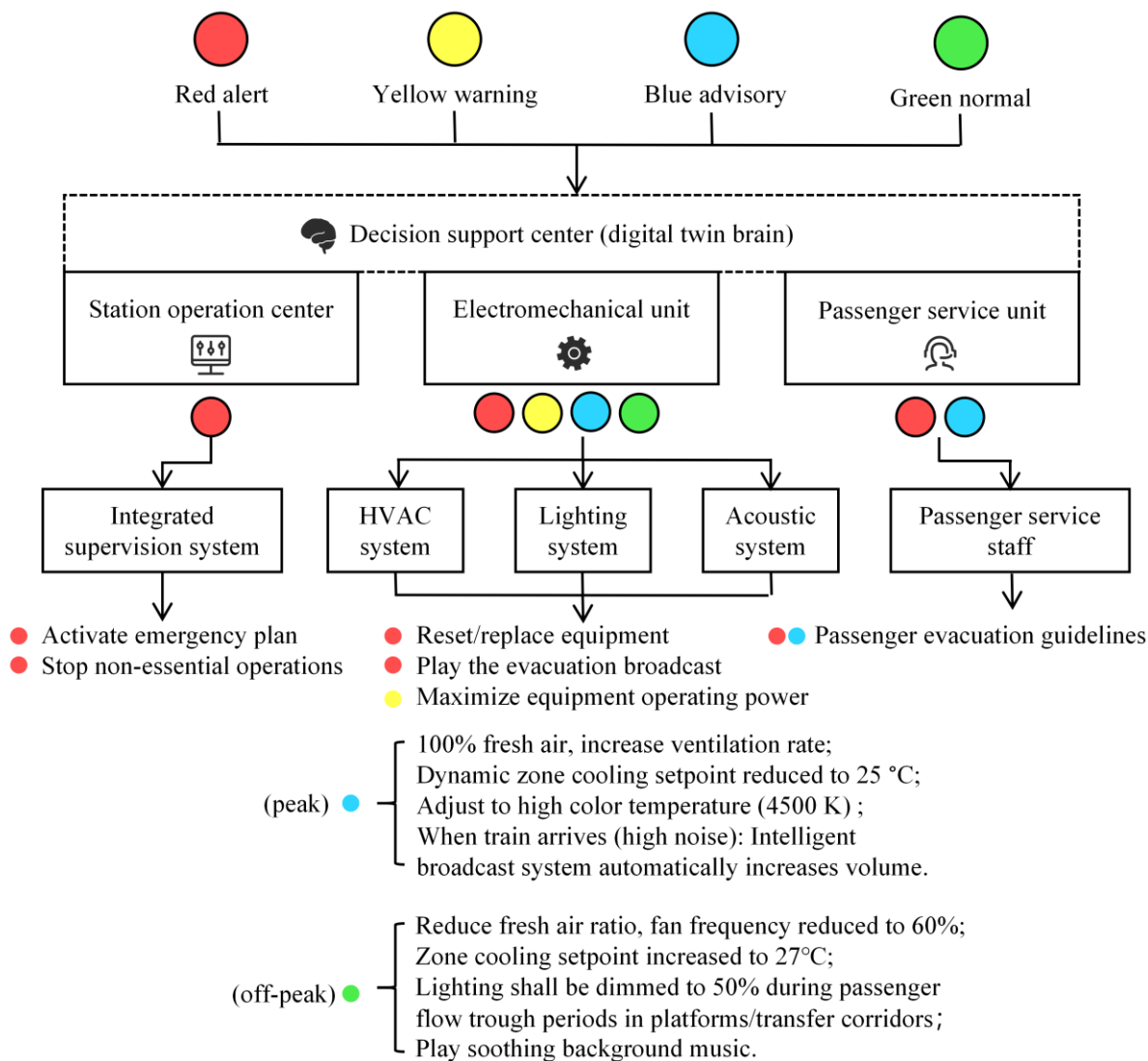


Fig. 9. Response plans of metro departments under different warning levels.

4. Case study and results

The case study was designed to assess the applicability and potential performance of the M_{UDT} framework in an existing subway station. The evaluation followed three steps: (i) establishing and calibrating the baseline simulation model using field measurements, (ii) identifying comfort and energy-performance weaknesses under typical operational scenarios, and (iii) comparing the existing and optimized strategies. Model calibration was evaluated using NMBE and CV(RMSE), while framework performance was assessed based on overall comfort scores, the proportions of comfort states, annual energy consumption, and operational carbon emissions. The underlying hypothesis was that context-aware adaptive

strategies could reduce modeled low-comfort periods and energy use compared with conventional fixed-schedule operation.

4.1. Case study

4.1.1. Basic information and passenger flow characteristics of the subway station

An early-built, medium-sized underground transfer station in Beijing was selected because it exhibits pronounced tidal passenger flow, fixed-schedule environmental control, peak-period comfort deficiencies, and off-peak energy waste. These characteristics are common challenges in existing urban rail stations and provide a suitable setting for evaluating the M_{UDT} framework under dynamic operational conditions. Opened in 2003, the station includes two station halls, platform levels for two transfer lines, and transfer corridors, as illustrated in Fig. 10.

Serving as a hub connecting large residential areas with the urban core, the station exhibits a pronounced “tidal” spatiotemporal distribution of passenger flow. During weekday peak hours, all functional areas experience extremely high passenger densities. To accurately quantify this dynamic load, real-time monitoring data from ToF terminals were combined with officially released passenger flow indices, allowing for grading of crowding levels and providing the data foundation for constructing scenario-adaptive comfort assessment models.

Regarding environmental and energy control systems, this station, being relatively old, primarily relies on conventional centralized mechanical ventilation and air conditioning (HVAC) systems for environmental regulation. Operational strategies are mostly based on fixed seasonal and time schedules, representing a typical passive response system, which leads to inadequate comfort during peak periods and energy waste during off-peak hours.

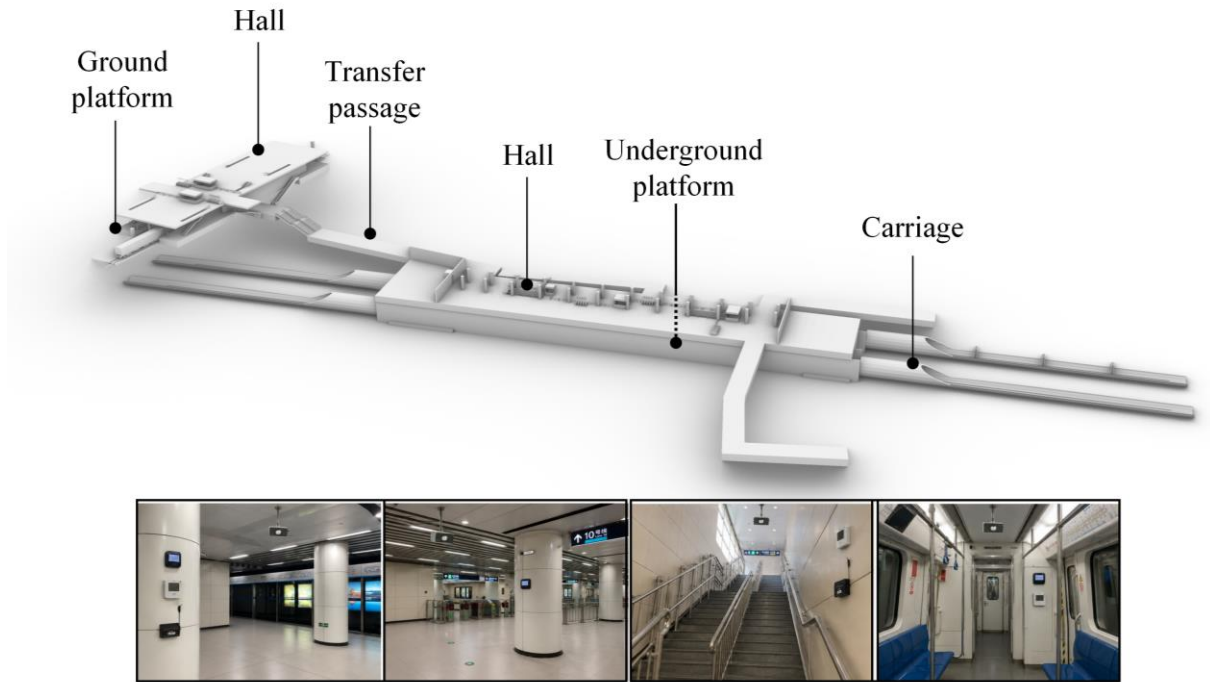


Fig. 10. Spatial layout and field views of the case subway station.

4.1.2. Data acquisition scheme

To comprehensively capture the spatiotemporal environmental characteristics of the transfer station, continuous on-site monitoring was conducted, covering morning and evening peak as well as off-peak operational periods. A total of 20 environmental monitoring nodes were deployed within the station, covering key areas including the station hall, platforms, transfer corridors, and main entrances/exits.

In the platform areas, considering that the two transfer lines are located at different underground levels and are significantly affected by train operations, screen door openings, and piston effects, measurement points were set at both core waiting areas and platform ends to capture spatial variations in thermal, humidity, and air quality conditions. Along long transfer corridors, CO₂ and PM_{2.5} monitoring nodes were primarily placed in mid-corridor sections and high-density areas to identify air quality changes under dense pedestrian movement. Main entrances and exits were equipped with reference nodes, coordinated with outdoor meteorological stations as external benchmarks. Passenger flow terminals sampled data at one-minute intervals, synchronized with environmental sensor measurements, ensuring correspondence between passenger density and environmental conditions.

4.2. Existing environmental comfort and energy consumption assessment

4.2.1. Simulation model analysis

To ensure that the digital model accurately reflects the operational status of the physical entity, hourly measured data from three consecutive summer and winter days were used to calibrate the EnergyPlus model, as shown in Fig. 11. Simulated air temperatures were compared with on-site sensor measurements. Normalized mean bias error (NMBE) and coefficient of variation of RMSE (CV(RMSE)) were used as evaluation metrics, as defined in Equations (5) and (6). The calibrated model yielded NMBE = -0.06% and CV(RMSE) = 1.29% for the typical summer period, and NMBE = 0.77% and CV(RMSE) = 1.84% for the typical winter period. All error metrics were well below the allowable limits defined by ASHRAE Guideline 14 ($|NMBE| < 10\%$, $CV(RMSE) < 30\%$) [42], indicating close agreement between simulated and measured air temperatures during the calibration periods and supporting the use of the calibrated model for subsequent evaluation. To further examine the robustness of the simulated energy-saving results, a one-at-a-time sensitivity analysis was conducted using the calibrated OpenStudio/EnergyPlus model and the original weather file. Passenger density, ventilation rate, and lighting power were varied by $\pm 10\%$, while cooling and heating setpoints were varied by $\pm 1^\circ\text{C}$, with all other inputs held constant. Across the tested cases, the annual energy-saving rate ranged from 14.7% to 15.8%, compared with 15.3% in the reference simulation, with ventilation rate producing the largest variation, as shown in Table B.5. The energy-saving tendency remained consistent across all tested input variations.

$$NMBE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{(n-1) \times \bar{y}} \times 100\% \quad (5)$$

$$CV(RMSE) = \frac{1}{\bar{y}} \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-1}} \times 100\% \quad (6)$$

where y_i and $\hat{y}_i$ are measured and simulated data, $\bar{y}$ is the mean of measured data, and n is the total number of data points.

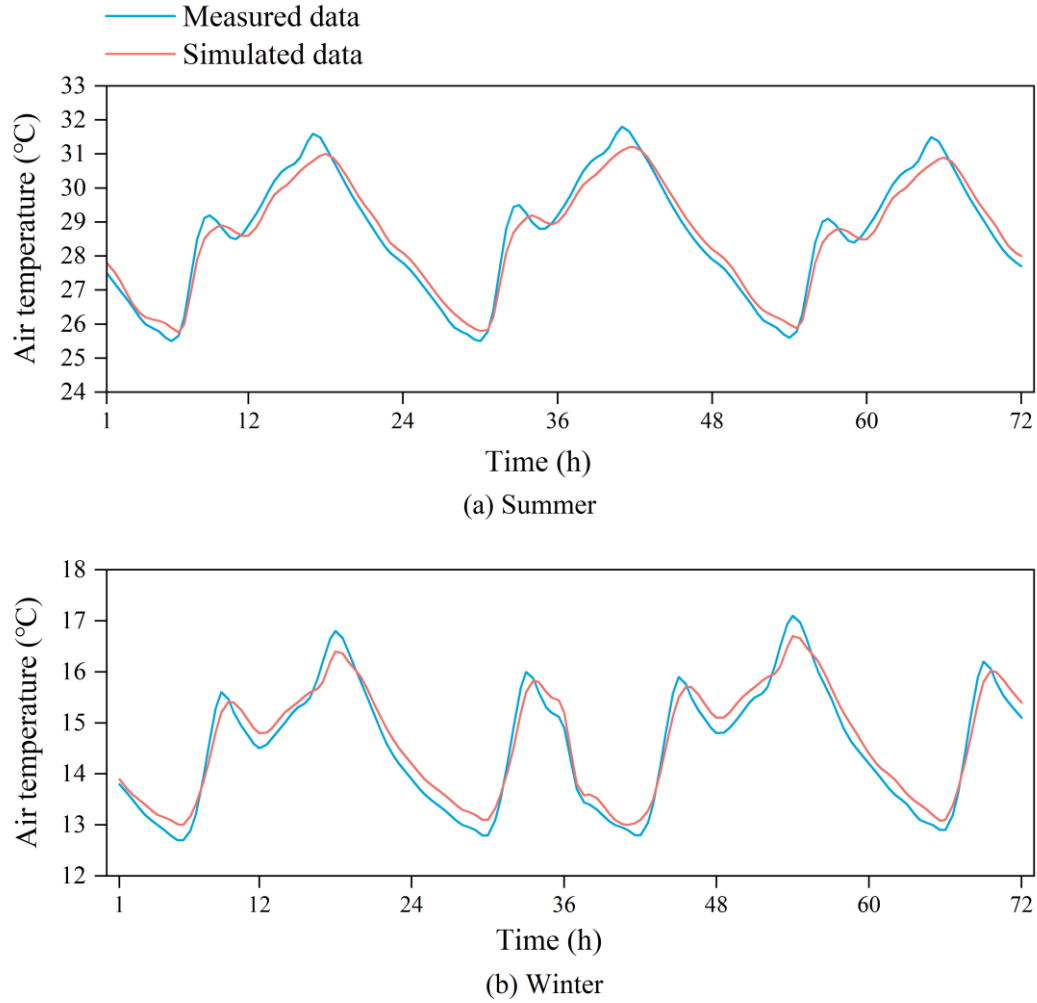


Fig. 11. Comparison of simulated and measured hourly air temperatures during typical summer and winter periods.

4.2.2. Identifying comfort weaknesses

Based on the comfort evaluation and adaptive optimization algorithm (U_{AO}), the current environmental control strategies of the case subway station were comprehensively assessed for overall comfort. Results reveal spatiotemporal differences in comfort under different operational scenarios. As shown in Table 1, temporally, the overall summer comfort score ($S_{overall}$) across all spaces is 0.48, with uncomfortable periods accounting for 36.2%—approximately 3.5 times higher than in winter (10.3%, with a score of 0.59). This seasonal difference reflects the combined challenge of high temperature, high humidity, and dense passenger flow in Beijing’s summer. Spatially, overall comfort follows the gradient:

station hall > transfer corridor > platform > carriage, with platforms and carriages identified as core targets for optimization.

Using SHAP dynamic weight attribution, during summer peak periods, the interaction of thermal environment and air quality showed the highest SHAP attribution, accounting for 35.3% in the summer-peak platform scenario. Air temperature reaches 28.7°C with 67.2% relative humidity and a CO₂ concentration of 1391 ppm, frequently causing overall comfort scores to fall below 0.42.

Further analysis combining factor contribution weights (Fig. 6) and the trends of environmental parameters and overall comfort scores (Fig. 12) indicates that lighting in the station hall exhibits hidden low-comfort conditions during off-peak periods. Between 10:00–12:00, lighting exceeds 400 lx, yet overall comfort ($S_{overall}$) decreases, creating a paradox of low passenger density but low comfort and energy waste. Uncomfortable periods are concentrated during peak hours, with delayed environmental control responses. This is due to passive control strategies, homogeneous mechanical ventilation, and HVAC power ramp-up lagging behind human heat generation and CO₂ release, highlighting the need for real-time monitoring and adaptive environmental management.

Table 1

Statistics on comfort characteristics of various functional areas.

Season	Functional area	$S_{overall}$ mean	Standard deviation	Proportion ($S_{overall} > 0.68$)	Proportion ($0.42 \leq S_{overall} \leq 0.68$)	Proportion ($S_{overall} < 0.42$)
Summer	Hall	0.56	0.08	24.2%	55.4%	20.4%
	Platform	0.44	0.11	12.1%	43.7%	44.2%
	Transfer passage	0.53	0.09	20.1%	53.6%	26.3%
	Carriage	0.40	0.12	8.8%	37.2%	54.0%
Winter	Hall	0.65	0.07	38.7%	56.2%	5.1%
	Platform	0.56	0.09	24.8%	63.5%	11.7%
	Transfer passage	0.62	0.08	33.5%	60.8%	5.7%
	Carriage	0.52	0.10	19.3%	62.1%	18.6%

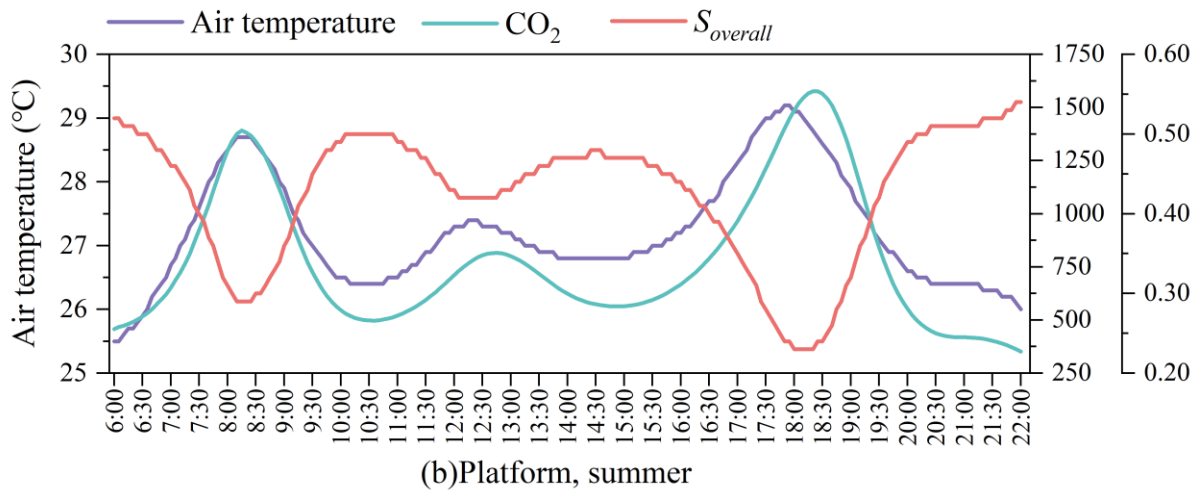
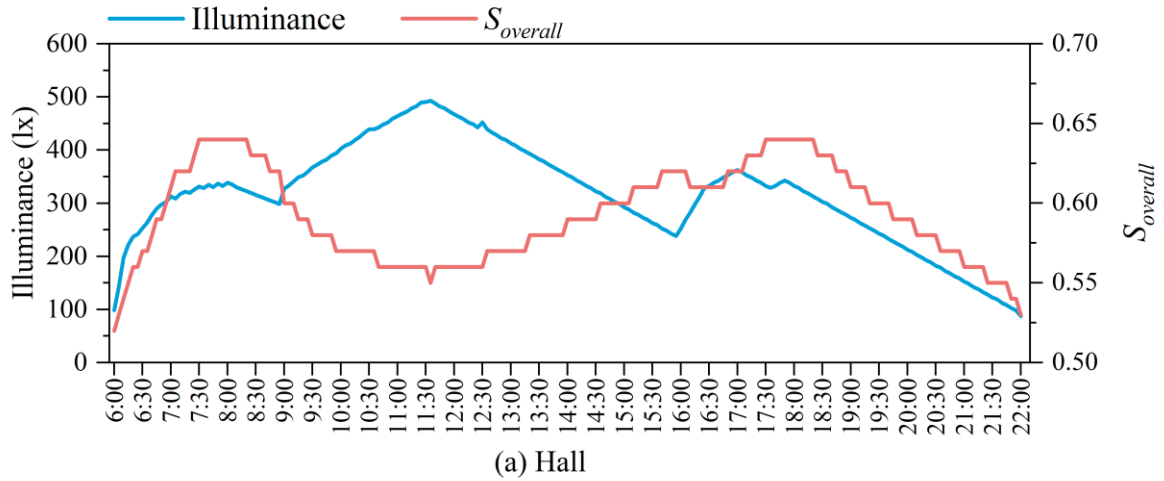


Fig. 12. Temporal evolution of environmental parameters and holistic comfort scores ($S_{overall}$) in functional zones during operation hours. (a) Station hall; (b) Platform in summer.

4.2.3. Energy consumption simulation

Using the technical workflow of the virtual simulation layer (U_{VE}), the energy consumption structure and potential savings of the case subway station were analyzed. Under the existing strategy, the environmental control system operates on a fixed timetable: fresh air and exhaust units run at constant full capacity, and lighting remains at 100% brightness. Simulation results (Fig. 13) show that mechanical power consumption (elevators, gates, etc.) accounts for the largest share of total annual energy use (26.0%), followed by the ventilation system (23.6%).

During off-peak hours, despite an approximately 60% reduction in passenger density compared to peak hours, energy consumption in ventilation and lighting does not decrease

due to the lack of dynamic adjustment, resulting in significant energy waste. Unit energy consumption per passenger during off-peak periods is 2.4 times that of peak periods. Monthly thermal balance analysis (Fig. 14) indicates a pronounced “high in winter, low in summer” asymmetry in energy demand. Winter is the peak consumption period, with January heating load reaching -159.4 kW. Average contributions from passenger heat, lighting, and electrical equipment are approximately 10.7 kW, 11.3 kW, and 16.8 kW, respectively, offsetting about 24% of infiltration heat loss. Although summer cooling loads are lower than winter heating, limited heat dissipation in underground spaces increases the precision requirement for environmental control.

These assessments reveal the core issues of the subway station’s environmental control system: static strategies cannot adapt to highly dynamic passenger flow and environmental load changes, providing a clear direction and quantitative benchmark for the adaptive optimization strategies described in subsequent sections.

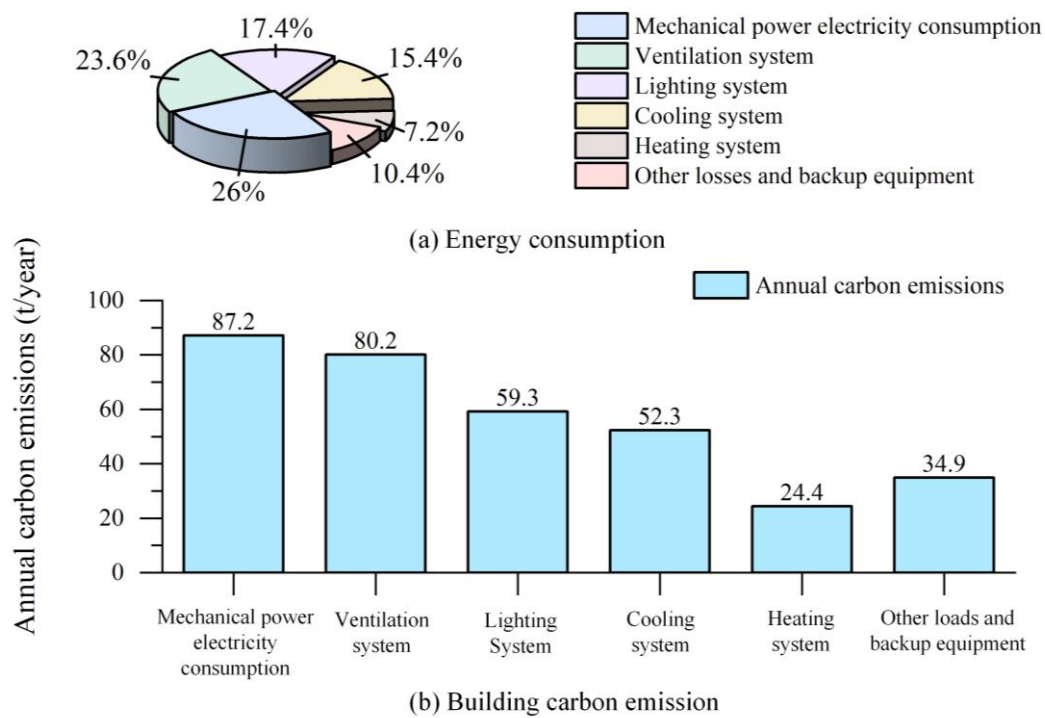


Fig. 13. Annual energy consumption and operational carbon emissions of the subway station.

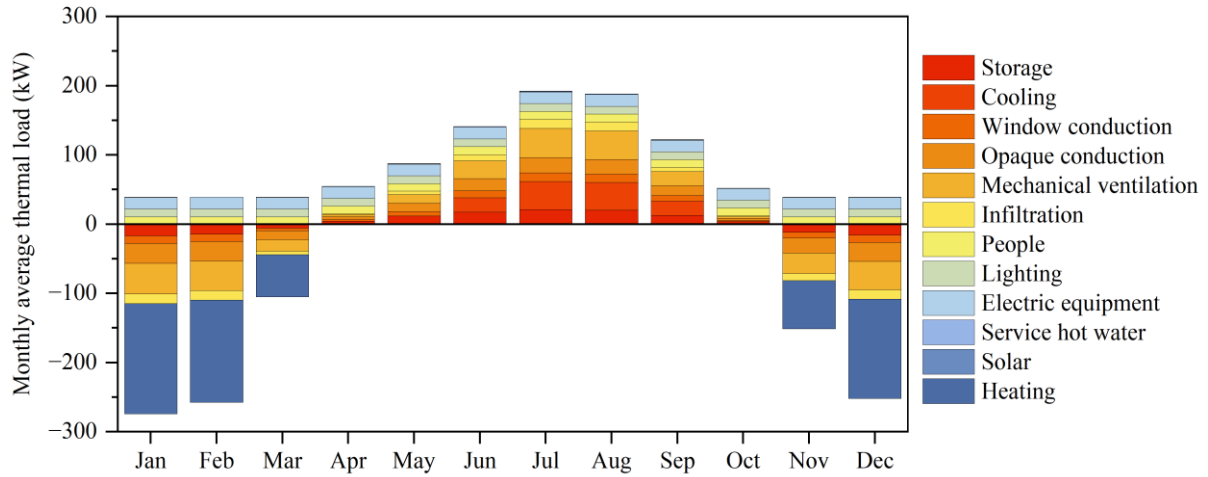


Fig. 14. Monthly thermal balance simulation of subway station.

4.3. Adaptive optimization scenarios and simulation results

4.3.1. Generation of adaptive optimization strategies

To evaluate the effectiveness of the proposed digital twin design strategy, adaptive optimization strategies balancing comfort and energy consumption were evaluated for the case subway station through simulation-based scenarios using the four-level warning decision framework constructed in Section 3.7 (S_s). The system adjusts environmental and energy strategies according to real-time passenger flow and environmental monitoring data, while simultaneously referencing three types of rigid constraints: safety limits, equipment rated operational ranges, and comfort baseline thresholds, under different passenger densities and operational scenarios (Fig. 9).

For the simulation-based evaluation, the four-level warning logic was translated into EnergyPlus schedules and setpoint assumptions. The red emergency mode was treated as a safety-override condition, and the yellow warning mode ($0.42 \leq S_{overall} \leq 0.68$) was represented by comfort-priority HVAC and ventilation operation within rated equipment limits. Under peak passenger-flow conditions, the blue dynamic-balance mode was simulated using full fresh-air operation, increased ventilation, and a reduced cooling setpoint of 25°C. Under low-load off-peak conditions, the green energy-saving mode was simulated using

reduced fresh-air supply, fan-frequency reduction to 60%, a relaxed cooling setpoint of 27°C, and lighting power reduction to 50% in platforms and transfer corridors.

The SHAP-informed prioritization strategy was used to identify the environmental features with relatively high model-attribution importance under each scenario, thereby supporting targeted adjustment of HVAC, ventilation, and lighting settings while contributing to simulated energy and emission reductions.

4.3.2. Simulated results of optimized comfort and energy consumption

In the simulation model, adaptive optimization simulations were conducted by adjusting fresh air supply, HVAC intensity, and electromechanical equipment operation power under different operational scenarios. As shown in Table 2 and Appendix Fig. A.3, environmental comfort ($S_{overall}$) improved across all functional zones. For the summer platform, the mean $S_{overall}$ increased to 0.58, the proportion of low-comfort periods with $S_{overall} < 0.42$ decreased to 12.3%, and the proportion of high-comfort periods with $S_{overall} > 0.68$ increased to 28.5%. Appendix Fig. A.3 shows that the optimized scenario reduces peak temperature and CO₂ concentration, while avoiding unnecessary over-conditioning during relatively low-load periods. Correspondingly, the modeled $S_{overall}$ remains higher and more stable throughout the day.

The simulated energy-saving effect of the adaptive optimization scenario is illustrated in Fig. 15, with the main reductions stemming from the lighting system (32.5%) and ventilation system (22.8%). Total annual station energy consumption and operational carbon emissions both decreased by 15.3%. These results demonstrate that the M_{UDT}-based comfort model can dynamically balance comfort and energy consumption, extracting significant energy-saving potential in underground spaces without compromising, and even enhancing, passenger experience, thereby providing quantifiable decision support for sustainable subway operations.

Table 2
Comfort comparison of functional areas before and after optimization.

Season	Functional area	Traditional strategy			Optimized adaptive strategy		
		$S_{overall}$	Proportion	Proportion	$S_{overall}$	Proportion	Proportion

		mean	(<i>S_{overall}</i> > 0.68)	(<i>S_{overall}</i> < 0.42)	mean	(<i>S_{overall}</i> > 0.68)	(<i>S_{overall}</i> < 0.42)
Summer	Hall	0.56	24.2%	20.4%	0.68	42.6%	3.1%
	Platform	0.44	12.1%	44.2%	0.58	28.5%	12.3%
	Transfer passage	0.53	20.1%	26.3%	0.63	37.2%	7.5%
	Carriage	0.40	8.8%	54.0%	0.52	21.4%	18.9%
Winter	Hall	0.65	38.7%	5.1%	0.71	52.3%	1.2%
	Platform	0.56	24.8%	11.7%	0.64	41.6%	2.8%
	Transfer passage	0.62	33.5%	5.7%	0.69	48.7%	1.5%
	Carriage	0.52	19.3%	18.6%	0.60	32.8%	5.6%

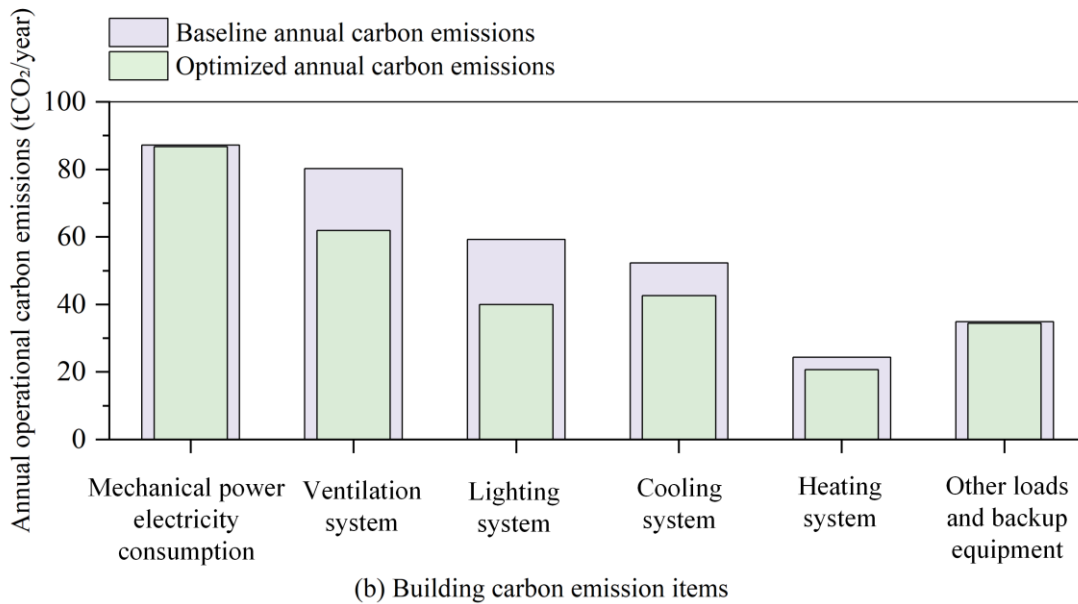
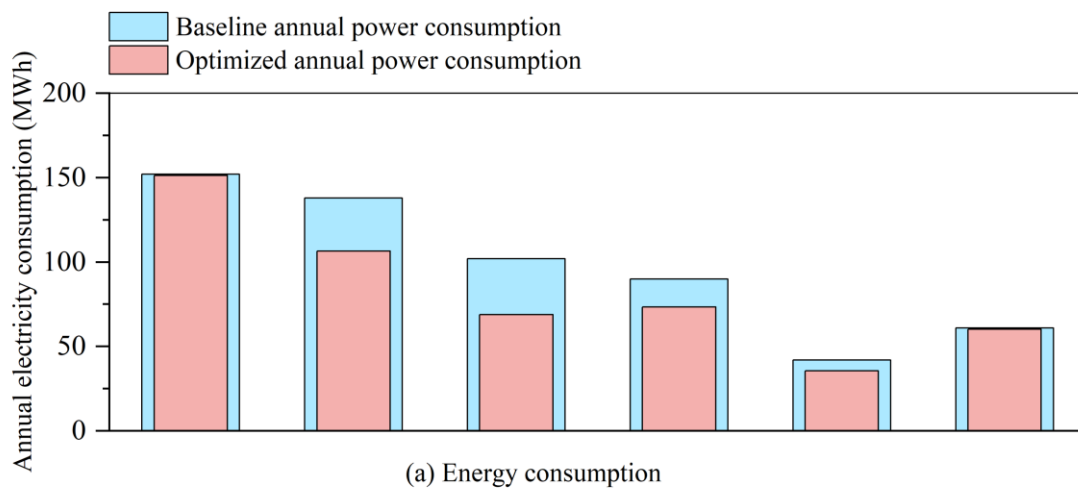


Fig. 15. Comparison of environmental regulation before and after optimization: (a) Energy consumption;
(b) Building carbon emission.

4.4. Digital twin-enabled visualization and decision support

The digital twin visualization platform serves as the terminal interface of the adaptive environmental management framework, integrating multi-dimensional sensor data, comfort-assessment results, and recommended optimization strategies. The platform visualizes monitored environmental states and comfort scores, while the comfort and energy performance of the recommended optimization strategies are evaluated in the calibrated virtual simulation layer and subsequently presented to operators.

Through a high-fidelity digital interface, operators can visually monitor the dynamic evolution of overall comfort scores across the entire station ($S_{overall}$) and visualize the real-time comfort state and the recommended adaptive operation strategies generated by the module (U_{AO}) under complex fluctuating loads. This supports the transition of environmental management from fixed-schedule operation to adaptive decision support.

As shown in Fig. 16, the platform consolidates monitored data and comfort information from the station hall, platform, transfer passage, and carriage, enabling operators to examine environmental states and identify spatial differences across functional zones. Recommended optimization strategies and their corresponding simulation-based performance information are displayed through the interface to support operational decision-making. This integrated mode allows complex environmental conditions to be quantified using intuitive ($S_{overall}$) indicators and enables operators to identify weak points in environmental quality in each sub-zone. Historical monitoring records can further be used to examine temporal environmental variations and support adaptive management under changing passenger-flow conditions.

Additionally, the visualization system supports decision-making for balancing passenger comfort and energy efficiency. By displaying sub-system energy consumption, it supports the coordinated management of passenger comfort and system energy consumption. This digital twin-based full-process monitoring supports adaptive decision-making under passenger-flow fluctuations and facilitates smarter regulation of complex environmental comfort and energy consumption in underground spaces.



(a) Hall



(b) Platform



(c) Transfer passage



(d) Carriage

Fig. 16. Digital twin visualization dashboard for environmental monitoring and decision support across different metro zones. (a) Hall; (b) Platform; (c) Transfer passage; (d) Carriage.

5. Discussion

5.1. Main findings and implications

Recent multi-domain IEQ studies have emphasized that environmental comfort results from the combined effects of thermal, visual, acoustic, and air-quality conditions rather than from a single physical variable [43,44]. The present findings support this view in underground transportation spaces, where environmental perception also varies with functional zone and passenger crowding. In particular, the thermal–air-quality interaction showed the highest SHAP attribution during summer peak periods, reaching 35.3%, as dense passenger flow simultaneously increased internal heat gain and CO₂ accumulation. The scenario-dependent variation in feature contributions further indicates that fixed environmental weights may not adequately represent comfort under dynamic subway operation. These SHAP-derived contributions should therefore be interpreted as

context-specific feature importance for comfort estimation and strategy prioritization rather than as causal control coefficients.

Compared with previous multi-domain studies that primarily focus on environmental assessment, the operation-oriented value of the proposed framework lies in identifying when, where, and why low-comfort states occur. The platform and carriage were more vulnerable to summer discomfort, whereas excessive off-peak lighting in the station hall provided limited comfort benefits while increasing energy use. These spatial and temporal differences demonstrate that uniform station-wide control may overlook distinct environmental weaknesses across functional zones. Integrating thermal, acoustic, lighting, air-quality, passenger-flow, and functional-zone variables therefore supports a shift toward zone-specific and scenario-sensitive environmental management.

The results are also consistent with occupancy-adaptive HVAC studies showing that passenger-flow or occupancy information can improve the coordination of environmental quality and energy use [14,45]. The present framework extends this direction from HVAC operation alone to the coordinated assessment of modeled comfort, energy consumption, and operational carbon emissions. Within the calibrated simulation, the proportion of uncomfortable periods in the summer station hall decreased from 20.4% to 3.1%, while annual station energy consumption and operational carbon emissions both decreased by 15.3%. Lighting and ventilation accounted for 32.5% and 22.8% of the total energy savings, respectively. These results suggest substantial operational potential in existing stations when equipment schedules and setpoints are adjusted according to passenger-flow and comfort-state information, without requiring large-scale equipment replacement.

Existing digital twin applications for indoor environmental management have mainly addressed personal thermal preference, equipment-comfort relationships, indoor air quality, or thermal-comfort assessment [29–32]. Recent studies on AI-assisted IEQ management, data-driven environmental assessment, and multi-domain performance simulation further demonstrate the value of integrating environmental information with energy-performance analysis, while also highlighting challenges in model integration and practical deployment

[46–48]. In comparison, the principal contribution of M_{UDT} is the integration of physical sensing, structured data management, context-aware holistic comfort evaluation, calibrated energy simulation, and graded decision support within a single workflow for underground transportation spaces. The four-level logic translates comfort, passenger-flow, safety, and equipment constraints into alternative HVAC, ventilation, and lighting schedules for scenario-based evaluation. Accordingly, the framework integrates real-time environmental monitoring and comfort-state identification with adaptive strategy generation, simulation-based performance evaluation, and operator-oriented decision support.

5.2. Limitations

Several validation boundaries should be considered when interpreting the reported benefits. Although the comfort model was evaluated using stratified cross-validation and a held-out test set, the reported comfort improvements represent model-estimated potential rather than passengers' actual subjective perceptions independently verified through post-intervention field surveys. In addition, the survey sample was dominated by young and frequent metro users, with 79% aged 30 years or younger, 60% female, and 80% travelling by metro at least three times per week. Field implementation, independent post-intervention surveys, and broader demographic sampling are needed to evaluate perceived comfort and improve model generalizability.

The energy and carbon results were also derived from scenario-based simulations and are sensitive to assumptions regarding passenger-flow schedules, HVAC setpoints, ventilation rates, lighting-power schedules, and the grid emission factor. In particular, the use of an annual-average constant electricity emission factor does not reflect hourly variations in grid carbon intensity. Time-varying emission factors may alter the estimated carbon benefits when adaptive operation shifts electricity demand across hours with different generation mixes [49]. Future work should therefore incorporate temporally resolved grid emission factors when evaluating carbon-oriented operational strategies. The carbon assessment covers operational emissions from station electricity consumption and does not include embodied emissions associated with sensors, equipment, retrofit, or maintenance. Moreover, the case study

represents one early-built transfer station in Beijing's cold climate. Application to other climates and underground-space typologies requires recalibration of meteorological boundaries, passenger-flow patterns, equipment characteristics, and comfort thresholds; thus, the transferability of the study lies primarily in its methodological workflow rather than in the reported numerical savings. Finally, the data-preprocessing procedures improve routine input quality but do not constitute fault-tolerant control. Future work should extend the present one-at-a-time sensitivity analysis to joint parameter uncertainty and interactions among multiple inputs, together with multi-station field validation, equipment-level closed-loop feedback, fail-safe operation under sensor or communication failure, and testing under schedule disruptions and extreme weather.

6. Conclusions

This study addresses the contradictions between insufficient peak-period comfort and off-peak energy waste in traditional fixed-schedule environmental control strategies by developing a digital twin-enabled adaptive optimization framework (M_{UDT}) for underground subway stations. The framework was evaluated through field measurements, subjective surveys, machine-learning-based comfort assessment, and calibrated energy simulation at a typical transfer station in Beijing. The core conclusions are:

(i) Overall subway passenger comfort results from the combined effects of multiple environmental factors and operational conditions. The interaction between thermal environment and air quality showed the highest SHAP attribution during summer peak periods, reaching 35.3%. The weights of the environmental dimensions dynamically adjust according to passenger crowding. Based on calculated comfort thresholds of 0.42 and 0.68, four-level warning grades and corresponding response strategies were established.

(ii) The simulation-based adaptive optimization results suggest simultaneous improvements in modeled comfort and energy performance. Simulation results indicate that the proposed strategy not only substantially reduces uncomfortable periods across all spaces during winter and summer, for example, the proportion of periods with $S_{overall} < 0.42$ in the summer station hall decreases from 20.4% to 3.1%, but also reduces the annual total station

energy consumption and carbon emissions by 15.3%. Lighting (32.5%) and ventilation (22.8%) contribute the majority of energy savings, demonstrating that significant energy-saving potential can be achieved through operational strategy optimization without hardware modifications.

(iii) The digital twin framework supports adaptive environmental management by linking real-time monitoring and comfort-state identification with graded strategy generation and simulation-based evaluation. The calibrated EnergyPlus model provides a basis for estimating seasonal thermal-load tendencies and evaluating alternative operation scenarios, while the four-level warning mechanism translates comfort and operational constraints into actionable management strategies. Together, these components support adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework.

(iv) The proposed technical framework does not require large-scale retrofitting of existing electromechanical systems and provides a replicable solution for intelligent environmental management in existing subway stations. Future research can expand the sample scope to include more cities and station types, and explore the influence of passenger psychological factors and individual differences on comfort perception, aiming to develop a more refined, personalized environmental control system.

Funding

This research was funded by the General Project of the National Social Science Foundation of China (Project No. 25BKX005) titled ‘Research on the Collaborative Optimization Mechanism and Path of Resilience and Comfort of Urban Underground Space Environment Driven by Digital Twin’.

Data availability

The data that support the findings of this study are openly available in the GitHub repository at https://github.com/DanielBoBo/paper_experiment_data.git.

References

- [1] Y. (Annie) Pan, Q. Cheng, A. Li, J. Zhang, J. Guo, Y. Chen, Analysis of congestion key parameters, dynamic discharge process, and capacity estimation at urban freeway bottlenecks: a case study in Beijing, China, *Transportation Letters* 17 (2025) 984–1003. <https://doi.org/10.1080/19427867.2024.2404349>.
- [2] C. Lin, L. Wu, H. Xia, M. Zhen, C. Shen, J. Zhu, X. Li, Characteristics of the thermal environment, air quality, and passenger comfort in the underground transfer space of metro stations in Beijing, *Journal of Building Engineering* 59 (2022) 105093. <https://doi.org/10.1016/j.jobe.2022.105093>.
- [3] T. Zou, J. Guan, Y. Wang, F. Zheng, Y. Lin, Y. Zhao, Research on the Thermal Comfort Experience of Metro Passengers Under Sustainable Transportation: Theory of Stimulus-Organism-Response Integration with a Technology Acceptance Model, *Sustainability* 17 (2025) 362. <https://doi.org/10.3390/su17010362>.
- [4] Y. Yu, S. You, H. Zhang, T. Ye, Y. Wang, S. Wei, A review on available energy saving strategies for heating, ventilation and air conditioning in underground metro stations, *Renewable and Sustainable Energy Reviews* 141 (2021) 110788. <https://doi.org/10.1016/j.rser.2021.110788>.
- [5] X. Dong, Y. Wu, X. Chen, H. Li, B. Cao, X. Zhang, X. Yan, Z. Li, Y. Long, X. Li, Effect of thermal, acoustic, and lighting environment in underground space on human comfort and work efficiency: A review, *Science of The Total Environment* 786 (2021) 147537. <https://doi.org/10.1016/j.scitotenv.2021.147537>.
- [6] Z. Fang, T. Guo, F. Li, X. Feng, Z. Zhai, Built environment and thermal comfort in metro systems: A case study in South China, *Tunnelling and Underground Space Technology* 162 (2025) 106621. <https://doi.org/10.1016/j.tust.2025.106621>.
- [7] X. Hu, N. Li, J. Gu, Y. He, A. Yongga, Lighting and thermal factors on human comfort, work performance, and sick building syndrome in the underground building environment, *Journal of Building Engineering* 79 (2023) 107878. <https://doi.org/10.1016/j.jobe.2023.107878>.
- [8] H.S. Mayes, M. Navarro, L.P. Satchell, M.J. Tipton, S. Ando, J.T. Costello, The effects of manipulating the visual environment on thermal perception: A structured narrative review, *Journal of Thermal Biology* 112 (2023) 103488. <https://doi.org/10.1016/j.jtherbio.2023.103488>.
- [9] M. Schweiker, E. Ampatzi, M.S. Andargie, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Berger, L. Bourikas, S. Carlucci, G. Chinazzo, L.P. Edappilly, M. Favero, S. Gauthier, A. Jamrozik, M. Kane, A. Mahdavi, C. Piselli, A.L. Pisello, A. Roetzel, A. Rysanek, K. Sharma, S. Zhang, Review of multi-domain approaches to indoor environmental perception and behaviour, *Building and Environment* 176 (2020) 106804. <https://doi.org/10.1016/j.buildenv.2020.106804>.
- [10] T. Liu, X. Shan, Q. Deng, Z. Zhou, G. Yang, J. Wang, Z. Ren, Thermal Perception and Physiological Responses under Different Protection States in Indoor

- Crowded Spaces during the COVID-19 Pandemic in Summer, *Sustainability* 14 (2022) 5477. <https://doi.org/10.3390/su14095477>.
- [11] Y. Zhao, D. Li, Multi-domain indoor environmental quality in buildings: a review of their interaction and combined effects on occupant satisfaction, *Build. Environ.* 228 (2023) 109844. <https://doi.org/10.1016/j.buildenv.2022.109844>.
- [12] M. Bavaresco, V. Gnecco, I. Pigliautile, C. Piselli, M. Bracht, R. Cureau, L. De Souza, M. Geraldi, N.G. Vasquez, C. Fabiani, E. Ghisi, R. Lamberts, A.P. Melo, A.L. Pisello, Multi-domain simulation for the holistic assessment of the indoor environment: a systematic review, *J. Build. Eng.* 84 (2024) 108612. <https://doi.org/10.1016/j.jobe.2024.108612>.
- [13] G. Huang, S.T. Ng, D. Li, Y. Zhang, State of the art review on the HVAC occupant-centric control in different commercial buildings, *J. Build. Eng.* 96 (2024) 110445. <https://doi.org/10.1016/j.jobe.2024.110445>.
- [14] Z. Ren, J.I. Kim, J. Kim, Assessment methodology for dynamic occupancy adaptive HVAC control in subway stations integrating passenger flow simulation into building energy modeling, *Energy and Buildings* 300 (2023) 113667. <https://doi.org/10.1016/j.enbuild.2023.113667>.
- [15] W. AlBalkhy, D. Karmaoui, L. Ducoulombier, Z. Lafhaj, T. Linner, Digital twins in the built environment: definition, applications, and challenges, *Autom. Constr.* 162 (2024) 105368. <https://doi.org/10.1016/j.autcon.2024.105368>.
- [16] N. Babanagar, B. Sheil, J. Ninić, Q. Zhang, S. Hardy, Digital twins for urban underground space, *Tunnelling and Underground Space Technology* 155 (2025) 106140. <https://doi.org/10.1016/j.tust.2024.106140>.
- [17] V.A. Arowoia, R.C. Moehler, Y. Fang, Digital twin technology for thermal comfort and energy efficiency in buildings: a state-of-the-art and future directions, *Energy Built Environ.* 5 (2024) 641–656. <https://doi.org/10.1016/j.enbenv.2023.05.004>.
- [18] S. Yeom, J. Kim, H. Kang, S. Jung, T. Hong, Digital twin (DT) and extended reality (XR) for building energy management, *Energy Build.* 323 (2024) 114746. <https://doi.org/10.1016/j.enbuild.2024.114746>.
- [19] F. Tao, M. Zhang, Y. Liu, A.Y.C. Nee, Digital twin driven prognostics and health management for complex equipment, *CIRP Annals* 67 (2018) 169–172. <https://doi.org/10.1016/j.cirp.2018.04.055>.
- [20] C. Semeraro, M. Lezoche, H. Panetto, M. Dassisti, Digital twin paradigm: a systematic literature review, *Comput. Ind.* 130 (2021) 103469. <https://doi.org/10.1016/j.compind.2021.103469>.
- [21] D.-G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, Digital twin application in the construction industry: a literature review, *J. Build. Eng.* 40 (2021) 102726. <https://doi.org/10.1016/j.jobe.2021.102726>.
- [22] Z. Ye, Y. Ye, C. Zhang, Z. Zhang, W. Li, X. Wang, L. Wang, L. Wang, A digital twin approach for tunnel construction safety early warning and management, *Computers in Industry* 144 (2023) 103783. <https://doi.org/10.1016/j.compind.2022.103783>.

- [23] B. Du, J. Ye, H. Zhu, L. Sun, Y. Du, Intelligent Monitoring System Based on Spatio-Temporal Data for Underground Space Infrastructure, *Engineering* 25 (2023) 194–203. <https://doi.org/10.1016/j.eng.2022.07.016>.
- [24] G. Chinazzo, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Becchio, L. Belussi, C. Berger, S. Carlucci, S.P. Corgnati, S. Crosby, L. Danza, L. De Castro, M. Favero, S. Gauthier, R.T. Hellwig, Q. Jin, J. Kim, M. Sarey Khanie, D. Khovalyg, C. Lingua, A. Luna-Navarro, A. Mahdavi, C. Miller, I. Mino-Rodriguez, I. Pigliautile, A.L. Pisello, R.F. Rupp, A.-M. Sadick, F. Salamone, M. Schweiker, M. Syndicus, G. Spiglantini, N.G. Vasquez, D. Vakalis, M. Vellei, S. Wei, Quality criteria for multi-domain studies in the indoor environment: critical review towards research guidelines and recommendations, *Build. Environ.* 226 (2022) 109719. <https://doi.org/10.1016/j.buildenv.2022.109719>.
- [25] E. Mamulova, M. Loomans, R. Loonen, M. Schweiker, H. Kort, Let's talk scalability: the current status of multi-domain thermal comfort models as support tools for the design of office buildings, *Build. Environ.* 242 (2023) 110502. <https://doi.org/10.1016/j.buildenv.2023.110502>.
- [26] M. Esrafilian-Najafabadi, F. Haghighat, Occupancy-based HVAC control systems in buildings: a state-of-the-art review, *Build. Environ.* 197 (2021) 107810. <https://doi.org/10.1016/j.buildenv.2021.107810>.
- [27] M. Kong, B. Dong, R. Zhang, Z. O'Neill, HVAC energy savings, thermal comfort and air quality for occupant-centric control through a side-by-side experimental study, *Appl. Energy* 306 (2022) 117987. <https://doi.org/10.1016/j.apenergy.2021.117987>.
- [28] Y. Hu, L. Qin, S. Li, X. Li, R. Zhou, Y. Li, W. Sheng, Adaptive corrected parameters algorithm applied in cooling load prediction based on black-box model: a case study for subway station, *Energy Build.* 297 (2023) 113429. <https://doi.org/10.1016/j.enbuild.2023.113429>.
- [29] M.M. Abdelrahman, A. Chong, C. Miller, Personal thermal comfort models using digital twins: preference prediction with BIM-extracted spatial-temporal proximity data from Build2Vec, *Build. Environ.* 207 (2022) 108532. <https://doi.org/10.1016/j.buildenv.2021.108532>.
- [30] H.H. Hosamo, H.K. Nielsen, D. Kraniotis, P.R. Svennevig, K. Svidt, Improving building occupant comfort through a digital twin approach: a bayesian network model and predictive maintenance method, *Energy Build.* 288 (2023) 112992. <https://doi.org/10.1016/j.enbuild.2023.112992>.
- [31] Y. Qian, J. Leng, K. Zhou, Y. Liu, How to measure and control indoor air quality based on intelligent digital twin platforms: a case study in China, *Build. Environ.* 253 (2024) 111349. <https://doi.org/10.1016/j.buildenv.2024.111349>.
- [32] Z. ElArwady, A. Kandil, M. Afiffy, M. Marzouk, Modeling indoor thermal comfort in buildings using digital twin and machine learning, *Developments in the Built Environment* 19 (2024) 100480. <https://doi.org/10.1016/j.dibe.2024.100480>.

- [33] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Code for design of metro (GB 50157-2013), (2013). (in Chinese).
- [34] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Acoustical requirement and measurement on station platform of urban rail transit, (2024).
- [35] C. Qian, Y. Guo, A. Hussaini, A. Musa, A. Sai, W. Yu, A New Layer Structure of Cyber-Physical Systems under the Era of Digital Twin, *ACM Trans. Internet Technol.* (2024) 3674974. <https://doi.org/10.1145/3674974>.
- [36] D. Zhu, G. Jiang, X. Liu, X. Zhao, Y. Shen, J. Luo, Digital twin-enabled subgrade monitoring: a BIM-integrated data management and visualization, *Eng. Res. Express* 7 (2025) 025129. <https://doi.org/10.1088/2631-8695/ade1a5>.
- [37] ISO 10551:2019, Ergonomics of the thermal environment — Assessment of the influence of the thermal environment using subjective judgement scales, (2019).
- [38] ASHRAE, Thermal Environmental Conditions for Human Occupancy, (2023).
- [39] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Design standard for energy efficiency of public buildings, (2015).
- [40] A. Fuller, Z. Fan, C. Day, C. Barlow, Digital twin: enabling technologies, challenges and open research, *IEEE Access* 8 (2020) 108952–108971. <https://doi.org/10.1109/ACCESS.2020.2998358>.
- [41] A. Afram, F. Janabi-Sharifi, Theory and applications of HVAC control systems – a review of model predictive control (MPC), *Build. Environ.* 72 (2014) 343–355. <https://doi.org/10.1016/j.buildenv.2013.11.016>.
- [42] A. Ashrae, Ashrae guideline 14: measurement of energy and demand savings, *Am. Soc. Heat. Refrig. Air-Cond. Eng.* 35 (2002) 41–63.
- [43] M.P. Braga, I.L. Niza, E.E. Broday, Assessment between indoor environmental quality aspects and productivity in buildings: a systematic literature review, *Build. Environ.* 278 (2025) 112983. <https://doi.org/10.1016/j.buildenv.2025.112983>.
- [44] M. Pellegatti, C. Visentin, S. Torresin, F. Babich, P. Wargocki, N. Prodi, Combined and cross-modal effects of acoustic and indoor air quality conditions on sensory and cognitive responses of university students, *Build. Environ.* 288 (2026) 114005. <https://doi.org/10.1016/j.buildenv.2025.114005>.
- [45] I. Qaisar, W. Liang, K. Sun, T. Xing, Q. Zhao, An experimental comparative study of energy saving based on occupancy-centric control in smart buildings, *Build. Environ.* 268 (2025) 112322. <https://doi.org/10.1016/j.buildenv.2024.112322>.
- [46] M. Trounce, D. Anderson, W. Bahnfleth, L. Bates, S. Bhangar, R. Bolin, W. Chen, S. Chwalek, S. Frank, J. Hartke, K. Kumagai, G. Lagoudas, E. Malmstrom, S. McCrady, C. Metzger, A. Mikszewski, D. Nall, B. Owens, J. Salas, S. Taylor, W. Vernon, L. Morawska, AI as a catalyst for synergistic gains in indoor air quality and energy efficiency, *Build. Environ.* 289 (2026) 114069. <https://doi.org/10.1016/j.buildenv.2025.114069>.

[47] R. Minassian, A.-S. Mihăiță, A. Shirazi, Optimizing indoor environmental prediction in smart buildings: a comparative analysis of deep learning models, Energy Build. 327 (2025) 115086. <https://doi.org/10.1016/j.enbuild.2024.115086>.

[48] B.D.C. Loeser, M. Bavaresco, A.P. Melo, A multi-domain simulation approach for enhanced indoor environmental quality in offices: challenges and opportunities, Energy Build. 341 (2025) 115829. <https://doi.org/10.1016/j.enbuild.2025.115829>.

[49] L. Álvarez Flórez, T. Péan, J. Salom, Hourly based methods to assess carbon footprint flexibility and primary energy use in decarbonized buildings, Energy Build. 294 (2023) 113213. <https://doi.org/10.1016/j.enbuild.2023.113213>.

Appendix A

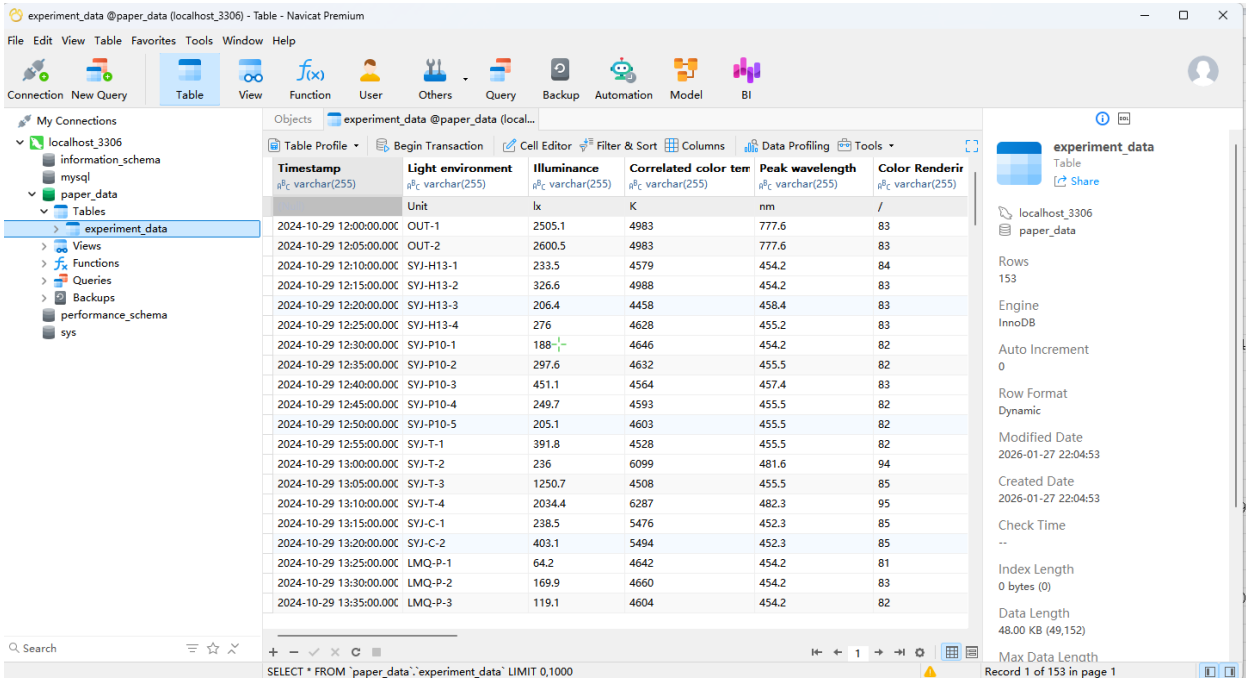


Fig. A.1. MySQL software interface.

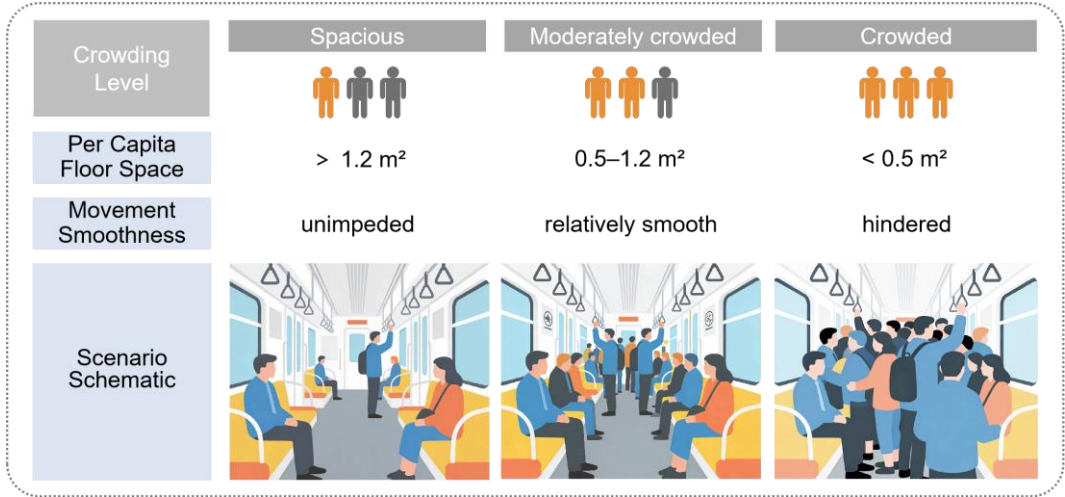
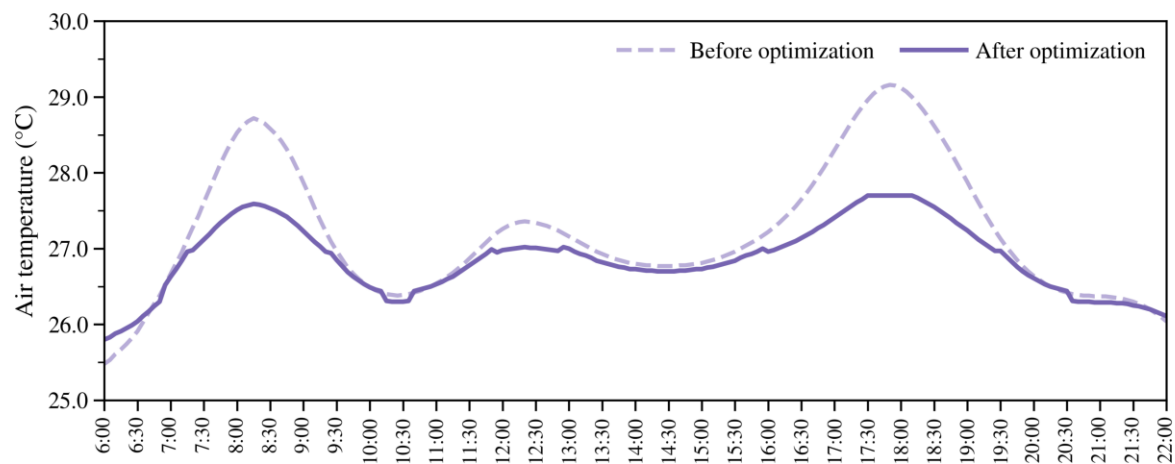
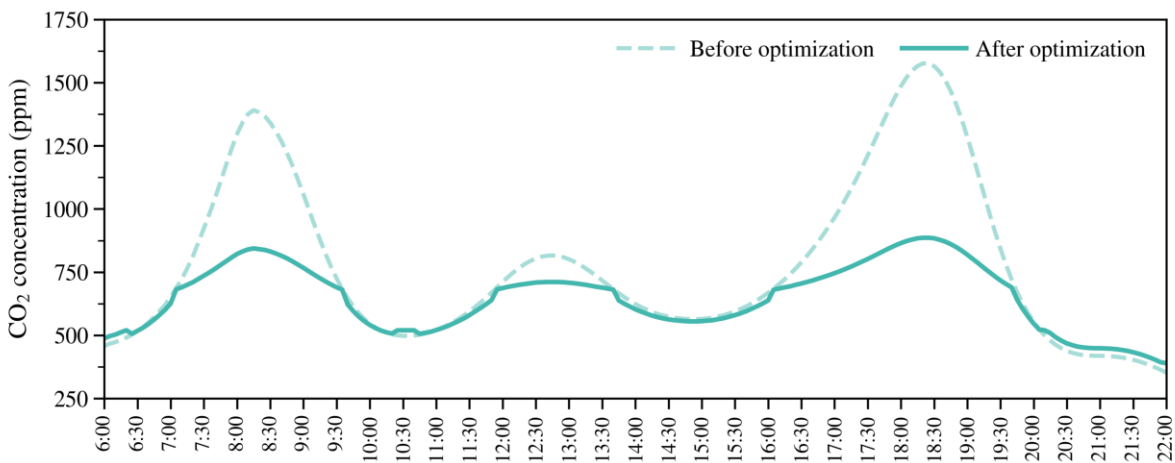


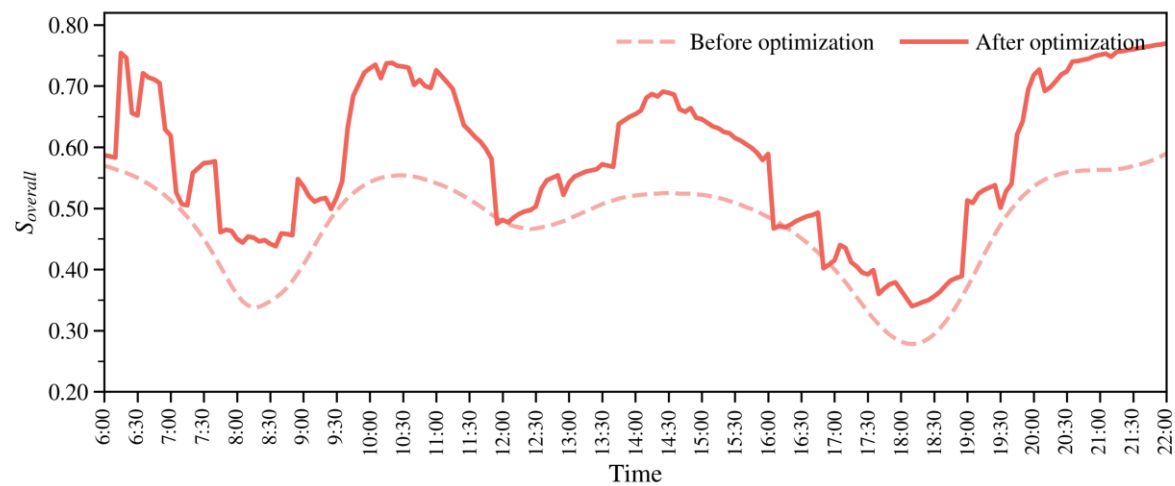
Fig. A.2. Classification rules for metro crowding degree.



(a) Air temperature



(b) CO₂ concentration



(c) $S_{overall}$

Fig. A.3. Time-series comparison of the summer platform under baseline and adaptive optimization scenarios: (a) air temperature; (b) CO₂ concentration; (c) $S_{overall}$.

Appendix B

Table B.1

Parameters of real-time monitoring equipment.

Instrument	Measurement content	Measurement range	Accuracy
Huapu MK630-TH	Air temperature	-40–120 °C	±0.2 °C
	Humidity	0–100% RH	±2% RH
Dual-color cloud spectrum HP250C spectrometer	Illuminance	5–200,000 lx	±4% of reading ±1 reading
	Correlated color temperature	1000–100000 K	Color rendering accuracy: ±1.5%; Chromaticity coordinate accuracy: ±0.0015
FN-TCE1860 environmental noise automatic monitor	Sound pressure level	30–130 dB	±3 dB
Hiciv air detector B6A	CO ₂	0–9999 ppm	± 40 ppm ± 3% reading
	PM _{2.5}	0–999 µg/m ³	± 15%
	PM ₁₀	0–999 µg/m ³	± 15%
	Formaldehyde	0–9.999 mg/m ³	± 0.03 mg/m ³ or 10%
	TVOC	0.22–9.99 mg/m ³	± 0.03 mg/m ³ or 10%
ToF passenger flow statistics terminal	Passenger flow	3.45 m × 2.52 m at an installation height of 3.2 m	≥ 98% under standard environment

Table B.2

Validation summary of the comfort evaluation model.

Item	Description
Dataset	581 synchronized subjective–objective samples
Data split	Stratified 8:2 training-test split by functional zone, crowding level, and overall comfort class
Class imbalance	SMOTE applied to the training set
Candidate models	RF and XGBoost
Hyperparameter optimization	RandomizedSearchCV followed by GridSearchCV using five-fold stratified cross-validation
Validation method	Five-fold stratified cross-validation and held-out test set
Best model	Random forest
Model performance	CV: Accuracy = 82.7%, Macro-F1 = 0.824 ± 0.020; Test: Accuracy = 67.5%, Macro-F1 = 0.618

Table B.3

Goodness-of-fit summary for Gaussian fitting.

Comfort class	Score range	n	Mean (μ)	SD (σ)	AD statistic	Q–Q plot R ²
Uncomfortable	$S_{overall} < 0.42$	197	0.31	0.07	0.462	0.986
Neutral	$0.42 \leq S_{overall} \leq 0.68$	240	0.55	0.08	0.389	0.991
Comfortable	$S_{overall} > 0.68$	28	0.77	0.06	0.571	0.978

Table B.4

Classification performance of the Gaussian-derived thresholds.

Metric	Value
Accuracy	0.7807
Macro-F1 Score	0.7602
Performance Improvement	+0.0664

Table B.5

One-at-a-time sensitivity analysis of key simulation inputs.

Simulation input	Variation	Annual energy-saving rate (%)	Change from reference (percentage points)
Reference case	—	15.25	—
Passenger density/ internal load	−10%	14.83	−0.42
	+10%	15.69	+0.44
Ventilation rate	−10%	14.73	−0.52
	+10%	15.80	+0.55
Lighting power	−10%	14.94	−0.31
	+10%	15.54	+0.30
Cooling setpoint	−1 °C	15.27	+0.03
	+1 °C	15.22	−0.03
Heating setpoint	−1 °C	15.23	−0.02
	+1 °C	15.27	+0.02

A Digital Twin-enabled Adaptive Optimization Framework for Balancing Holistic Comfort and Energy Efficiency in Underground Spaces

Zijian Ye ^{a, b, c*}, Tianlong Liang ^{a,c}, Yu Chen ^a

^a School of Art and Design, Beijing Institute of Fashion Technology, Beijing 100029, China

^b Key Laboratory of Urban Underground Engineering of Ministry of Education, Beijing Jiaotong University,
Beijing 100044, China

^c These authors contributed equally to this work and should be considered co-first authors.

* Corresponding author: E-mail: zijianye@bift.edu.cn

Abstract

Underground space environmental control often faces the dual challenge of insufficient passenger comfort and high system energy consumption. To balance holistic comfort and energy efficiency, this study proposes a digital twin-enabled adaptive optimization framework for underground environments. Environmental data were collected automatically through distributed thermal, acoustic, and lighting sensors, and a structured database was established. Field measurements, subjective comfort surveys, machine learning, and virtual simulations were integrated to construct a digital twin-enabled adaptive optimization framework balancing energy consumption and environmental comfort. A four-level early-warning and strategy-selection mechanism was evaluated in a calibrated simulation environment to coordinate comfort status and energy use under different operational scenarios. A case study was conducted at a Beijing subway station in a cold region to assess the potential effectiveness of the framework. Simulation results indicate that the strategy effectively reduced low-comfort periods across functional areas; uncomfortable periods in the station hall during summer decreased from 20.4% to 3.1%. Total annual energy consumption and operational carbon emissions both decreased by 15.3%, with lighting and ventilation contributing 32.5% and 22.8% of the energy savings, respectively. This study provides a

methodological reference for comfort-oriented and low-carbon environmental management in existing underground transportation spaces.

Keywords: Digital twin; Adaptive environmental management; Holistic comfort; Machine learning; Energy efficiency.

1. Introduction

Rapid urbanization has intensified surface traffic congestion, making subway systems a critical backbone in major cities for alleviating transportation pressure [1,2]. However, explosive growth in passenger flow poses challenges for underground stations, which are enclosed environments. High-density crowds and confined structures lead to heat and humidity accumulation, degraded air quality, and noise pollution, threatening passenger comfort and health [3]. Additionally, HVAC and lighting systems consume substantial energy to maintain indoor conditions, significantly impacting operating costs [4].

To improve the environmental quality of underground spaces, numerous studies have focused on the assessment and control of indoor environmental quality (IEQ). Many scholars have conducted in-depth investigations into individual physical factors, such as thermal conditions, lighting, air quality, or acoustic environment, and have established baseline comfort thresholds for these indicators [5]. For instance, Fang et al. analyzed measured data and survey responses from subway platforms in hot and humid regions, identifying an optimal comfort temperature range of 21.6–28.8 °C [6]. Hu et al. emphasized lighting comfort in underground spaces, showing that the optimal illuminance levels for long-term and short-term occupants are greater than 300 lx and 494 lx, respectively, which helps reduce the incidence of sick building syndrome (SBS) [7].

Despite the maturity of single-factor studies, current assessment systems have two notable limitations. First, they lack a holistic perspective on overall comfort. Environmental psychology research indicates that human perception of the environment is the result of complex interactions between multi-dimensional physical stimuli (sound, light, thermal, and air quality) and psychological factors such as crowding and perceived safety [8,9]. Focusing on individual environmental dimensions alone may not fully reflect passengers' overall

satisfaction. Second, existing models generally overlook the dynamic influence of operational contexts. Most models are constructed under static experimental conditions and do not fully consider the real-time impact of high dynamic passenger flow on comfort thresholds. For example, Liu et al. found that individuals' thermal tolerance changes significantly under crowded conditions, highlighting the need to further examine comfort differences under dynamic operational scenarios [10].

Recent studies further confirm that these limitations are not isolated issues. Multi-domain IEQ research has shown that the interactions among thermal, acoustic, visual, and air-quality domains remain insufficiently quantified, especially when overall satisfaction is considered [11]. Multi-domain simulation studies have also indicated that validated models integrating all IEQ domains are still limited [12]. In parallel, occupant-centric control research has emphasized that building environmental regulation should respond to occupancy information rather than rely solely on predefined schedules [13]. This issue is particularly pronounced in subway stations, where passenger flow is highly uneven and dynamic; passenger-flow information therefore needs to be integrated with building energy modeling to evaluate occupancy-adaptive HVAC strategies [14]. Although recent digital twin studies have demonstrated the potential of real-time sensing, virtual simulation, and decision support for built-environment management [15–18], most applications still focus on monitoring, prediction, facility management, or single-domain environmental control rather than an integrated framework that jointly considers multi-domain comfort, passenger-flow context, and energy performance. Therefore, a methodological gap remains in developing a digital twin-enabled framework that links multi-source sensing, context-aware comfort evaluation, and simulation-based energy optimization for underground transportation spaces.

To address this gap, this study proposes a digital twin-enabled adaptive optimization framework for underground space environmental management, aiming to coordinate overall passenger comfort with system energy consumption. The methodology combines field measurements, survey assessments, machine learning evaluation, visualization modeling, and energy consumption simulation to develop strategies for identifying and optimizing

environmental states in underground spaces. The effectiveness of the proposed framework is assessed through energy consumption simulations of a real-world case. The main objectives of this study are: (i) to integrate multi-source data including thermal, lighting, acoustic, air quality, and passenger crowding metrics in underground spaces; (ii) to develop a system framework that merges geometric modeling, digital twin visualization, and performance simulation; and (iii) to apply the framework to a representative underground transfer station and, through a comparative analysis of conventional and optimized control strategies, evaluate the potential benefits of the proposed digital twin framework.

2. Literature review

This section reviews the progress and remaining gaps in digital twin-enabled environmental management for underground transportation spaces, focusing on three aspects: (1) digital twins in underground environments, (2) holistic comfort assessment under dynamic operational contexts, and (3) integrated comfort–energy optimization and decision support.

2.1. Digital twins in underground environments

Digital twin research has gradually evolved from virtual representations of individual industrial assets toward integrated systems involving physical entities, virtual models, data exchange, connection mechanisms, and service-oriented functions. The five-dimensional digital twin architecture proposed by Tao et al. provides a structured representation of these components through the physical entity, virtual entity, data, connection, and service dimensions [19]. Semeraro et al. further identified continuous physical–virtual synchronization, multi-source data integration, model updating, and decision support as central characteristics of digital twins, while also noting that inconsistent definitions and limited interoperability remain common implementation barriers [20]. In the architecture, engineering, construction, operation, and facility-management fields, existing DT applications have mainly supported lifecycle information management, construction monitoring, facility operation, and asset maintenance, with relatively limited integration across the complete operational workflow [21].

Within underground environments, digital twin applications have primarily concentrated on construction safety, structural condition assessment, and infrastructure monitoring. Ye et al. developed a digital twin approach for tunnel-construction safety warning and management by integrating construction information, monitoring data, and virtual visualization [22]. Du et al. proposed an intelligent underground-infrastructure monitoring system based on spatio-temporal data, demonstrating how heterogeneous sensing records can be organized and visualized to support infrastructure-state identification [23]. These studies establish important technical foundations for underground-space sensing, data management, and virtual representation. However, their principal objects are structural or construction states rather than passenger-oriented environmental quality, and the interaction between comfort evaluation, operational context, and station energy performance is generally outside their scope.

2.2. Holistic comfort assessment under dynamic occupancy conditions

Holistic environmental comfort requires the simultaneous consideration of thermal, visual, acoustic, and air-quality stimuli. A critical review by Chinazzo et al. found substantial heterogeneity in the experimental design, terminology, analysis, and reporting of multi-domain indoor-environment studies, which restricts comparison and generalization across different settings [24]. Mamulova et al. similarly showed that the practical scalability of multi-domain comfort models is limited by differences in model scope, resolution, portability, and data requirements [25]. These limitations are particularly relevant to underground transportation spaces, because passenger exposure is transient, functional zones differ substantially, and environmental conditions change rapidly with train operation and crowding.

Occupancy-responsive environmental control provides a possible means of addressing this temporal variability. Esrafilian-Najafabadi and Haghighat showed that occupancy-based HVAC control can use current or historical occupancy information to adjust system schedules and setpoints, but its effectiveness depends on occupancy-model accuracy, sensing resolution, and the selected control logic [26]. A side-by-side experimental study by Kong et al. further

demonstrated that occupancy sensing can support simultaneous energy savings and the maintenance of thermal comfort and perceived air quality, while also showing that control performance is sensitive to sensor accuracy [27]. For subway stations, passenger-flow simulation has been coupled with building energy modeling to assess occupancy-adaptive HVAC strategies [14], and data-driven models have been developed to improve dynamic cooling-load prediction under changing station conditions [28]. Nevertheless, these studies mainly address occupancy, thermal load, ventilation, or HVAC energy use. Their outputs are not generally connected to an overall comfort model that simultaneously represents thermal, acoustic, lighting, air-quality, functional-zone, and crowding effects.

2.3. Digital twin-enabled comfort and energy management

Recent studies have begun to extend digital twins from infrastructure monitoring toward occupant-related building operation. Abdelrahman et al. incorporated BIM-derived spatial-temporal proximity data into digital twins to predict individual thermal preferences, demonstrating the potential of DT-supported personal comfort modeling [29]. Hosamo et al. integrated BIM, real-time sensor data, occupant feedback, and Bayesian networks to identify discomfort conditions and examine the relationship between equipment operation and occupant comfort [30]. Qian et al. developed an intelligent DT platform combining IoT, BIM, and machine learning for indoor-air-quality monitoring, rapid prediction, visualization, and feedback-oriented management [31]. ElArwady et al. also combined digital twins and machine-learning models to estimate indoor thermal comfort, illustrating the value of linking virtual building representations with data-driven comfort prediction [32].

Collectively, these studies demonstrate that digital twins can integrate sensing, virtual models, occupant-related data, environmental prediction, and operational decision support. However, the reviewed applications generally emphasize a particular component, such as personal thermal preference, indoor air quality, equipment diagnosis, or thermal-comfort prediction. Underground-space DT studies, meanwhile, remain primarily focused on construction and infrastructure states. A methodological gap therefore remains between three research streams: underground digital-twin monitoring, multi-domain and context-sensitive

comfort assessment, and occupancy-responsive energy management. The proposed M_{UDT} framework is positioned at this intersection. It extends the five-dimensional DT architecture by incorporating a dedicated comfort-evaluation and adaptive-optimization component, thereby linking multi-source sensing, operational-context identification, holistic comfort assessment, calibrated performance simulation, and graded decision support within a unified methodological workflow for underground transportation spaces.

3. Research methods

3.1. Design framework of the platform (M_{UDT})

The five-dimensional model proposed by Tao et al. was selected because its physical entity, virtual entity, data, connection, and service dimensions align closely with the sensing, simulation, data integration, and decision-support requirements of underground environmental control [19]. Compared with application-specific BIM-IoT or monitoring-oriented architectures, this model provides clearer cross-layer relationships and greater extensibility. Therefore, to achieve precise evaluation and adaptive optimization of underground space environments, this study builds a hybrid model for multidimensional comfort in underground spaces (M_{UDT}), integrating multi-dimensional coupled characteristics of acoustic, lighting, thermal, and air quality factors with dynamic passenger flow patterns. The resulting six-dimensional model is expressed in Equation (1) as follows:

$$M_{UDT} = (U_{PE}, U_{VE}, D_D, U_{AO}, S_S, C_N) \quad (1)$$

where U_{PE} represents the physical entity of the underground space; U_{VE} is the high-fidelity mapping of the physical entity in the digital space, encompassing not only geometric models but also physics-based performance simulation models for real-time environment state simulation; D_D represents the multi-source core data and knowledge base; C_N denotes the communication and data processing mechanisms connecting all components; U_{AO} is the core extended dimension of the model, representing the comfort evaluation and adaptive optimization algorithms, serving as the computational core; and S_S refers to the visual decision service interface.

As shown in Fig. 1, the M_{UDT} model is based on the collaborative relationships among the physical entity, virtual model, data layer, connection mechanism, evaluation and optimization algorithms, and service interface. Environmental and passenger flow data from the physical space are collected, cleaned, and temporally and spatially matched through the connection layer, then managed in the data layer. The virtual layer characterizes the underground environment using geometric and performance simulation models. The evaluation and optimization algorithm layer further computes comfort levels and energy performance, with results fed back to the service layer to support environmental control strategy formulation.

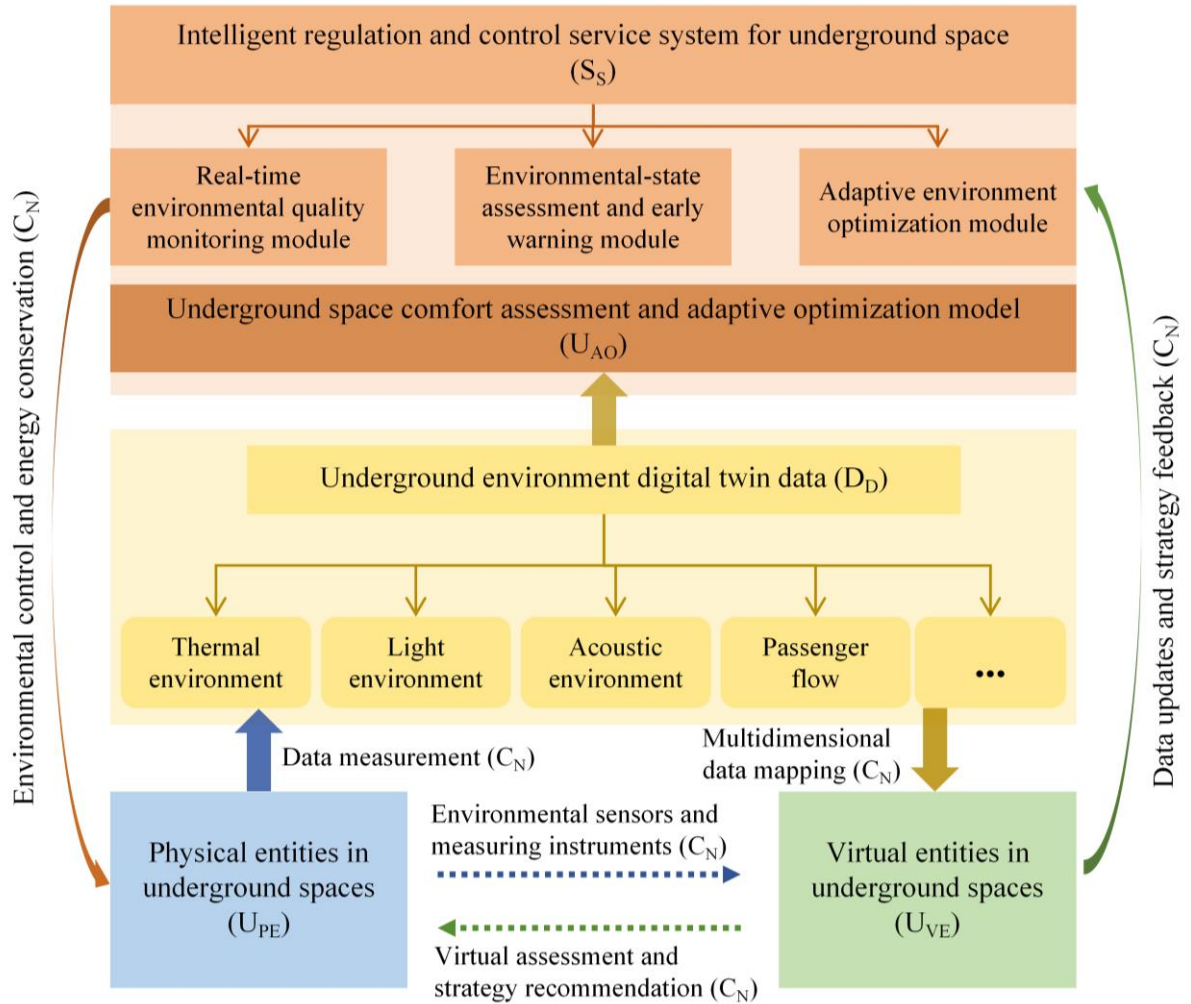


Fig. 1. Hybrid model of multidimensional comfort in underground space based on digital twin.

Based on this theoretical model, a hierarchical and modular six-layer digital twin system architecture is further developed, as shown in Fig. 2, translating the components of Equation

207 (1) into interoperable technical modules that support data acquisition, comfort assessment,
208 energy simulation, optimization strategy generation, and case-based evaluation. Construction
209 methods for each layer are detailed in Sections 3.2–3.7.

210

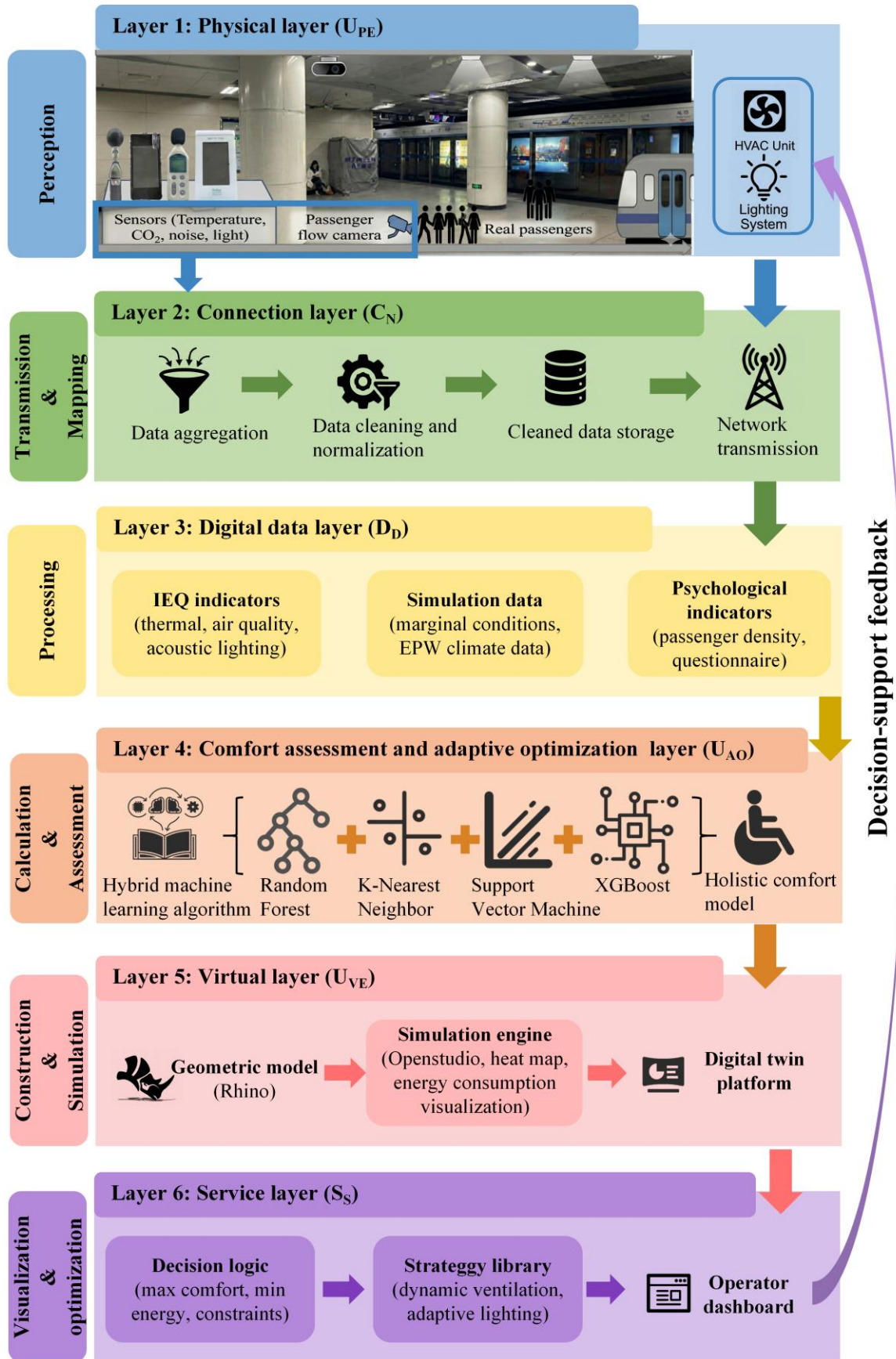


Fig. 2. System framework for environmental regulation and comfort improvement in underground spaces.

3.2. Physical layer

The physical perception layer (U_{PE}) provides fundamental data support for the M_{UDT} model, enabling high-precision mapping of environmental fields and dynamic passenger flow through a sensor network. Sensor deployment follows the guidelines of the *Code for Design of Metro* [33] and *Acoustical Requirement and Measurement on Station Platform of Urban Rail Transit* [34], covering four functional zones: station hall, platform, transfer corridor, and carriage, while also synchronously collecting outdoor meteorological data for reference. Measurement points are deployed on a 10–15 m grid, positioned at least 0.5 m away from HVAC outlets, heat sources, and walls, with monitoring heights corresponding to standard positions for human breathing, visual, and auditory perception. Passenger flow terminals are ceiling-mounted above gates and carriages, collecting data from a vertical perspective to avoid occlusion.

Based on existing specifications of physical environmental monitoring instruments, a detailed layout and deployment method for station sensors are proposed, as illustrated in Fig. 3. Detailed parameters and equipment specifications of each sensor are listed in Appendix Table B.1.

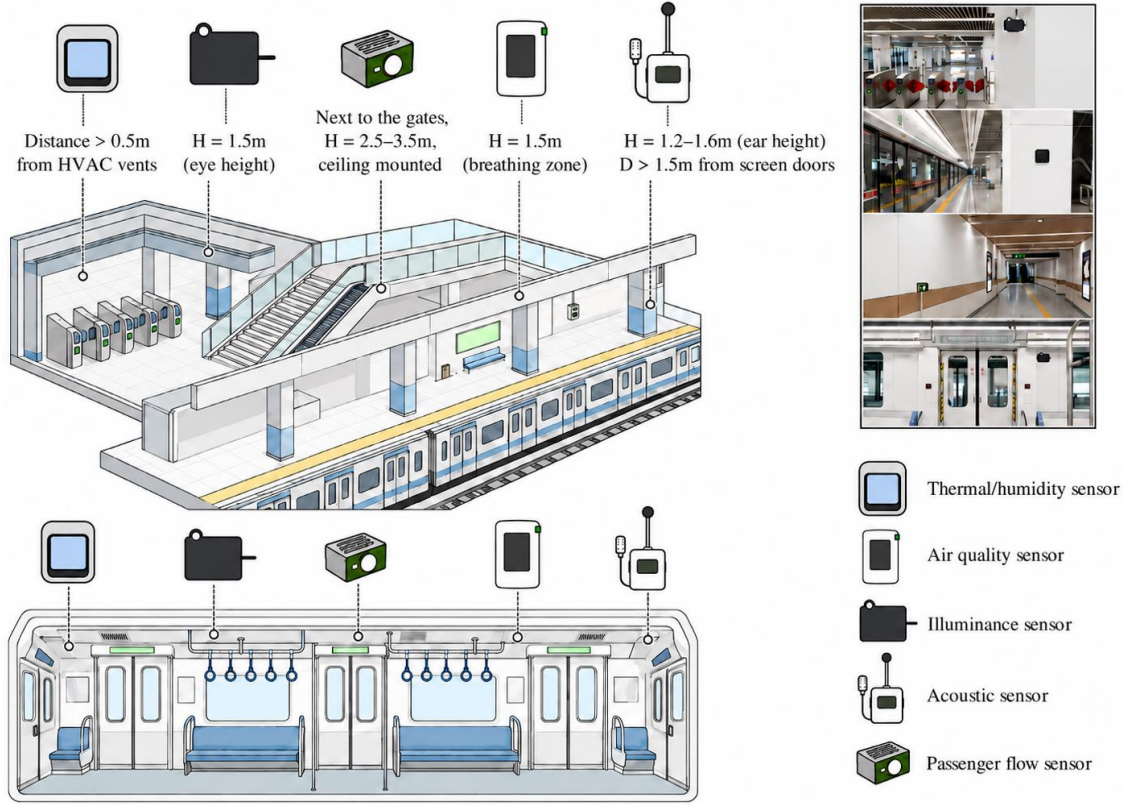


Fig. 3. Deployment strategy for monitoring instruments.

3.3. Connection layer

The connection layer (C_N) receives data collected from the Physical Perception Layer (U_{PE}) and converts it into standardized data that can be stored and accessed in the digital twin data layer (D_D) [35]. Environmental sensors and ToF passenger flow terminals deployed in the underground space are connected to on-site acquisition terminals according to their measurement point identifiers. Data are uploaded via wireless Wi-Fi to regional gateways and then transmitted through the station's local area network to the backend database.

To ensure consistency across multi-source data, the connection layer standardizes timestamps, functional zone identifiers, and sensor type fields, and performs unit conversions, outlier detection, missing data imputation, and format normalization. To further evaluate the data quality of the connection layer, the raw monitoring records were checked before being used for comfort assessment and simulation input. Missing records accounted for 1.8% of the environmental-monitoring dataset, and outliers identified by physical range checks and the 3σ criterion accounted for 0.7%. Short gaps no longer than three consecutive sampling intervals

were filled using linear interpolation, whereas longer gaps were replaced using the median value of the same functional zone and corresponding operating period. For real-time environmental and passenger flow data, the connection layer continuously writes updates to the database at fixed intervals. Non-real-time data, such as survey responses, geometric models, and meteorological boundary conditions, are linked using unified identifiers with temporal or spatial fields.

Thus, the connection layer not only facilitates the transmission of physical perception data to the digital twin platform but also establishes correspondences among physical measurement points, database records, and virtual model nodes, providing reliable input for subsequent environmental state identification, comfort evaluation, and adaptive optimization strategy generation.

3.4. Digital data layer

The digital data layer (D_D) is responsible for storing and managing multi-source heterogeneous data [36]. Based on the post-processed transmitted data, this layer implements multi-source data storage through a relational database (MySQL) and cloud storage solutions. For large-volume, unstructured static data, such as 3D geometric model files of subway stations and electromechanical equipment, the study indexes and manages them via link addresses, with the actual files stored in the cloud.

For real-time environmental and passenger flow data, as shown in Appendix Fig. A.1, structured storage and management are carried out using MySQL, supporting retrieval and backtracking of extensive historical monitoring data. By implementing unified temporal and spatial coding mapping, the system supports real-time visualization of underground space environmental states and identification of comfort weaknesses [23], laying a solid data foundation for generating adaptive environmental control strategies.

3.5. Comfort evaluation and adaptive optimization layer

3.5.1. Data sources

To construct an underground space environmental comfort assessment model, this study establishes a foundational dataset based on three types of data: on-site environmental

monitoring, passenger flow statistics, and subjective passenger surveys. Field monitoring was conducted at 11 representative Beijing subway stations, covering both winter heating and summer cooling periods. Monitored variables included thermal and humidity conditions, lighting, acoustic environment, air quality, and passenger crowding. Sensors were deployed following the principles described in Section 3.2, and the grading rules for passenger crowding are shown in Appendix Fig. A.2.

Subjective comfort data were collected simultaneously via on-site questionnaires. The surveys were designed in accordance with ISO 10551:2019 [37] and ASHRAE Standard 55 [38], using a five-level Likert scale (-2 to +2) to quantify passengers' satisfaction with thermal, lighting, acoustic, air quality, and overall environmental conditions, as illustrated in Fig. 4. After lie-detection question checks, invalid responses were removed, and response time filtering was applied, resulting in 581 valid samples with a response rate of 81.7%. The respondent profile was characterized by a high proportion of young and frequent metro users. Specifically, 79% of respondents were 30 years old or younger, 60% were female, and 80% were medium-to-high frequency passengers taking at least three trips per week. These valid samples provided reliable data support for subsequent machine learning model training, dynamic comfort assessment, and optimization strategy generation.

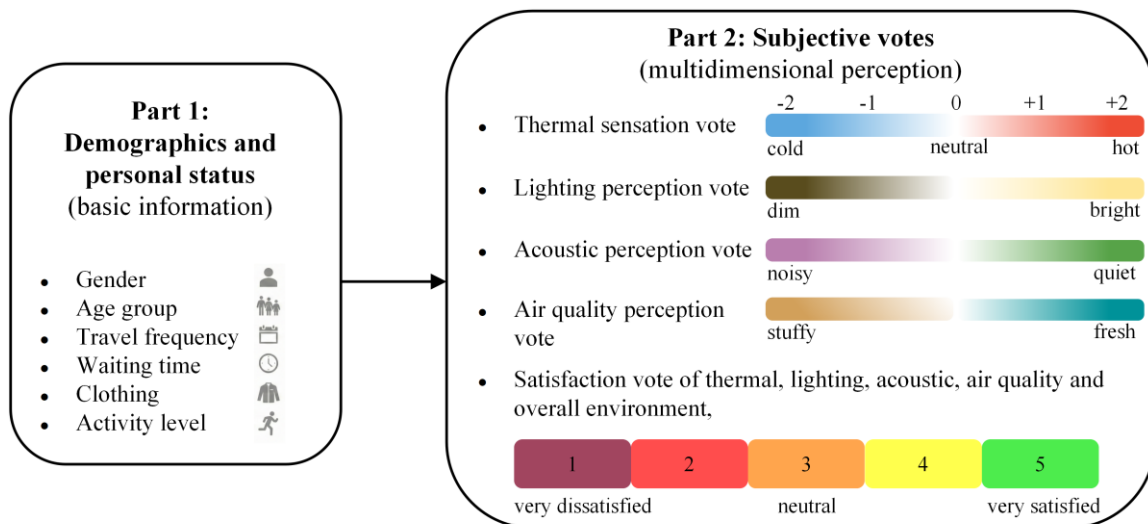


Fig. 4. Logic structure of the subjective questionnaire for passenger comfort assessment.

3.5.2. Dynamic comfort evaluation framework

To achieve precise environmental perception inference within the digital twin system, an operation scenario-adaptive comfort evaluation and decision-support framework (U_{AO}) was developed. This framework incorporates spatial functional attributes (e.g., platform, station hall) and real-time passenger crowding as key situational variables in the assessment logic.

At the computational architecture level, the system integrates machine learning algorithms such as Random Forest (RF) and Extreme Gradient Boosting (XGBoost), as shown in Fig. 5. Using synchronized subjective and objective data, passenger subjective comfort scores are used as training labels, while monitored environmental indicators, space types, and passenger crowding are input features to establish the mapping between environmental states and overall comfort. The comfort model was trained using a stratified 8:2 training–test split based on overall comfort class, functional zone, and crowding level. To address class imbalance, SMOTE was applied only to the training set. RF and XGBoost models were compared using five-fold stratified cross-validation, with macro-F1 as the primary evaluation metric. The validation procedure and model performance are summarized in Appendix Table B.2.

To enhance model interpretability, SHAP attribution analysis was applied to the trained machine-learning model. SHAP assigns feature-attribution values to individual model predictions, while aggregated mean absolute SHAP values can be used to characterize model-level feature and interaction importance. In this study, these attribution results were used to identify the relative importance of environmental features and selected interaction terms under different operational scenarios, with the results illustrated in Fig. 6. This analysis framework helps explain how environmental conditions and their interactions are associated with overall comfort under different operational scenarios, providing methodological support for generating precise quantitative evaluation benchmarks in the digital twin system.

Comfort Assessment and Adaptive Optimization Model (U_{AO})

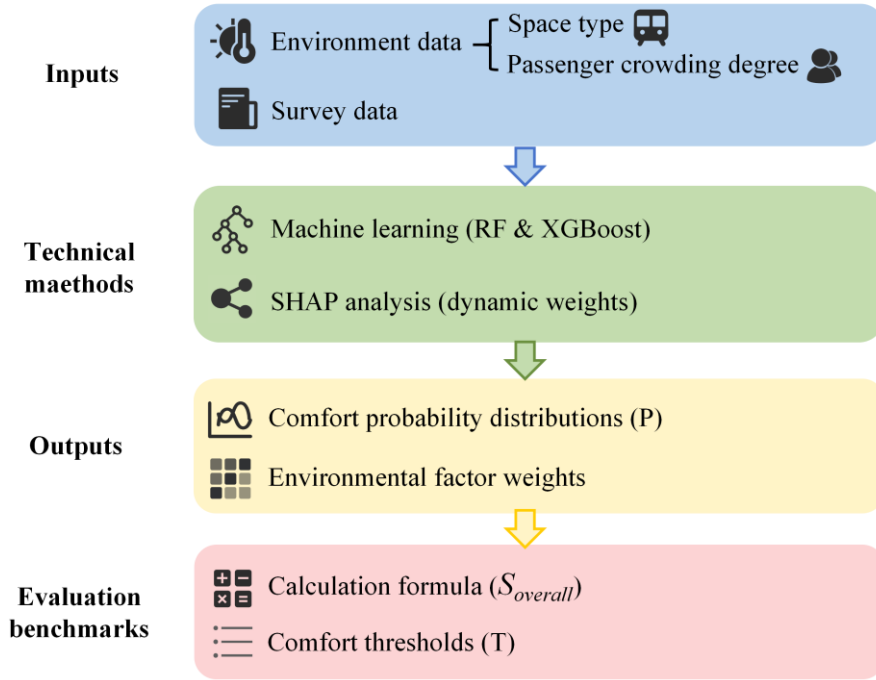


Fig. 5. Comfort evaluation framework.

Operational scenario	S_L	S_S	S_W	S_A	$S_L \times S_S$	$S_L \times S_W$	$S_L \times S_A$	$S_S \times S_W$	$S_S \times S_A$	$S_W \times S_A$
Hall										
- Spacious	0.0	7.5	11.6	5.9	5.7	5.0	0.0	31.0	7.1	16.2
- Moderately Crowded	3.1	4.0	35.3	5.4	0.0	10.0	7.7	7.5	0.0	22.6
- Crowded	0.0	19.8	16.0	3.3	14.4	13.7	0.0	6.2	12.8	12.4
Passageway										
- Spacious	16.8	4.8	14.6	16.8	0.0	8.4	5.1	0.0	6.3	22.2
- Moderately Crowded	0.0	0.0	6.0	12.6	5.1	5.7	14.0	5.0	6.1	36.2
- Crowded	5.6	9.8	22.0	8.0	10.6	0.0	8.4	16.5	0.0	14.1
Platform										
- Spacious	5.3	0.0	8.9	20.9	0.0	10.6	16.5	5.8	5.8	20.8
- Moderately Crowded	3.8	0.0	8.7	15.3	0.0	11.1	8.7	5.1	6.2	35.3
- Crowded	12.5	0.0	6.6	8.2	4.4	10.9	12.6	0.0	4.8	32.9
Carriage										
- Spacious	0.0	15.0	8.8	11.1	6.7	0.0	5.4	16.0	15.3	18.4
- Moderately Crowded	5.4	0.0	5.4	0.0	8.5	15.0	6.9	21.8	7.1	22.1
- Crowded	5.9	8.1	0.0	8.8	13.0	6.1	15.1	0.0	10.8	24.4

S_L : lighting environment
 S_S : acoustic environment

S_W : thermal environment
 S_A : air quality

SHAP attribution (%)
 0 10 20 30 40

Fig. 6. SHAP attribution percentages of environmental factors under various operational scenarios.

3.5.3. Dynamic evaluation model and grading standards

The overall comfort state was learned using nonlinear machine-learning models, in which the input features included sub-dimensional comfort scores, operational context

variables, and interaction terms among comfort dimensions. To support rapid comfort-state estimation in the digital twin platform, the trained model was further interpreted using SHAP analysis. For each operational scenario, the mean absolute SHAP values of the selected comfort features and interaction terms were normalized to obtain scenario-specific relative aggregation weights for the overall comfort calculation. This use of SHAP-derived weights provides a model-informed representation of the relative importance learned by the fitted comfort model. Based on the scenario-specific feature weights output by the machine learning framework as shown in Fig. 6, the overall comfort score is calculated as follows:

$$S_{overall} = \sum_{i=1}^4 w_{i,c} S_i + \sum_{i<j} w_{ij,c} S_i S_j \quad (2)$$

Where $S_{overall}$ denotes the overall comfort score; S_i denotes the sub-dimensional comfort score; $S_i S_j$ denotes the interaction term between two comfort dimensions; $w_{i,c}$ and $w_{ij,c}$ denote the scenario-specific normalized weights; and c denotes the operational context defined by functional zone and crowding level.

To support physical-parameter range extraction at the sub-dimensional level, a comfort-probability criterion of $P \geq 0.6$ was used to identify parameter intervals associated with a relatively high probability of being perceived as comfortable. For the overall comfort score $S_{overall}$, which serves as the primary decision variable, the classification thresholds were determined separately. Based on the distributions of $S_{overall}$ in the training set, Gaussian fitting and intersection analysis were applied to adjacent subjective evaluation classes, with the Gaussian probability density function given in Equation (3), yielding the optimal classification thresholds $T1 = 0.42$, $T2 = 0.68$. These mathematical boundaries classify environmental states into three levels: “uncomfortable” ($S_{overall} < 0.42$), “neutral” ($0.42 \leq S_{overall} \leq 0.68$) and “comfortable” ($S_{overall} > 0.68$). The class-wise goodness-of-fit summary for the Gaussian fitting is reported in Appendix Table B.3, and the classification performance of the Gaussian-derived thresholds on the test set is presented in Appendix Table

B.4. This validation provides a solid grading basis for subsequent operation-scenario decision support.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad (3)$$

Where x represents the overall comfort score $S_{overall}$. By solving the intersection equations of the adjacent evaluation level distribution curves, $f_i(x) = f_{i+1}(x)$, the optimal classification thresholds T_n are obtained.

3.6. Virtual layer

The virtual simulation layer (U_{VE}) provides physical simulation support for the Comfort Evaluation and Adaptive Optimization Layer (U_{AO}). This layer replicates the geometric structure of the subway station in the digital space and maps thermal and energy performance. The output environmental and energy consumption data are used to complement and validate the comfort evaluation results.

Based on CAD drawings of the station, 3D geometric models were constructed in Rhino, with thermal zoning applied to station halls, platforms, and transfer corridors in accordance with thermal consistency requirements, while non-critical building components were appropriately simplified. The geometric model is then imported into OpenStudio, where building envelope thermal parameters are defined according to current national standards, such as *Design Standard for Energy Efficiency of Public Buildings* [39], and on-site measured data. Soil temperature boundary conditions are set to accurately represent the thermal exchange characteristics between underground structures and surrounding soil.

Using the Grasshopper parametric platform and the Honeybee plugin, the model integration and simulation workflow is constructed. Typical local meteorological data, dynamic passenger loads, lighting, and equipment operation parameters are input as boundary conditions. EnergyPlus is used as the simulation engine to perform hourly simulations throughout the year, producing key indicators such as air temperature, mean radiant temperature, and HVAC system energy consumption. The annual operational carbon

emissions were calculated from the simulated site electricity consumption using the following equation:

$$C_{\text{total}} = E_{\text{total}} \times EF_{\text{grid}} \quad (4)$$

Where C_{total} denotes annual total carbon emissions (kgCO₂), E_{total} denotes annual total electricity consumption (kWh), and the latest official Beijing provincial grid emission factor, $EF_{\text{grid}} = 0.5554$ kgCO₂/kWh, was adopted for operational carbon-emission accounting. The Beijing provincial factor was selected because the case station is located in Beijing; the corresponding national and North China regional average factors are 0.5306 and 0.6361 kgCO₂/kWh, respectively. Different local or national average factors would affect the absolute carbon-emission estimate but not the relative reduction when the same constant factor is applied to both scenarios. The annual-average factor adopted here does not capture temporal variations in grid carbon intensity.

The simulation workflow is illustrated in Fig. 7. Simulation results are fused with comfort evaluation model data within the digital twin platform, presenting the spatial distribution of environmental parameters and comfort in 3D visualization, providing direct data support for the generation and evaluation of adaptive environmental management strategies.

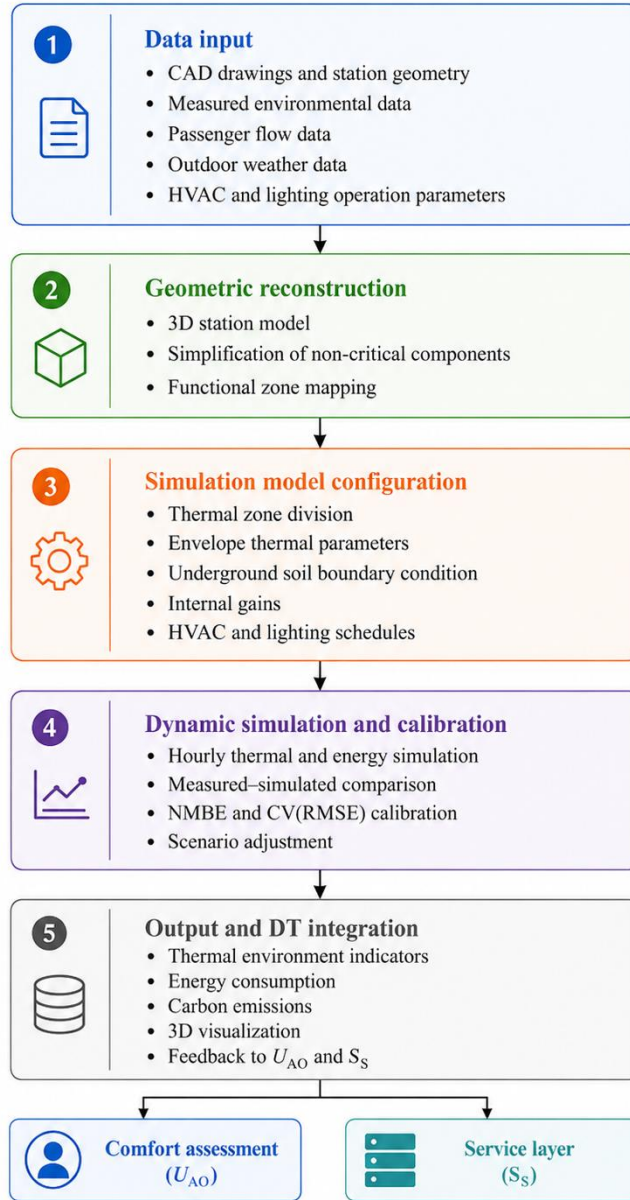


Fig. 7. Technical route for energy and comfort simulation.

3.7. Service layer

The decision service layer (S_S) transforms the comfort assessment and energy simulation results from the virtual simulation layer (U_{VE}) into environmental control strategies for operational management. As shown in Fig. 8, this layer takes real-time environmental parameters, passenger crowding, overall comfort scores, and system energy consumption as inputs, and evaluates the current operational status and corresponding response levels based on three types of constraints: safety limits, equipment operational ranges, and comfort thresholds. Accordingly, the adaptive optimization process is achieved by selecting

context-dependent environmental control strategies according to the identified operational states and generating corresponding scenarios for subsequent performance evaluation [40].

As shown in Fig. 9, the service layer jointly considers two performance goals: enhancing passenger comfort and reducing system energy consumption. Their coordination follows the graded decision logic shown in Fig. 8, in which safety and equipment constraints are prioritized, followed by comfort status and passenger-flow conditions. When environmental parameters exceed safety or equipment limits, the system enters a red emergency mode; when the overall comfort score ranges from 0.42 to 0.68, a yellow warning mode is triggered; when comfort requirements are met but passenger flow is high, a blue dynamic balance mode is activated; and when passenger flow is low and the environment is stable, a green energy-saving mode is adopted. Based on these graded results, the service layer further matches response strategies for subsystems such as ventilation, air conditioning, lighting, and broadcasting, as illustrated in Fig. 9, providing a strategic basis for scenario-based adaptive optimization simulations in subsequent case studies. These response strategies constitute the adaptive optimization scenarios, in which alternative environmental configurations are evaluated through subsequent comfort and energy simulations to identify suitable operation strategies under different conditions [41].

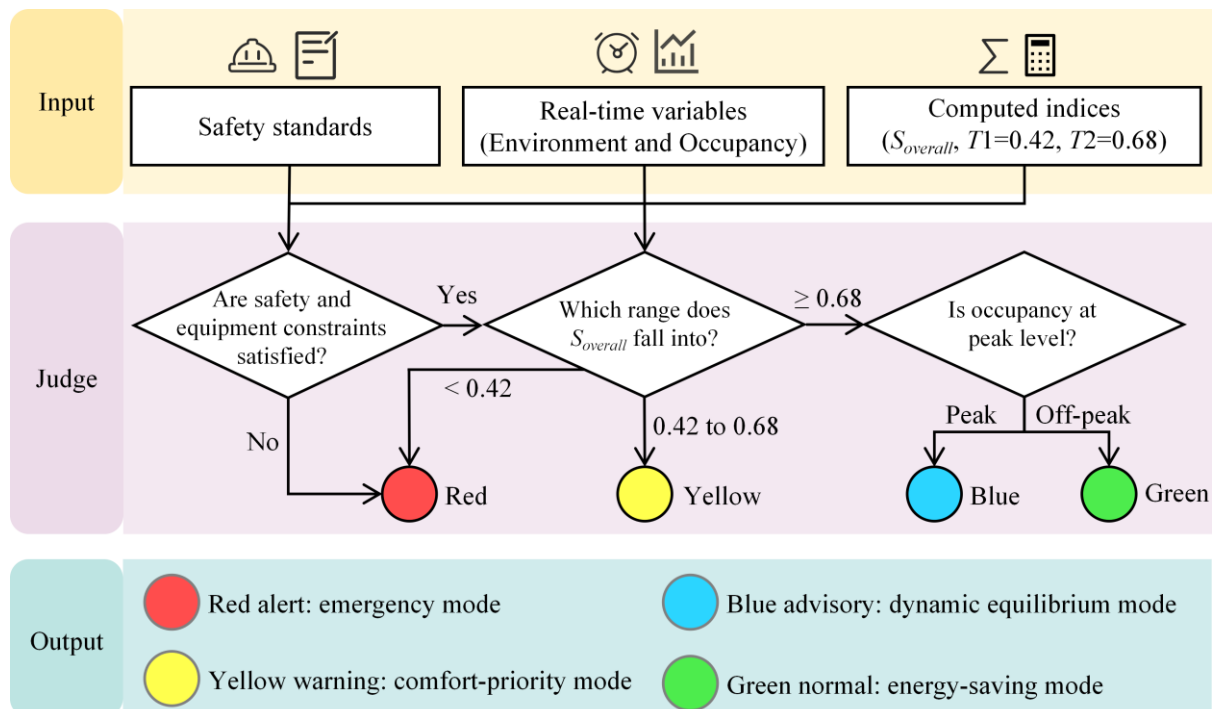


Fig. 8. Decision framework for balancing safety, comfort, and energy consumption in subway stations.

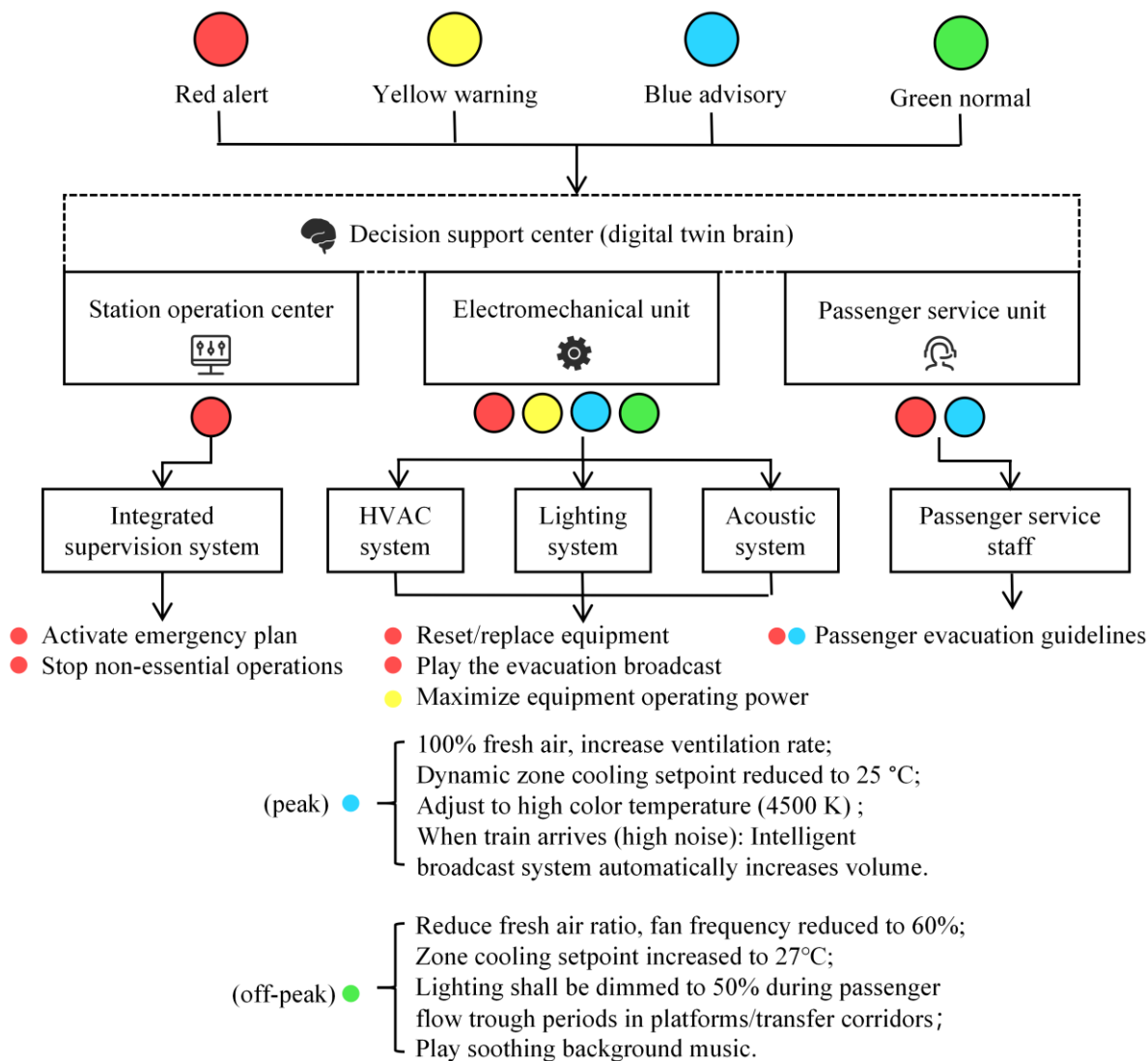


Fig. 9. Response plans of metro departments under different warning levels.

4. Case study and results

The case study was designed to assess the applicability and potential performance of the M_{UDT} framework in an existing subway station. The evaluation followed three steps: (i) establishing and calibrating the baseline simulation model using field measurements, (ii) identifying comfort and energy-performance weaknesses under typical operational scenarios, and (iii) comparing the existing and optimized strategies. Model calibration was evaluated using NMBE and CV(RMSE), while framework performance was assessed based on overall comfort scores, the proportions of comfort states, annual energy consumption, and operational carbon emissions. The underlying hypothesis was that context-aware adaptive

strategies could reduce modeled low-comfort periods and energy use compared with conventional fixed-schedule operation.

4.1. Case study

4.1.1. Basic information and passenger flow characteristics of the subway station

An early-built, medium-sized underground transfer station in Beijing was selected because it exhibits pronounced tidal passenger flow, fixed-schedule environmental control, peak-period comfort deficiencies, and off-peak energy waste. These characteristics are common challenges in existing urban rail stations and provide a suitable setting for evaluating the M_{UDT} framework under dynamic operational conditions. Opened in 2003, the station includes two station halls, platform levels for two transfer lines, and transfer corridors, as illustrated in Fig. 10.

Serving as a hub connecting large residential areas with the urban core, the station exhibits a pronounced “tidal” spatiotemporal distribution of passenger flow. During weekday peak hours, all functional areas experience extremely high passenger densities. To accurately quantify this dynamic load, real-time monitoring data from ToF terminals were combined with officially released passenger flow indices, allowing for grading of crowding levels and providing the data foundation for constructing scenario-adaptive comfort assessment models.

Regarding environmental and energy control systems, this station, being relatively old, primarily relies on conventional centralized mechanical ventilation and air conditioning (HVAC) systems for environmental regulation. Operational strategies are mostly based on fixed seasonal and time schedules, representing a typical passive response system, which leads to inadequate comfort during peak periods and energy waste during off-peak hours.

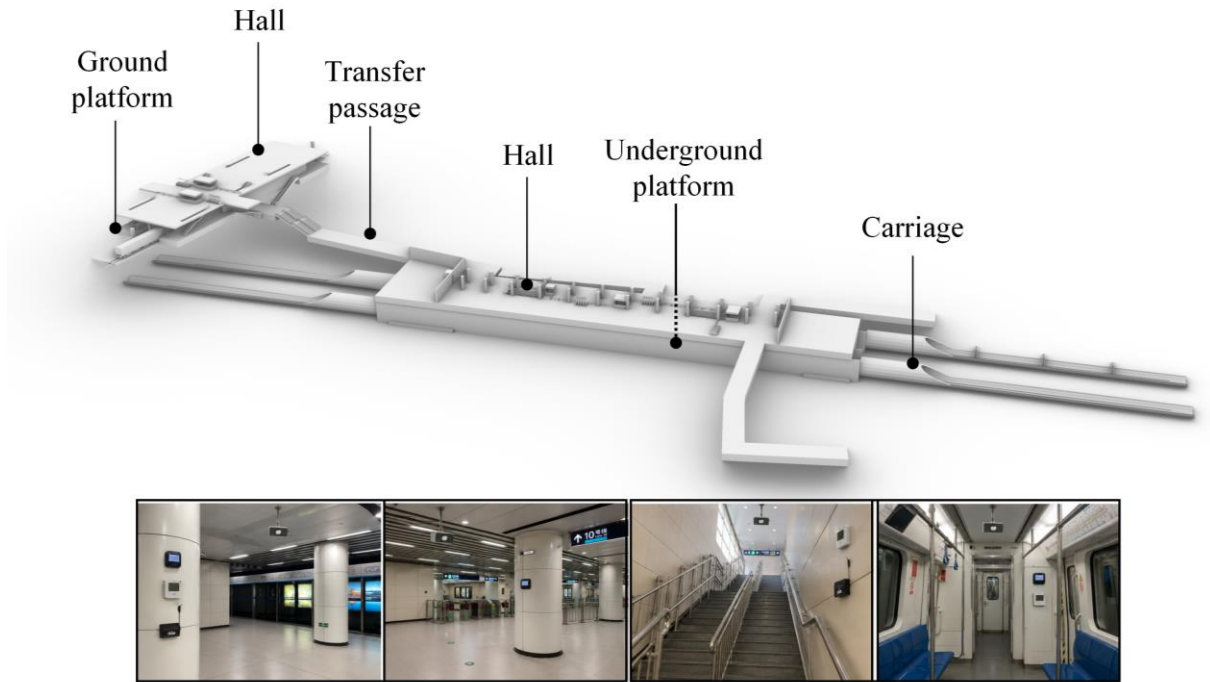


Fig. 10. Spatial layout and field views of the case subway station.

4.1.2. Data acquisition scheme

To comprehensively capture the spatiotemporal environmental characteristics of the transfer station, continuous on-site monitoring was conducted, covering morning and evening peak as well as off-peak operational periods. A total of 20 environmental monitoring nodes were deployed within the station, covering key areas including the station hall, platforms, transfer corridors, and main entrances/exits.

In the platform areas, considering that the two transfer lines are located at different underground levels and are significantly affected by train operations, screen door openings, and piston effects, measurement points were set at both core waiting areas and platform ends to capture spatial variations in thermal, humidity, and air quality conditions. Along long transfer corridors, CO₂ and PM_{2.5} monitoring nodes were primarily placed in mid-corridor sections and high-density areas to identify air quality changes under dense pedestrian movement. Main entrances and exits were equipped with reference nodes, coordinated with outdoor meteorological stations as external benchmarks. Passenger flow terminals sampled data at one-minute intervals, synchronized with environmental sensor measurements, ensuring correspondence between passenger density and environmental conditions.

4.2. Existing environmental comfort and energy consumption assessment

4.2.1. Simulation model analysis

To ensure that the digital model accurately reflects the operational status of the physical entity, hourly measured data from three consecutive summer and winter days were used to calibrate the EnergyPlus model, as shown in Fig. 11. Simulated air temperatures were compared with on-site sensor measurements. Normalized mean bias error (NMBE) and coefficient of variation of RMSE (CV(RMSE)) were used as evaluation metrics, as defined in Equations (5) and (6). The calibrated model yielded $NMBE = -0.06\%$ and $CV(RMSE) = 1.29\%$ for the typical summer period, and $NMBE = 0.77\%$ and $CV(RMSE) = 1.84\%$ for the typical winter period. All error metrics were well below the allowable limits defined by ASHRAE Guideline 14 ($|NMBE| < 10\%$, $CV(RMSE) < 30\%$) [42], indicating close agreement between simulated and measured air temperatures during the calibration periods and supporting the use of the calibrated model for subsequent evaluation. To further examine the robustness of the simulated energy-saving results, a one-at-a-time sensitivity analysis was conducted using the calibrated OpenStudio/EnergyPlus model and the original weather file. Passenger density, ventilation rate, and lighting power were varied by $\pm 10\%$, while cooling and heating setpoints were varied by $\pm 1^\circ\text{C}$, with all other inputs held constant. Across the tested cases, the annual energy-saving rate ranged from 14.7% to 15.8%, compared with 15.3% in the reference simulation, with ventilation rate producing the largest variation, as shown in Table B.5. The energy-saving tendency remained consistent across all tested input variations.

$$NMBE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)}{(n-1) \times \bar{y}} \times 100\% \quad (5)$$

$$CV(RMSE) = \frac{1}{\bar{y}} \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-1}} \times 100\% \quad (6)$$

where y_i and $\hat{y}_i$ are measured and simulated data, $\bar{y}$ is the mean of measured data, and n is the total number of data points.

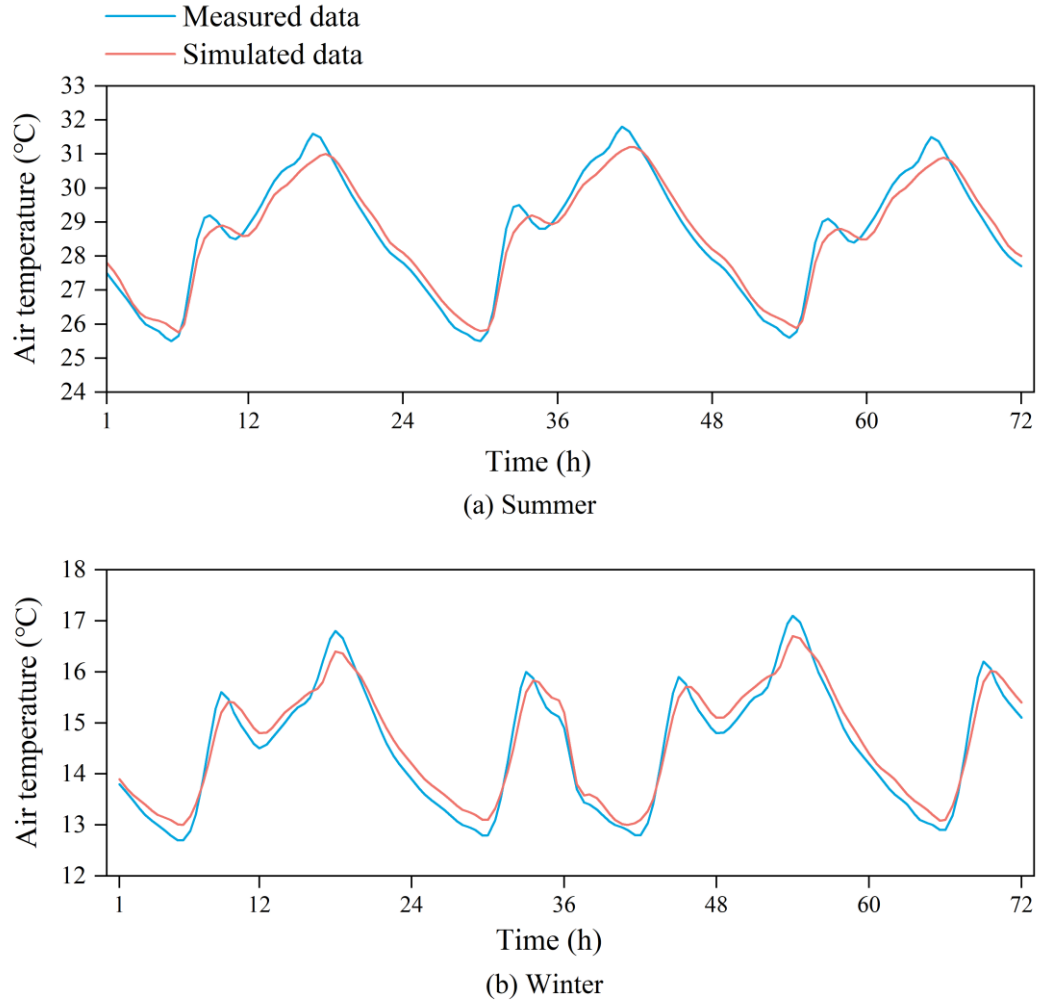


Fig. 11. Comparison of simulated and measured hourly air temperatures during typical summer and winter periods.

4.2.2. Identifying comfort weaknesses

Based on the comfort evaluation and adaptive optimization algorithm (U_{AO}), the current environmental control strategies of the case subway station were comprehensively assessed for overall comfort. Results reveal spatiotemporal differences in comfort under different operational scenarios. As shown in Table 1, temporally, the overall summer comfort score ($S_{overall}$) across all spaces is 0.48, with uncomfortable periods accounting for 36.2%—approximately 3.5 times higher than in winter (10.3%, with a score of 0.59). This seasonal difference reflects the combined challenge of high temperature, high humidity, and dense passenger flow in Beijing’s summer. Spatially, overall comfort follows the gradient:

station hall > transfer corridor > platform > carriage, with platforms and carriages identified as core targets for optimization.

Using SHAP dynamic weight attribution, during summer peak periods, the interaction of thermal environment and air quality showed the highest SHAP attribution, accounting for 35.3% in the summer-peak platform scenario. Air temperature reaches 28.7°C with 67.2% relative humidity and a CO₂ concentration of 1391 ppm, frequently causing overall comfort scores to fall below 0.42.

Further analysis combining factor contribution weights (Fig. 6) and the trends of environmental parameters and overall comfort scores (Fig. 12) indicates that lighting in the station hall exhibits hidden low-comfort conditions during off-peak periods. Between 10:00–12:00, lighting exceeds 400 lx, yet overall comfort ($S_{overall}$) decreases, creating a paradox of low passenger density but low comfort and energy waste. Uncomfortable periods are concentrated during peak hours, with delayed environmental control responses. This is due to passive control strategies, homogeneous mechanical ventilation, and HVAC power ramp-up lagging behind human heat generation and CO₂ release, highlighting the need for real-time monitoring and adaptive environmental management.

Table 1
Statistics on comfort characteristics of various functional areas.

Season	Functional area	$S_{overall}$ mean	Standard deviation	Proportion ($S_{overall} > 0.68$)	Proportion ($0.42 \leq S_{overall} \leq 0.68$)	Proportion ($S_{overall} < 0.42$)
Summer	Hall	0.56	0.08	24.2%	55.4%	20.4%
	Platform	0.44	0.11	12.1%	43.7%	44.2%
	Transfer passage	0.53	0.09	20.1%	53.6%	26.3%
	Carriage	0.40	0.12	8.8%	37.2%	54.0%
Winter	Hall	0.65	0.07	38.7%	56.2%	5.1%
	Platform	0.56	0.09	24.8%	63.5%	11.7%
	Transfer passage	0.62	0.08	33.5%	60.8%	5.7%
	Carriage	0.52	0.10	19.3%	62.1%	18.6%

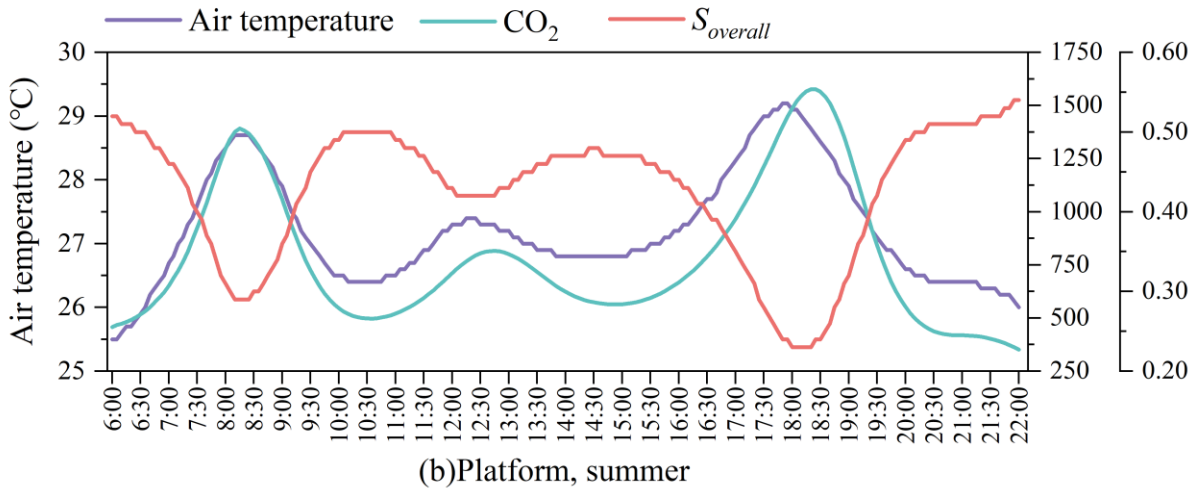
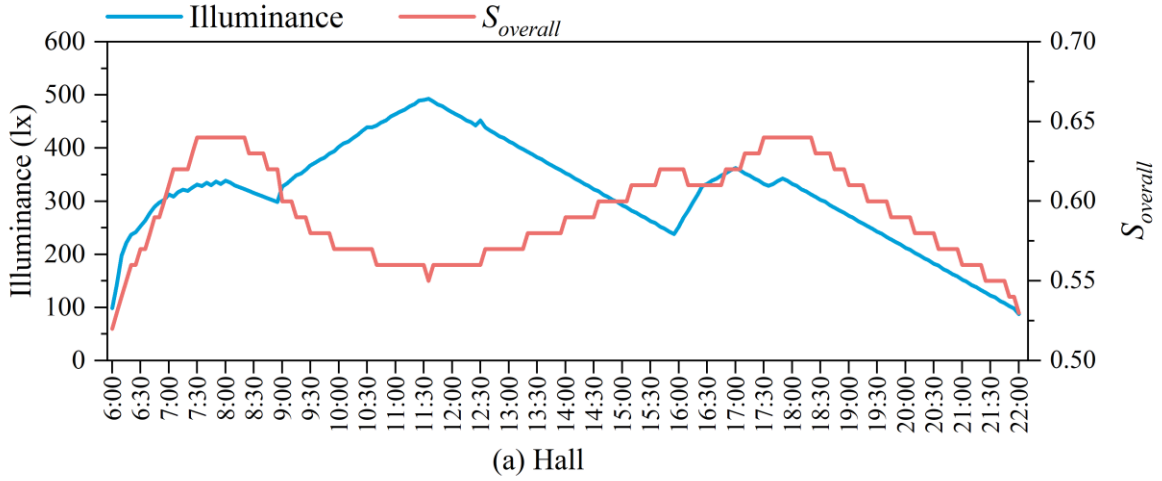


Fig. 12. Temporal evolution of environmental parameters and holistic comfort scores ($S_{overall}$) in functional zones during operation hours. (a) Station hall; (b) Platform in summer.

4.2.3. Energy consumption simulation

Using the technical workflow of the virtual simulation layer (U_{VE}), the energy consumption structure and potential savings of the case subway station were analyzed. Under the existing strategy, the environmental control system operates on a fixed timetable: fresh air and exhaust units run at constant full capacity, and lighting remains at 100% brightness. Simulation results (Fig. 13) show that mechanical power consumption (elevators, gates, etc.) accounts for the largest share of total annual energy use (26.0%), followed by the ventilation system (23.6%).

During off-peak hours, despite an approximately 60% reduction in passenger density compared to peak hours, energy consumption in ventilation and lighting does not decrease

due to the lack of dynamic adjustment, resulting in significant energy waste. Unit energy consumption per passenger during off-peak periods is 2.4 times that of peak periods. Monthly thermal balance analysis (Fig. 14) indicates a pronounced “high in winter, low in summer” asymmetry in energy demand. Winter is the peak consumption period, with January heating load reaching -159.4 kW. Average contributions from passenger heat, lighting, and electrical equipment are approximately 10.7 kW, 11.3 kW, and 16.8 kW, respectively, offsetting about 24% of infiltration heat loss. Although summer cooling loads are lower than winter heating, limited heat dissipation in underground spaces increases the precision requirement for environmental control.

These assessments reveal the core issues of the subway station’s environmental control system: static strategies cannot adapt to highly dynamic passenger flow and environmental load changes, providing a clear direction and quantitative benchmark for the adaptive optimization strategies described in subsequent sections.

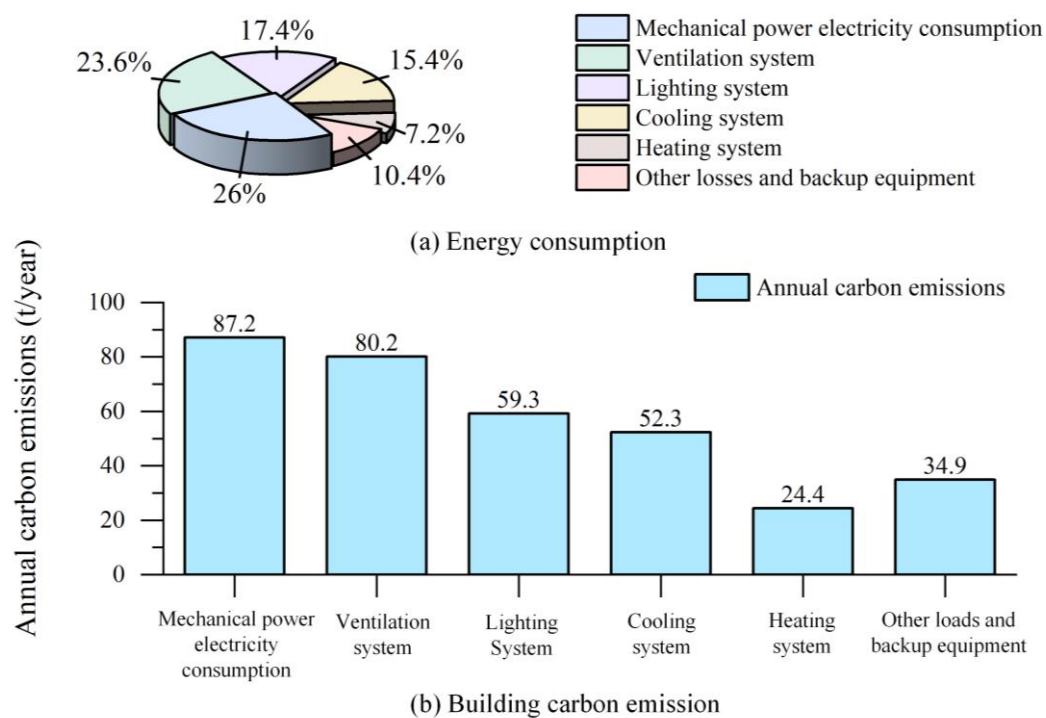


Fig. 13. Annual energy consumption and operational carbon emissions of the subway station.

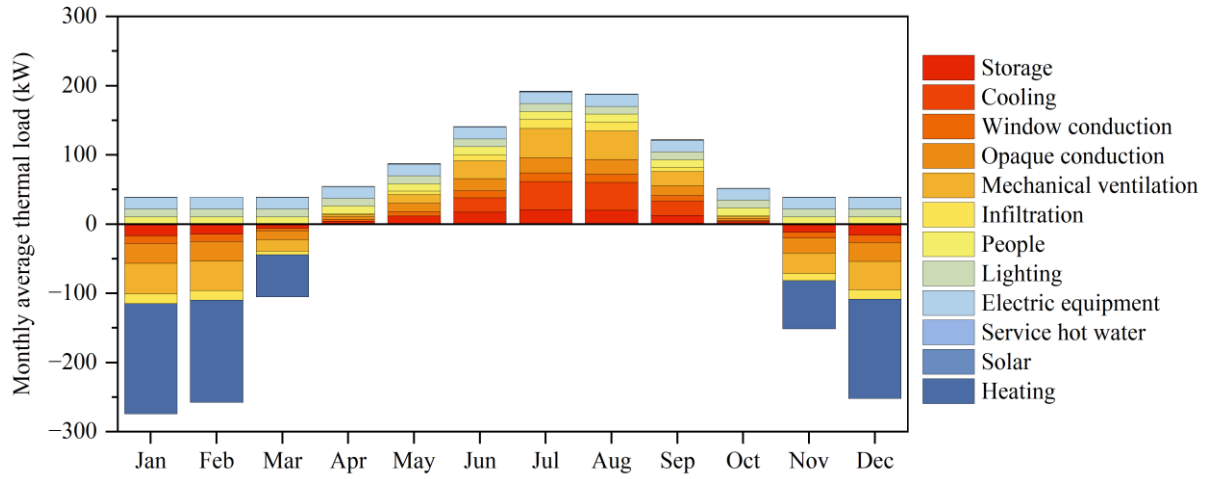


Fig. 14. Monthly thermal balance simulation of subway station.

4.3. Adaptive optimization scenarios and simulation results

4.3.1. Generation of adaptive optimization strategies

To evaluate the effectiveness of the proposed digital twin design strategy, adaptive optimization strategies balancing comfort and energy consumption were evaluated for the case subway station through simulation-based scenarios using the four-level warning decision framework constructed in Section 3.7 (S_s). The system adjusts environmental and energy strategies according to real-time passenger flow and environmental monitoring data, while simultaneously referencing three types of rigid constraints: safety limits, equipment rated operational ranges, and comfort baseline thresholds, under different passenger densities and operational scenarios (Fig. 9).

For the simulation-based evaluation, the four-level warning logic was translated into EnergyPlus schedules and setpoint assumptions. The red emergency mode was treated as a safety-override condition, and the yellow warning mode ($0.42 \leq S_{overall} \leq 0.68$) was represented by comfort-priority HVAC and ventilation operation within rated equipment limits. Under peak passenger-flow conditions, the blue dynamic-balance mode was simulated using full fresh-air operation, increased ventilation, and a reduced cooling setpoint of 25°C. Under low-load off-peak conditions, the green energy-saving mode was simulated using

reduced fresh-air supply, fan-frequency reduction to 60%, a relaxed cooling setpoint of 27°C, and lighting power reduction to 50% in platforms and transfer corridors.

The SHAP-informed prioritization strategy was used to identify the environmental features with relatively high model-attribution importance under each scenario, thereby supporting targeted adjustment of HVAC, ventilation, and lighting settings while contributing to simulated energy and emission reductions.

4.3.2. Simulated results of optimized comfort and energy consumption

In the simulation model, adaptive optimization simulations were conducted by adjusting fresh air supply, HVAC intensity, and electromechanical equipment operation power under different operational scenarios. As shown in Table 2 and Appendix Fig. A.3, environmental comfort ($S_{overall}$) improved across all functional zones. For the summer platform, the mean $S_{overall}$ increased to 0.58, the proportion of low-comfort periods with $S_{overall} < 0.42$ decreased to 12.3%, and the proportion of high-comfort periods with $S_{overall} > 0.68$ increased to 28.5%. Appendix Fig. A.3 shows that the optimized scenario reduces peak temperature and CO₂ concentration, while avoiding unnecessary over-conditioning during relatively low-load periods. Correspondingly, the modeled $S_{overall}$ remains higher and more stable throughout the day.

The simulated energy-saving effect of the adaptive optimization scenario is illustrated in Fig. 15, with the main reductions stemming from the lighting system (32.5%) and ventilation system (22.8%). Total annual station energy consumption and operational carbon emissions both decreased by 15.3%. These results demonstrate that the M_{UDT}-based comfort model can dynamically balance comfort and energy consumption, extracting significant energy-saving potential in underground spaces without compromising, and even enhancing, passenger experience, thereby providing quantifiable decision support for sustainable subway operations.

Table 2

Comfort comparison of functional areas before and after optimization.

Season	Functional area	Traditional strategy			Optimized adaptive strategy		
		$S_{overall}$	Proportion	Proportion	$S_{overall}$	Proportion	Proportion

		mean	(<i>S_{overall}</i> > 0.68)	(<i>S_{overall}</i> < 0.42)	mean	(<i>S_{overall}</i> > 0.68)	(<i>S_{overall}</i> < 0.42)
Summer	Hall	0.56	24.2%	20.4%	0.68	42.6%	3.1%
	Platform	0.44	12.1%	44.2%	0.58	28.5%	12.3%
	Transfer passage	0.53	20.1%	26.3%	0.63	37.2%	7.5%
	Carriage	0.40	8.8%	54.0%	0.52	21.4%	18.9%
Winter	Hall	0.65	38.7%	5.1%	0.71	52.3%	1.2%
	Platform	0.56	24.8%	11.7%	0.64	41.6%	2.8%
	Transfer passage	0.62	33.5%	5.7%	0.69	48.7%	1.5%
	Carriage	0.52	19.3%	18.6%	0.60	32.8%	5.6%

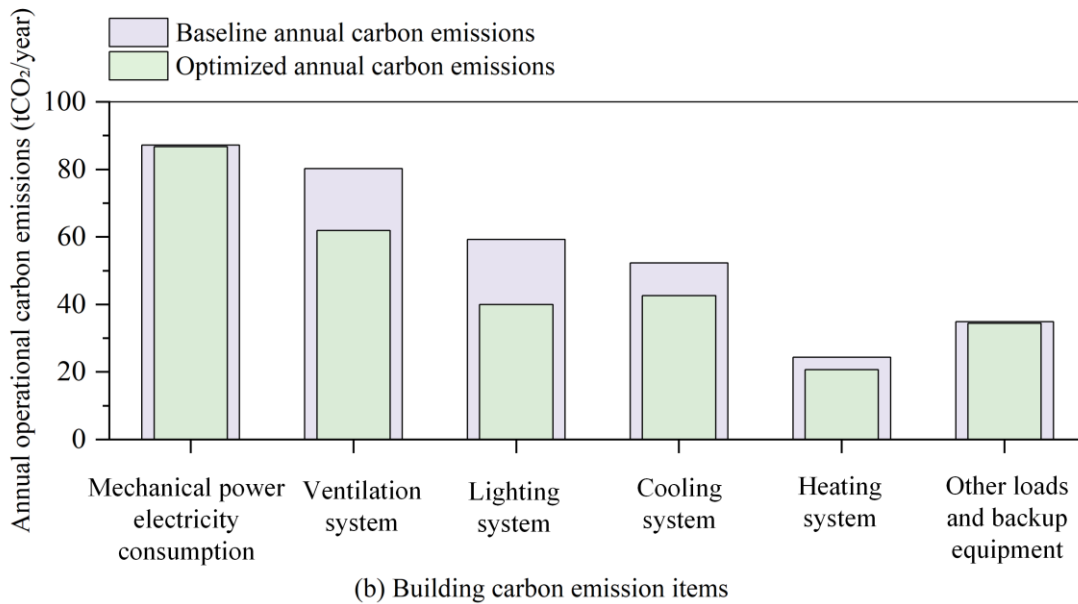
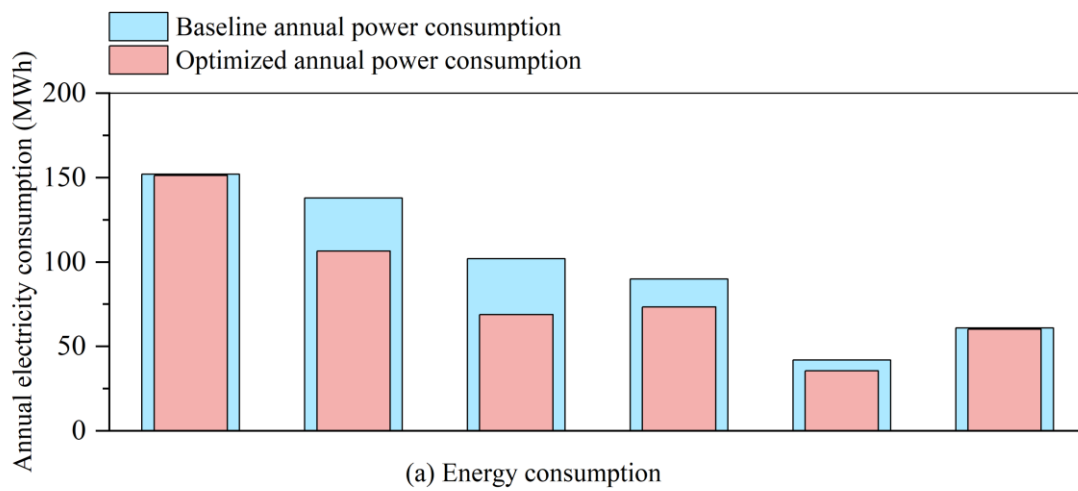


Fig. 15. Comparison of environmental regulation before and after optimization: (a) Energy consumption;
(b) Building carbon emission.

4.4. Digital twin-enabled visualization and decision support

The digital twin visualization platform serves as the terminal interface of the adaptive environmental management framework, integrating multi-dimensional sensor data, comfort-assessment results, and recommended optimization strategies. The platform visualizes monitored environmental states and comfort scores, while the comfort and energy performance of the recommended optimization strategies are evaluated in the calibrated virtual simulation layer and subsequently presented to operators.

Through a high-fidelity digital interface, operators can visually monitor the dynamic evolution of overall comfort scores across the entire station ($S_{overall}$) and visualize the real-time comfort state and the recommended adaptive operation strategies generated by the module (U_{AO}) under complex fluctuating loads. This supports the transition of environmental management from fixed-schedule operation to adaptive decision support.

As shown in Fig. 16, the platform consolidates monitored data and comfort information from the station hall, platform, transfer passage, and carriage, enabling operators to examine environmental states and identify spatial differences across functional zones. Recommended optimization strategies and their corresponding simulation-based performance information are displayed through the interface to support operational decision-making. This integrated mode allows complex environmental conditions to be quantified using intuitive ($S_{overall}$) indicators and enables operators to identify weak points in environmental quality in each sub-zone. Historical monitoring records can further be used to examine temporal environmental variations and support adaptive management under changing passenger-flow conditions.

Additionally, the visualization system supports decision-making for balancing passenger comfort and energy efficiency. By displaying sub-system energy consumption, it supports the coordinated management of passenger comfort and system energy consumption. This digital twin-based full-process monitoring supports adaptive decision-making under passenger-flow fluctuations and facilitates smarter regulation of complex environmental comfort and energy consumption in underground spaces.



(a) Hall



(b) Platform



(c) Transfer passage



(d) Carriage

Fig. 16. Digital twin visualization dashboard for environmental monitoring and decision support across different metro zones. (a) Hall; (b) Platform; (c) Transfer passage; (d) Carriage.

5. Discussion

5.1. Main findings and implications

Recent multi-domain IEQ studies have emphasized that environmental comfort results from the combined effects of thermal, visual, acoustic, and air-quality conditions rather than from a single physical variable [43,44]. The present findings support this view in underground transportation spaces, where environmental perception also varies with functional zone and passenger crowding. In particular, the thermal–air-quality interaction showed the highest SHAP attribution during summer peak periods, reaching 35.3%, as dense passenger flow simultaneously increased internal heat gain and CO₂ accumulation. The scenario-dependent variation in feature contributions further indicates that fixed environmental weights may not adequately represent comfort under dynamic subway operation. These SHAP-derived contributions should therefore be interpreted as

context-specific feature importance for comfort estimation and strategy prioritization rather than as causal control coefficients.

Compared with previous multi-domain studies that primarily focus on environmental assessment, the operation-oriented value of the proposed framework lies in identifying when, where, and why low-comfort states occur. The platform and carriage were more vulnerable to summer discomfort, whereas excessive off-peak lighting in the station hall provided limited comfort benefits while increasing energy use. These spatial and temporal differences demonstrate that uniform station-wide control may overlook distinct environmental weaknesses across functional zones. Integrating thermal, acoustic, lighting, air-quality, passenger-flow, and functional-zone variables therefore supports a shift toward zone-specific and scenario-sensitive environmental management.

The results are also consistent with occupancy-adaptive HVAC studies showing that passenger-flow or occupancy information can improve the coordination of environmental quality and energy use [14,45]. The present framework extends this direction from HVAC operation alone to the coordinated assessment of modeled comfort, energy consumption, and operational carbon emissions. Within the calibrated simulation, the proportion of uncomfortable periods in the summer station hall decreased from 20.4% to 3.1%, while annual station energy consumption and operational carbon emissions both decreased by 15.3%. Lighting and ventilation accounted for 32.5% and 22.8% of the total energy savings, respectively. These results suggest substantial operational potential in existing stations when equipment schedules and setpoints are adjusted according to passenger-flow and comfort-state information, without requiring large-scale equipment replacement.

Existing digital twin applications for indoor environmental management have mainly addressed personal thermal preference, equipment-comfort relationships, indoor air quality, or thermal-comfort assessment [29–32]. Recent studies on AI-assisted IEQ management, data-driven environmental assessment, and multi-domain performance simulation further demonstrate the value of integrating environmental information with energy-performance analysis, while also highlighting challenges in model integration and practical deployment

[46–48]. In comparison, the principal contribution of M_{UDT} is the integration of physical sensing, structured data management, context-aware holistic comfort evaluation, calibrated energy simulation, and graded decision support within a single workflow for underground transportation spaces. The four-level logic translates comfort, passenger-flow, safety, and equipment constraints into alternative HVAC, ventilation, and lighting schedules for scenario-based evaluation. Accordingly, the framework integrates real-time environmental monitoring and comfort-state identification with adaptive strategy generation, simulation-based performance evaluation, and operator-oriented decision support.

5.2. Limitations

Several validation boundaries should be considered when interpreting the reported benefits. Although the comfort model was evaluated using stratified cross-validation and a held-out test set, the reported comfort improvements represent model-estimated potential rather than passengers' actual subjective perceptions independently verified through post-intervention field surveys. In addition, the survey sample was dominated by young and frequent metro users, with 79% aged 30 years or younger, 60% female, and 80% travelling by metro at least three times per week. Field implementation, independent post-intervention surveys, and broader demographic sampling are needed to evaluate perceived comfort and improve model generalizability.

The energy and carbon results were also derived from scenario-based simulations and are sensitive to assumptions regarding passenger-flow schedules, HVAC setpoints, ventilation rates, lighting-power schedules, and the grid emission factor. In particular, the use of an annual-average constant electricity emission factor does not reflect hourly variations in grid carbon intensity. Time-varying emission factors may alter the estimated carbon benefits when adaptive operation shifts electricity demand across hours with different generation mixes [49]. Future work should therefore incorporate temporally resolved grid emission factors when evaluating carbon-oriented operational strategies. The carbon assessment covers operational emissions from station electricity consumption and does not include embodied emissions associated with sensors, equipment, retrofit, or maintenance. Moreover, the case study

represents one early-built transfer station in Beijing's cold climate. Application to other climates and underground-space typologies requires recalibration of meteorological boundaries, passenger-flow patterns, equipment characteristics, and comfort thresholds; thus, the transferability of the study lies primarily in its methodological workflow rather than in the reported numerical savings. Finally, the data-preprocessing procedures improve routine input quality but do not constitute fault-tolerant control. Future work should extend the present one-at-a-time sensitivity analysis to joint parameter uncertainty and interactions among multiple inputs, together with multi-station field validation, equipment-level closed-loop feedback, fail-safe operation under sensor or communication failure, and testing under schedule disruptions and extreme weather.

6. Conclusions

This study addresses the contradictions between insufficient peak-period comfort and off-peak energy waste in traditional fixed-schedule environmental control strategies by developing a digital twin-enabled adaptive optimization framework (M_{UDT}) for underground subway stations. The framework was evaluated through field measurements, subjective surveys, machine-learning-based comfort assessment, and calibrated energy simulation at a typical transfer station in Beijing. The core conclusions are:

(i) Overall subway passenger comfort results from the combined effects of multiple environmental factors and operational conditions. The interaction between thermal environment and air quality showed the highest SHAP attribution during summer peak periods, reaching 35.3%. The weights of the environmental dimensions dynamically adjust according to passenger crowding. Based on calculated comfort thresholds of 0.42 and 0.68, four-level warning grades and corresponding response strategies were established.

(ii) The simulation-based adaptive optimization results suggest simultaneous improvements in modeled comfort and energy performance. Simulation results indicate that the proposed strategy not only substantially reduces uncomfortable periods across all spaces during winter and summer, for example, the proportion of periods with $S_{overall} < 0.42$ in the summer station hall decreases from 20.4% to 3.1%, but also reduces the annual total station

energy consumption and carbon emissions by 15.3%. Lighting (32.5%) and ventilation (22.8%) contribute the majority of energy savings, demonstrating that significant energy-saving potential can be achieved through operational strategy optimization without hardware modifications.

(iii) The digital twin framework supports adaptive environmental management by linking real-time monitoring and comfort-state identification with graded strategy generation and simulation-based evaluation. The calibrated EnergyPlus model provides a basis for estimating seasonal thermal-load tendencies and evaluating alternative operation scenarios, while the four-level warning mechanism translates comfort and operational constraints into actionable management strategies. Together, these components support adaptive optimization of environmental operation strategies by jointly considering passenger comfort and energy performance within the graded decision framework.

(iv) The proposed technical framework does not require large-scale retrofitting of existing electromechanical systems and provides a replicable solution for intelligent environmental management in existing subway stations. Future research can expand the sample scope to include more cities and station types, and explore the influence of passenger psychological factors and individual differences on comfort perception, aiming to develop a more refined, personalized environmental control system.

Funding

This research was funded by the General Project of the National Social Science Foundation of China (Project No. 25BKX005) titled ‘Research on the Collaborative Optimization Mechanism and Path of Resilience and Comfort of Urban Underground Space Environment Driven by Digital Twin’.

Data availability

The data that support the findings of this study are openly available in the GitHub repository at https://github.com/DanielBoBo/paper_experiment_data.git.

References

- [1] Y. (Annie) Pan, Q. Cheng, A. Li, J. Zhang, J. Guo, Y. Chen, Analysis of congestion key parameters, dynamic discharge process, and capacity estimation at urban freeway bottlenecks: a case study in Beijing, China, *Transportation Letters* 17 (2025) 984–1003. <https://doi.org/10.1080/19427867.2024.2404349>.
- [2] C. Lin, L. Wu, H. Xia, M. Zhen, C. Shen, J. Zhu, X. Li, Characteristics of the thermal environment, air quality, and passenger comfort in the underground transfer space of metro stations in Beijing, *Journal of Building Engineering* 59 (2022) 105093. <https://doi.org/10.1016/j.jobe.2022.105093>.
- [3] T. Zou, J. Guan, Y. Wang, F. Zheng, Y. Lin, Y. Zhao, Research on the Thermal Comfort Experience of Metro Passengers Under Sustainable Transportation: Theory of Stimulus-Organism-Response Integration with a Technology Acceptance Model, *Sustainability* 17 (2025) 362. <https://doi.org/10.3390/su17010362>.
- [4] Y. Yu, S. You, H. Zhang, T. Ye, Y. Wang, S. Wei, A review on available energy saving strategies for heating, ventilation and air conditioning in underground metro stations, *Renewable and Sustainable Energy Reviews* 141 (2021) 110788. <https://doi.org/10.1016/j.rser.2021.110788>.
- [5] X. Dong, Y. Wu, X. Chen, H. Li, B. Cao, X. Zhang, X. Yan, Z. Li, Y. Long, X. Li, Effect of thermal, acoustic, and lighting environment in underground space on human comfort and work efficiency: A review, *Science of The Total Environment* 786 (2021) 147537. <https://doi.org/10.1016/j.scitotenv.2021.147537>.
- [6] Z. Fang, T. Guo, F. Li, X. Feng, Z. Zhai, Built environment and thermal comfort in metro systems: A case study in South China, *Tunnelling and Underground Space Technology* 162 (2025) 106621. <https://doi.org/10.1016/j.tust.2025.106621>.
- [7] X. Hu, N. Li, J. Gu, Y. He, A. Yongga, Lighting and thermal factors on human comfort, work performance, and sick building syndrome in the underground building environment, *Journal of Building Engineering* 79 (2023) 107878. <https://doi.org/10.1016/j.jobe.2023.107878>.
- [8] H.S. Mayes, M. Navarro, L.P. Satchell, M.J. Tipton, S. Ando, J.T. Costello, The effects of manipulating the visual environment on thermal perception: A structured narrative review, *Journal of Thermal Biology* 112 (2023) 103488. <https://doi.org/10.1016/j.jtherbio.2023.103488>.
- [9] M. Schweiker, E. Ampatzi, M.S. Andargie, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Berger, L. Bourikas, S. Carlucci, G. Chinazzo, L.P. Edappilly, M. Favero, S. Gauthier, A. Jamrozik, M. Kane, A. Mahdavi, C. Piselli, A.L. Pisello, A. Roetzel, A. Rysanek, K. Sharma, S. Zhang, Review of multi-domain approaches to indoor environmental perception and behaviour, *Building and Environment* 176 (2020) 106804. <https://doi.org/10.1016/j.buildenv.2020.106804>.
- [10] T. Liu, X. Shan, Q. Deng, Z. Zhou, G. Yang, J. Wang, Z. Ren, Thermal Perception and Physiological Responses under Different Protection States in Indoor

- Crowded Spaces during the COVID-19 Pandemic in Summer, *Sustainability* 14 (2022) 5477. <https://doi.org/10.3390/su14095477>.
- [11] Y. Zhao, D. Li, Multi-domain indoor environmental quality in buildings: a review of their interaction and combined effects on occupant satisfaction, *Build. Environ.* 228 (2023) 109844. <https://doi.org/10.1016/j.buildenv.2022.109844>.
- [12] M. Bavaresco, V. Gnecco, I. Pigliautile, C. Piselli, M. Bracht, R. Cureau, L. De Souza, M. Geraldi, N.G. Vasquez, C. Fabiani, E. Ghisi, R. Lamberts, A.P. Melo, A.L. Pisello, Multi-domain simulation for the holistic assessment of the indoor environment: a systematic review, *J. Build. Eng.* 84 (2024) 108612. <https://doi.org/10.1016/j.jobe.2024.108612>.
- [13] G. Huang, S.T. Ng, D. Li, Y. Zhang, State of the art review on the HVAC occupant-centric control in different commercial buildings, *J. Build. Eng.* 96 (2024) 110445. <https://doi.org/10.1016/j.jobe.2024.110445>.
- [14] Z. Ren, J.I. Kim, J. Kim, Assessment methodology for dynamic occupancy adaptive HVAC control in subway stations integrating passenger flow simulation into building energy modeling, *Energy and Buildings* 300 (2023) 113667. <https://doi.org/10.1016/j.enbuild.2023.113667>.
- [15] W. AlBalkhy, D. Karmaoui, L. Ducoulombier, Z. Lafhaj, T. Linner, Digital twins in the built environment: definition, applications, and challenges, *Autom. Constr.* 162 (2024) 105368. <https://doi.org/10.1016/j.autcon.2024.105368>.
- [16] N. Babanagar, B. Sheil, J. Ninić, Q. Zhang, S. Hardy, Digital twins for urban underground space, *Tunnelling and Underground Space Technology* 155 (2025) 106140. <https://doi.org/10.1016/j.tust.2024.106140>.
- [17] V.A. Arowoia, R.C. Moehler, Y. Fang, Digital twin technology for thermal comfort and energy efficiency in buildings: a state-of-the-art and future directions, *Energy Built Environ.* 5 (2024) 641–656. <https://doi.org/10.1016/j.enbenv.2023.05.004>.
- [18] S. Yeom, J. Kim, H. Kang, S. Jung, T. Hong, Digital twin (DT) and extended reality (XR) for building energy management, *Energy Build.* 323 (2024) 114746. <https://doi.org/10.1016/j.enbuild.2024.114746>.
- [19] F. Tao, M. Zhang, Y. Liu, A.Y.C. Nee, Digital twin driven prognostics and health management for complex equipment, *CIRP Annals* 67 (2018) 169–172. <https://doi.org/10.1016/j.cirp.2018.04.055>.
- [20] C. Semeraro, M. Lezoche, H. Panetto, M. Dassisti, Digital twin paradigm: a systematic literature review, *Comput. Ind.* 130 (2021) 103469. <https://doi.org/10.1016/j.compind.2021.103469>.
- [21] D.-G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, Digital twin application in the construction industry: a literature review, *J. Build. Eng.* 40 (2021) 102726. <https://doi.org/10.1016/j.jobe.2021.102726>.
- [22] Z. Ye, Y. Ye, C. Zhang, Z. Zhang, W. Li, X. Wang, L. Wang, L. Wang, A digital twin approach for tunnel construction safety early warning and management, *Computers in Industry* 144 (2023) 103783. <https://doi.org/10.1016/j.compind.2022.103783>.

- [23] B. Du, J. Ye, H. Zhu, L. Sun, Y. Du, Intelligent Monitoring System Based on Spatio-Temporal Data for Underground Space Infrastructure, *Engineering* 25 (2023) 194–203. <https://doi.org/10.1016/j.eng.2022.07.016>.
- [24] G. Chinazzo, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Becchio, L. Belussi, C. Berger, S. Carlucci, S.P. Corgnati, S. Crosby, L. Danza, L. De Castro, M. Favero, S. Gauthier, R.T. Hellwig, Q. Jin, J. Kim, M. Sarey Khanie, D. Khovalyg, C. Lingua, A. Luna-Navarro, A. Mahdavi, C. Miller, I. Mino-Rodriguez, I. Pigliautile, A.L. Pisello, R.F. Rupp, A.-M. Sadick, F. Salamone, M. Schweiker, M. Syndicus, G. Spigliantini, N.G. Vasquez, D. Vakalis, M. Vellei, S. Wei, Quality criteria for multi-domain studies in the indoor environment: critical review towards research guidelines and recommendations, *Build. Environ.* 226 (2022) 109719. <https://doi.org/10.1016/j.buildenv.2022.109719>.
- [25] E. Mamulova, M. Loomans, R. Loonen, M. Schweiker, H. Kort, Let's talk scalability: the current status of multi-domain thermal comfort models as support tools for the design of office buildings, *Build. Environ.* 242 (2023) 110502. <https://doi.org/10.1016/j.buildenv.2023.110502>.
- [26] M. Esrafilian-Najafabadi, F. Haghighat, Occupancy-based HVAC control systems in buildings: a state-of-the-art review, *Build. Environ.* 197 (2021) 107810. <https://doi.org/10.1016/j.buildenv.2021.107810>.
- [27] M. Kong, B. Dong, R. Zhang, Z. O'Neill, HVAC energy savings, thermal comfort and air quality for occupant-centric control through a side-by-side experimental study, *Appl. Energy* 306 (2022) 117987. <https://doi.org/10.1016/j.apenergy.2021.117987>.
- [28] Y. Hu, L. Qin, S. Li, X. Li, R. Zhou, Y. Li, W. Sheng, Adaptive corrected parameters algorithm applied in cooling load prediction based on black-box model: a case study for subway station, *Energy Build.* 297 (2023) 113429. <https://doi.org/10.1016/j.enbuild.2023.113429>.
- [29] M.M. Abdelrahman, A. Chong, C. Miller, Personal thermal comfort models using digital twins: preference prediction with BIM-extracted spatial-temporal proximity data from Build2Vec, *Build. Environ.* 207 (2022) 108532. <https://doi.org/10.1016/j.buildenv.2021.108532>.
- [30] H.H. Hosamo, H.K. Nielsen, D. Kraniotis, P.R. Svennevig, K. Svidt, Improving building occupant comfort through a digital twin approach: a bayesian network model and predictive maintenance method, *Energy Build.* 288 (2023) 112992. <https://doi.org/10.1016/j.enbuild.2023.112992>.
- [31] Y. Qian, J. Leng, K. Zhou, Y. Liu, How to measure and control indoor air quality based on intelligent digital twin platforms: a case study in China, *Build. Environ.* 253 (2024) 111349. <https://doi.org/10.1016/j.buildenv.2024.111349>.
- [32] Z. ElArwady, A. Kandil, M. Afiffy, M. Marzouk, Modeling indoor thermal comfort in buildings using digital twin and machine learning, *Developments in the Built Environment* 19 (2024) 100480. <https://doi.org/10.1016/j.dibe.2024.100480>.

- [33] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Code for design of metro (GB 50157-2013), (2013). (in Chinese).
- [34] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Acoustical requirement and measurement on station platform of urban rail transit, (2024).
- [35] C. Qian, Y. Guo, A. Hussaini, A. Musa, A. Sai, W. Yu, A New Layer Structure of Cyber-Physical Systems under the Era of Digital Twin, *ACM Trans. Internet Technol.* (2024) 3674974. <https://doi.org/10.1145/3674974>.
- [36] D. Zhu, G. Jiang, X. Liu, X. Zhao, Y. Shen, J. Luo, Digital twin-enabled subgrade monitoring: a BIM-integrated data management and visualization, *Eng. Res. Express* 7 (2025) 025129. <https://doi.org/10.1088/2631-8695/ade1a5>.
- [37] ISO 10551:2019, Ergonomics of the thermal environment — Assessment of the influence of the thermal environment using subjective judgement scales, (2019).
- [38] ASHRAE, Thermal Environmental Conditions for Human Occupancy, (2023).
- [39] Ministry of Housing and Urban-Rural Development of the People's Republic of China, Design standard for energy efficiency of public buildings, (2015).
- [40] A. Fuller, Z. Fan, C. Day, C. Barlow, Digital twin: enabling technologies, challenges and open research, *IEEE Access* 8 (2020) 108952–108971. <https://doi.org/10.1109/ACCESS.2020.2998358>.
- [41] A. Afram, F. Janabi-Sharifi, Theory and applications of HVAC control systems – a review of model predictive control (MPC), *Build. Environ.* 72 (2014) 343–355. <https://doi.org/10.1016/j.buildenv.2013.11.016>.
- [42] A. Ashrae, Ashrae guideline 14: measurement of energy and demand savings, *Am. Soc. Heat. Refrig. Air-Cond. Eng.* 35 (2002) 41–63.
- [43] M.P. Braga, I.L. Niza, E.E. Broday, Assessment between indoor environmental quality aspects and productivity in buildings: a systematic literature review, *Build. Environ.* 278 (2025) 112983. <https://doi.org/10.1016/j.buildenv.2025.112983>.
- [44] M. Pellegatti, C. Visentin, S. Torresin, F. Babich, P. Wargocki, N. Prodi, Combined and cross-modal effects of acoustic and indoor air quality conditions on sensory and cognitive responses of university students, *Build. Environ.* 288 (2026) 114005. <https://doi.org/10.1016/j.buildenv.2025.114005>.
- [45] I. Qaisar, W. Liang, K. Sun, T. Xing, Q. Zhao, An experimental comparative study of energy saving based on occupancy-centric control in smart buildings, *Build. Environ.* 268 (2025) 112322. <https://doi.org/10.1016/j.buildenv.2024.112322>.
- [46] M. Trounce, D. Anderson, W. Bahnfleth, L. Bates, S. Bhangar, R. Bolin, W. Chen, S. Chwalek, S. Frank, J. Hartke, K. Kumagai, G. Lagoudas, E. Malmstrom, S. McCrady, C. Metzger, A. Mikszewski, D. Nall, B. Owens, J. Salas, S. Taylor, W. Vernon, L. Morawska, AI as a catalyst for synergistic gains in indoor air quality and energy efficiency, *Build. Environ.* 289 (2026) 114069. <https://doi.org/10.1016/j.buildenv.2025.114069>.

[47] R. Minassian, A.-S. Mihăiță, A. Shirazi, Optimizing indoor environmental prediction in smart buildings: a comparative analysis of deep learning models, Energy Build. 327 (2025) 115086. <https://doi.org/10.1016/j.enbuild.2024.115086>.

[48] B.D.C. Loeser, M. Bavaresco, A.P. Melo, A multi-domain simulation approach for enhanced indoor environmental quality in offices: challenges and opportunities, Energy Build. 341 (2025) 115829. <https://doi.org/10.1016/j.enbuild.2025.115829>.

[49] L. Álvarez Flórez, T. Péan, J. Salom, Hourly based methods to assess carbon footprint flexibility and primary energy use in decarbonized buildings, Energy Build. 294 (2023) 113213. <https://doi.org/10.1016/j.enbuild.2023.113213>.

Appendix A

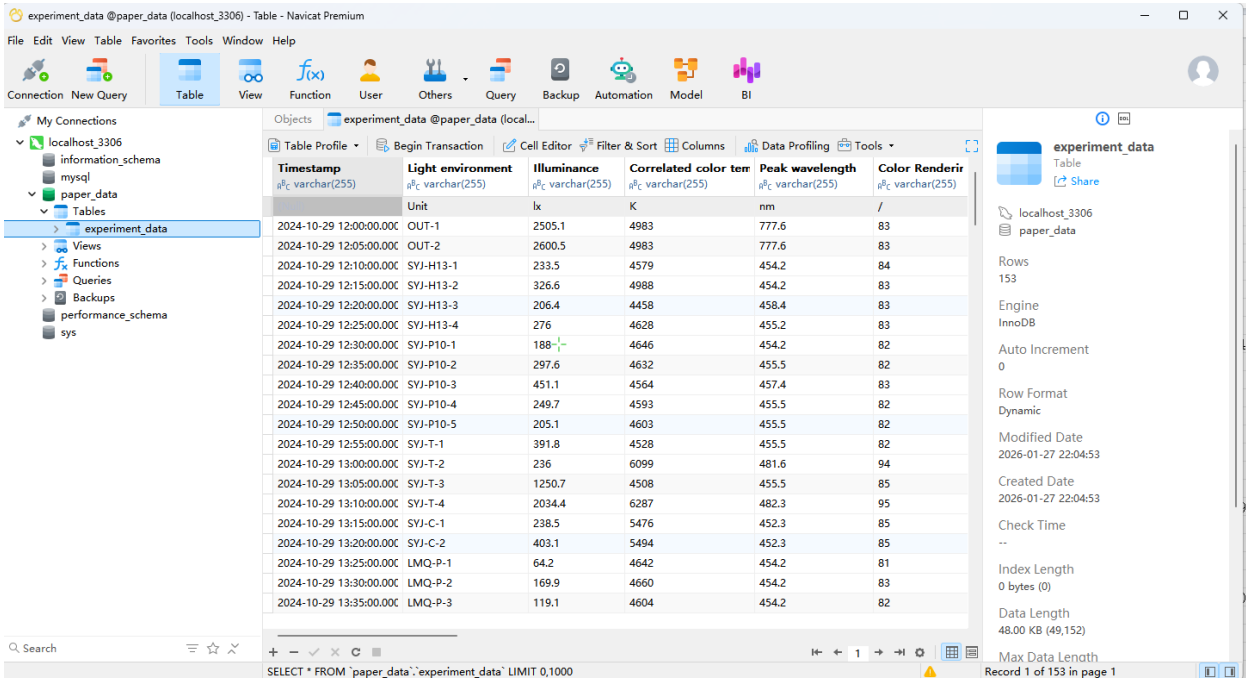


Fig. A.1. MySQL software interface.

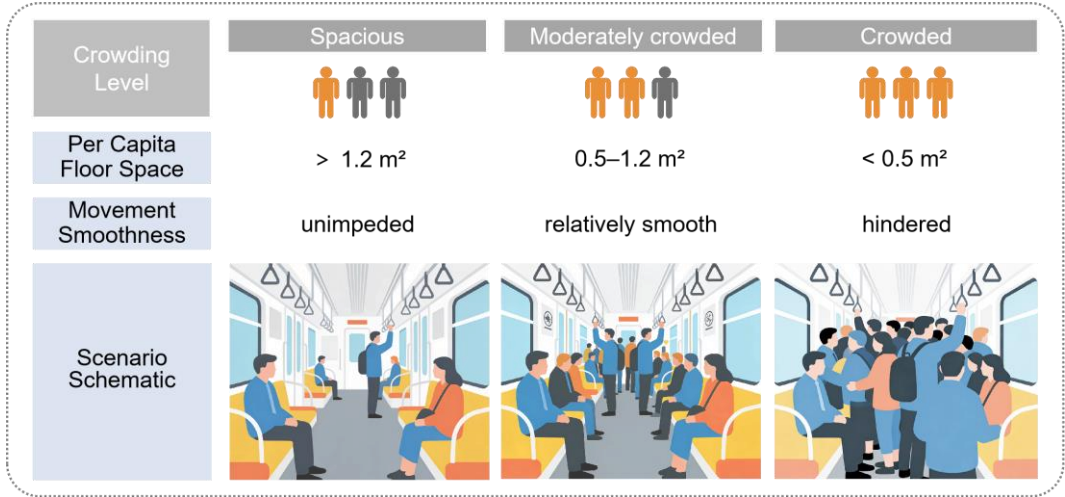
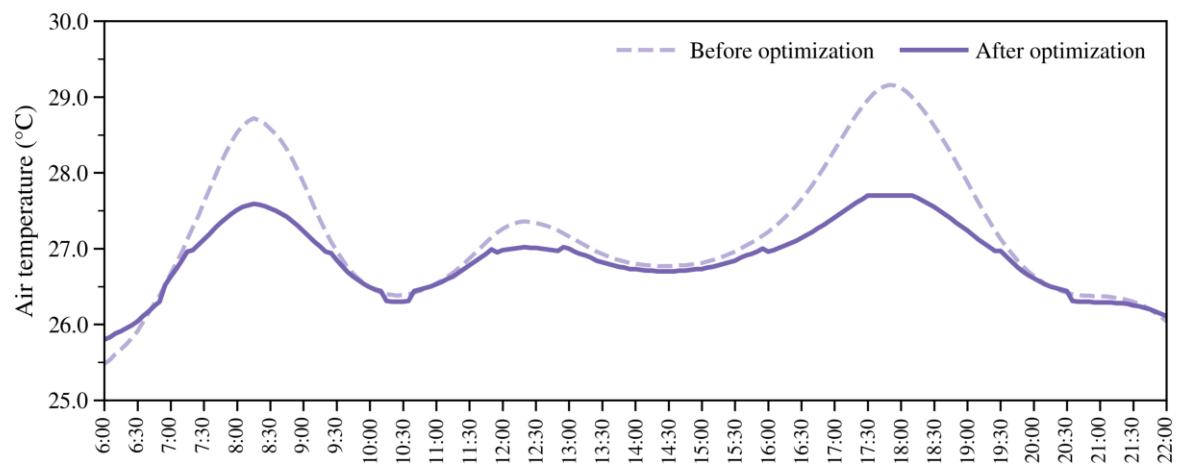
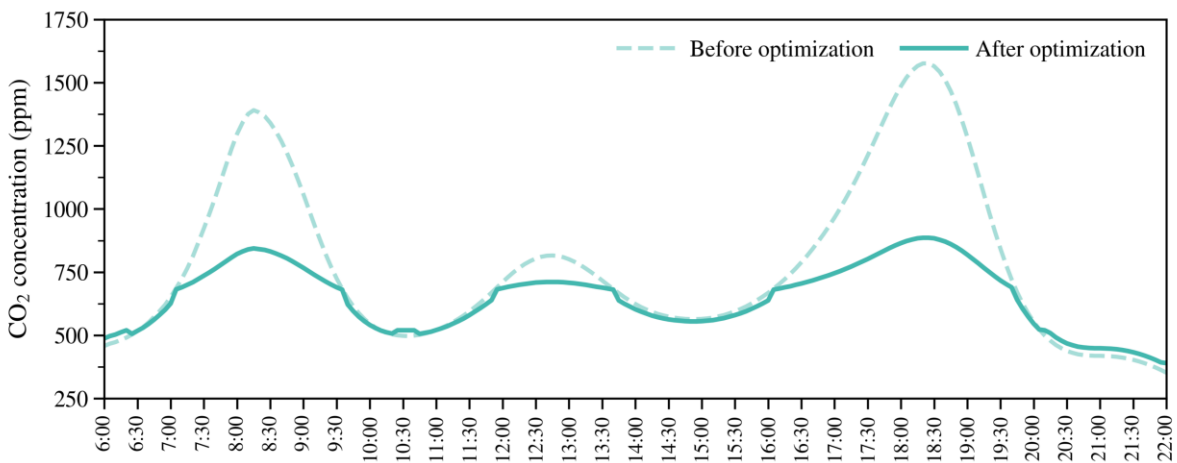


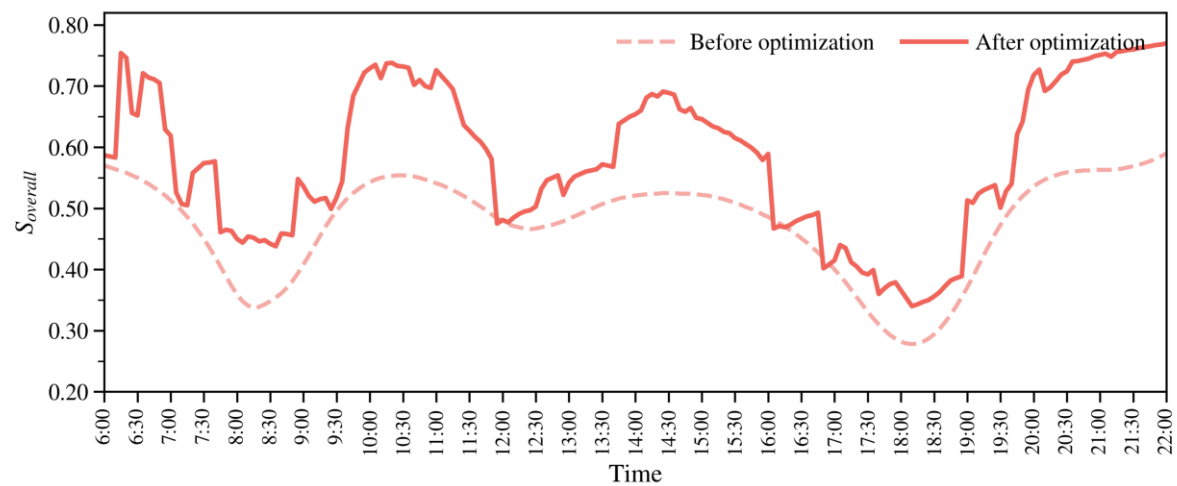
Fig. A.2. Classification rules for metro crowding degree.



(a) Air temperature



(b) CO₂ concentration



(c) $S_{overall}$

Fig. A.3. Time-series comparison of the summer platform under baseline and adaptive optimization scenarios: (a) air temperature; (b) CO₂ concentration; (c) $S_{overall}$.

Appendix B

Table B.1

Parameters of real-time monitoring equipment.

Instrument	Measurement content	Measurement range	Accuracy
Huapu MK630-TH	Air temperature	-40–120 °C	±0.2 °C
	Humidity	0–100% RH	±2% RH
Dual-color cloud spectrum HP250C spectrometer	Illuminance	5–200,000 lx	±4% of reading ±1 reading
	Correlated color temperature	1000–100000 K	Color rendering accuracy: ±1.5%; Chromaticity coordinate accuracy: ±0.0015
FN-TCE1860 environmental noise automatic monitor	Sound pressure level	30–130 dB	±3 dB
Hiciv air detector B6A	CO ₂	0–9999 ppm	± 40 ppm ± 3% reading
	PM _{2.5}	0–999 µg/m ³	± 15%
	PM ₁₀	0–999 µg/m ³	± 15%
	Formaldehyde	0–9.999 mg/m ³	± 0.03 mg/m ³ or 10%
	TVOC	0.22–9.99 mg/m ³	± 0.03 mg/m ³ or 10%
ToF passenger flow statistics terminal	Passenger flow	3.45 m × 2.52 m at an installation height of 3.2 m	≥ 98% under standard environment

Table B.2

Validation summary of the comfort evaluation model.

Item	Description
Dataset	581 synchronized subjective–objective samples
Data split	Stratified 8:2 training-test split by functional zone, crowding level, and overall comfort class
Class imbalance	SMOTE applied to the training set
Candidate models	RF and XGBoost
Hyperparameter optimization	RandomizedSearchCV followed by GridSearchCV using five-fold stratified cross-validation
Validation method	Five-fold stratified cross-validation and held-out test set
Best model	Random forest
Model performance	CV: Accuracy = 82.7%, Macro-F1 = 0.824 ± 0.020; Test: Accuracy = 67.5%, Macro-F1 = 0.618

Table B.3

Goodness-of-fit summary for Gaussian fitting.

Comfort class	Score range	n	Mean (μ)	SD (σ)	AD statistic	Q–Q plot R^2
Uncomfortable	$S_{overall} < 0.42$	197	0.31	0.07	0.462	0.986
Neutral	$0.42 \leq S_{overall} \leq 0.68$	240	0.55	0.08	0.389	0.991
Comfortable	$S_{overall} > 0.68$	28	0.77	0.06	0.571	0.978

Table B.4

Classification performance of the Gaussian-derived thresholds.

Metric	Value
Accuracy	0.7807
Macro-F1 Score	0.7602
Performance Improvement	+0.0664

Table B.5

One-at-a-time sensitivity analysis of key simulation inputs.

Simulation input	Variation	Annual energy-saving rate (%)	Change from reference (percentage points)
Reference case	—	15.25	—
Passenger density/ internal load	–10%	14.83	–0.42
	+10%	15.69	+0.44
Ventilation rate	–10%	14.73	–0.52
	+10%	15.80	+0.55
Lighting power	–10%	14.94	–0.31
	+10%	15.54	+0.30
Cooling setpoint	–1 °C	15.27	+0.03
	+1 °C	15.22	–0.03
Heating setpoint	–1 °C	15.23	–0.02
	+1 °C	15.27	+0.02



Click here to download Research data
https://github.com/danielbobo/paper_experiment_data.git

